Waves in Thermoelastic and Multiphysics Continua: Generalized Heat Conduction, Coupled Fields and Size Effects

A Unified Review with Verified Reductions, Dispersion Analysis and Benchmark Problems

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Abstract

Wave propagation in solids that conduct heat, deform, and carry further fields such as electric polarization, magnetic induction, diffusing species or microstructural degrees of freedom is governed by a family of coupled models that is usually presented one member at a time. This review organises that family around a single heat-flux template with three lags and one memory term, shows how the classical, Lord–Shulman, Green–Lindsay, Green–Naghdi, dual-phase-lag, three-phase-lag, Moore–Gibson–Thompson, fractional and memory-dependent theories arise (or fail to arise) as parameter choices, and records exactly which limits are regular and which are singular, including a correction to a widely repeated stability statement for the dual-phase-lag model. For the coupled problem we derive and symbolically verify a single dispersion polynomial, and use it to compare phase speed, attenuation and thermoelastic damping across theories; we then treat transient problems (heat pulse, thermal shock), Rayleigh waves, reflection and interface energy partition, piezoelectric, magneto-thermoelastic, diffusive and microstructural couplings, and nonlocal and gradient size effects against lattice dispersion. We discuss analytical and numerical solution techniques with a benchmark, applications from laser ultrasonics to microresonators and tissue heating, and close with a comparative synthesis, a pitfalls checklist, a decision guide and a research agenda. We also provide flexural and plate thermoelastic damping validated against exact solutions, anisotropic and free-surface reflection results, a finite-difference benchmark for hyperbolic heat conduction, worked examples and reporting standards. All quantitative figures are generated by accompanying scripts and the key identities are checked symbolically or numerically.

Keywords: generalized thermoelasticity; second sound; dual-phase-lag; Moore–Gibson–Thompson; fractional heat conduction; memory-dependent derivative; nonlocal elasticity; piezoelectricity; Rayleigh waves; thermoelastic damping; interface energy partition.

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1 Introduction

1.1 The problem that generalized thermoelasticity addresses

Heat conduction in solids is described, to an excellent approximation under ordinary conditions, by Fourier's law: the heat flux is proportional to the temperature gradient at the same instant. Combined with energy balance this gives a parabolic equation, whose solutions have a property that is physically uncomfortable: a temperature disturbance applied at one point changes the temperature everywhere else at the same instant, however small the change may be. For slow processes in bulk matter the discrepancy between this instantaneous response and what a real, finite-speed carrier of heat would do is immeasurably small. It becomes measurable, and sometimes dominant, when the heating or loading is very fast (femtosecond to nanosecond laser pulses), when the structure is very small or very cold so that phonon mean free paths approach the structure size, when the medium is strongly heterogeneous, or when the observed quantity is a high-frequency elastic wave whose period is comparable to a thermal relaxation time.

Thermoelasticity couples this thermal field to deformation: heating produces thermal stress through thermal expansion, and deformation changes the temperature through the dilatation term in the energy balance. The classical coupled theory of Biot [1], building on the thermomechanical stress–strain law of Duhamel and Neumann [2] and the early dynamic solutions of Danilovskaya [3] and Chadwick and Sneddon [4, 5], inherits the infinite thermal signal speed. The extensions that remove it, collectively called *generalized* or *non-classical* thermoelasticity [6–10], now form one of the largest strands of continuum mechanics, with a very large body of theoretical work. At the same time two further developments widened the field. First, devices exploit the coupling of deformation and temperature to additional physical fields, above all electric polarization (piezoelectricity and pyroelectricity), but also magnetic induction, diffusing species, fluid-filled pores and microstructural kinematics [11–14]. Second, at small scales the local stress–strain relation itself fails to describe dispersion, and nonlocal and gradient enrichments are used [15–18].

1.2 Why another review

Reviews of generalized thermoelasticity exist and are excellent [6–9, 19, 20]. What has been lacking is a treatment that serves a reader who wants to *use* the theories for wave problems and compare them reliably. Three observations motivate the present organisation.

1. **Proliferation without a common language.** Over the past three decades the number of named heat-conduction models for thermoelastic waves has grown very large, many of them combinations (two-temperature fractional Lord–Shulman, three-phase-lag with memory, nonlocal Moore–Gibson–Thompson, and so on). Each paper introduces its own constants and notation. Section 3.4 introduces a single heat-flux function $G(s)$ in the Laplace domain through which every linear model passes, so that a new combination is understood as a new choice of G.
2. **Unreliable folklore about limits and stability.** Several statements are repeated across papers without re-derivation: which parameter limits reduce one model to another, which parameter ranges are stable, how the wave branches should be labelled. We re-derive these and check them numerically. One widely repeated statement—that dual-phase-lag thermoelasticity requires $\tau_T \geq \tau_q/2$ to be stable—turns out to depend on the order of the Taylor

expansion that defines the model (Section 9).

3. **Figures that cannot be compared.** Published dispersion and response curves use different nondimensionalisations, parameter sets and branch conventions, so that the visual impression of "the effect of the fractional order" or "the effect of nonlocality" is hard to calibrate. We use one nondimensionalisation throughout, quantified by a small number of material-independent groups.

1.3 Contributions and organisation

This review does not report new experiments. Its contributions are:

- a unified heat-flux template, a table of *exact* reductions and singular limits, and a taxonomy that places the main models in relation to one another (Sections 3.4–11);
- a coupled dispersion polynomial valid for the whole template, verified symbolically, and comparative dispersion, damping and speed results (Sections 6–13);
- benchmark transient problems, namely the heat pulse and the thermal shock of a half-space, each with a Laplace-domain closed form and a numerical inversion (Section 16);
- thermoelastic Rayleigh waves, reflection and interface energy partition, with limit checks (Sections 18–21);
- a treatment of size effects contrasted with lattice dispersion, and of the principal multi-physics couplings (Sections 22–23);
- a kinetic basis for the Cattaneo and Guyer–Krumhansl laws with the dispersion of the latter (Section 4), and a quantitative test of how well Taylor-truncated lag laws represent the delay law they approximate (Section 10);
- flexural thermoelastic damping, including a plate (Lamb-wave) computation validated against the Rayleigh–Lamb equation and the Lifshitz–Roukes solution, resonant modes of a bar, anisotropy in cubic crystals, free-surface reflection and piezoelectric surface waves (Sections 7, 14, 15, 19, 20 and 24);
- a discussion of solution methods with finite-difference and inversion benchmarks, experimental evidence and parameter identification, worked examples, application regimes, reporting standards, a comparative synthesis with a pitfalls checklist and a decision guide, and a research agenda (Sections 25 to 33).

Section 2 gives the historical and literature background. Section 3 derives the thermomechanical foundations that the remaining sections rely on. Readers primarily interested in choosing a model can go directly to Section 31.

1.4 Scope and conventions

We treat *linear* wave propagation in homogeneous solids about a uniform reference state at temperature T_0 unless stated otherwise; nonlinear thermoelasticity, large deformation, radiative transport and chemically reacting media are outside our scope. The following notation is used throughout.

The strain tensor is always written with two indices (ε_{ij}) so that it is not confused with the scalar coupling ε. Plane waves are $\exp[\mathrm{i}(kx-\omega t)]$ with $\operatorname{Im} k \geq 0$ for spatial decay in the propagation direction, and a "branch" of the dispersion relation is a root $k(\omega)$ followed by continuity in ω.

Table 1: Principal notation. Indices follow the Einstein summation convention; a comma denotes a spatial derivative and a dot a time derivative.

u_i, ε_{ij}	displacement, strain	σ_{ij}	stress
$\theta = T - T_0$	temperature increment	T_0	reference temperature
q_i	heat flux	v	thermal displacement, $\dot{v} = \theta$
ρ	mass density	c	specific heat at constant strain
λ, μ	Lamé constants	α_T	linear expansion coefficient
$\beta = (3\lambda + 2\mu)\alpha_T$	thermal-stress coefficient	k, k^*	conductivities (Fourier, GN)
τ_q, τ_T, τ_v	relaxation time and lags	$\alpha = k/\rho c$	thermal diffusivity
$c_1 = \sqrt{(\lambda + 2\mu)/\rho}$	isothermal P-wave speed	$c_2 = \sqrt{\mu/\rho}$	shear wave speed
$\varepsilon = \beta^2 T_0/(\rho^2 c c_1^2)$	thermoelastic coupling	$\omega_* = \rho c c_1^2/k$	thermal characteristic frequency
$G(s)$	heat-flux function (Eq. (15))	$s = -\mathrm{i}\omega$	Laplace variable / complex frequency
c_T	thermal wave speed	K^2	electromechanical coupling
ξ, ℓ	nonlocal and gradient lengths	$Z = \rho c$	mechanical impedance

1.5 Reproducibility

All figures in this review that display numerical results were produced by Python scripts that accompany the manuscript, and the identities used repeatedly (the coupled dispersion polynomial, the Debye form of weak-coupling damping, the Green–Naghdi type-II wave speeds, the reductions of Table 4) are checked either symbolically or against independent numerical roots in those scripts (Appendix C). Where a quantity depends on a choice of model variant, for example the form of a memory-dependent derivative, the choice is stated next to the result.

2 Historical development and the literature landscape

2.1 A short history

Origins (1830s–1950s). Duhamel's memoir of 1837 [2] established the stress–strain–temperature relation that still bears his name (with Neumann's later contribution): the stress depends on strain and on the temperature increment through an expansion coefficient. The dynamic coupling appeared much later. Biot [1] showed that the classical thermoelastic equations follow from the thermodynamics of irreversible processes with a Rayleigh-type dissipation function, and Danilovskaya [3] solved the thermal shock of an elastic half-space, in the uncoupled approximation. Chadwick and Sneddon [4] analysed plane waves in the coupled theory; Chadwick's chapter [5] became the standard reference for the dynamical theory; compact textbook accounts followed [21–23]. The core results of that period are that the coupled longitudinal wave is attenuated and dispersed, that shear waves decouple in isotropic media, and that the thermal disturbance is a diffusive wave. Zener's earlier analysis of thermoelastic internal friction in vibrating reeds [24] had already shown the same relaxation behaviour from a different direction.

Finite-speed heat conduction (1940s–1970s). The conceptual starting point is Cattaneo's 1948 note [25] and Vernotte's independent discussion [26], which add a relaxation term to Fourier's law, making the heat equation hyperbolic. Experimental support from low-temperature solids, namely second sound in solid helium and in sodium fluoride [27, 28], and a kinetic theory of phonon hydrodynamics [29–31], supplied physical contexts in which heat actually propagates as a wave. Lord and Shulman [32] then built Cattaneo's law into a complete thermoelastic theory, and Green and Lindsay [33] proposed a different extension with two relaxation times in the constitutive relations. The relation between these theories and thermodynamics was the subject of extensive discussion; reviews are [6, 34, 35].

Energy-based theories (1990s). Green and Naghdi [36–38] introduced a family of three theories (types I, II, III) based on an entropy balance rather than an entropy inequality for the heat flux, in which the thermal displacement plays the role of a potential. Type II is a nondissipative theory with undamped thermal waves; type III contains classical damping and thermal waves. Uniqueness and stability theory developed around these [39].

Lagging models (1995–2010). Tzou [40, 41] proposed that the heat flux and the temperature gradient are separated by delay times, giving the dual-phase-lag (DPL) model; Roy Choudhuri [42] added a third lag associated with the thermal displacement. These models quickly spread because Taylor expansion of the lag equation gives a differential equation of finite order whose parameters can be chosen to fit data, and the stability and thermodynamic admissibility of those truncations has been studied extensively [43–46].

Nonlocal, fractional and memory-dependent extensions (1970s–present). Eringen's nonlocal elasticity [15, 16, 47] and its thermoelastic extension [48] predate the generalized heat theories but converge with them in micro- and nanosystems; gradient elasticity [17, 18, 49, 50] is the other main enrichment. Fractional calculus entered heat conduction through the Caputo derivative [51], was developed into fractional thermoelasticity [52–58], and was joined by the memory-dependent derivative of Wang and Li [59, 60]. Two-temperature theories [61, 62] and their hyperbolic variants [63–65] introduced a distinction between conductive and thermodynamic temperature.

Mathematical thermoelasticity of third order (2010s). The Moore–Gibson–Thompson (MGT) equation, which arose in nonlinear acoustics [66], was studied systematically as an evolution problem [67–70] and then brought into thermoelasticity by Quintanilla [71, 72], producing a model that fits the template of Section 3.4.

Multiphysics and structure (1960s–present). Micropolar and microcontinuum theories [13, 73, 74], materials with voids [75–77], thermopiezoelectricity [11, 12, 78], and thermoelastic diffusion and magneto-thermoelasticity extended the mechanical content, usually with the heat law as an independent module.

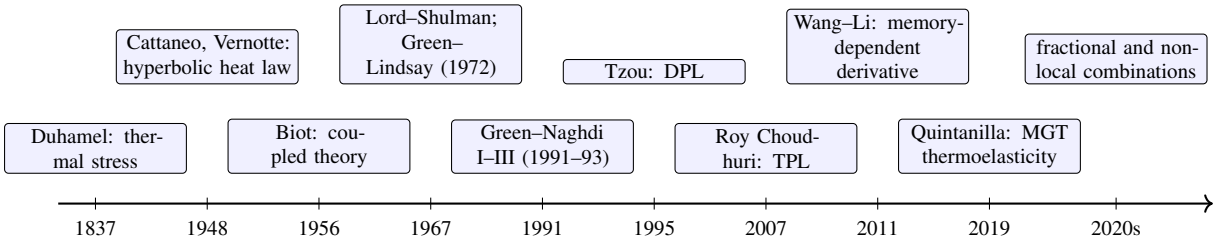


Figure 1: Milestones in the theory of thermoelastic waves. Dates are those of the cited original publications (Cattaneo 1948, Vernotte 1958, Biot 1956, Lord–Shulman 1967, Green–Lindsay 1972, Green–Naghdi 1991–1993, Tzou 1995, Roy Choudhuri 2007, Wang–Li 2011, Quintanilla 2019).

2.2 A map of the contemporary literature

The recent literature on wave problems in thermoelastic and multiphysics media can be sorted along three largely independent axes: the *heat-flux law*, the *constitutive enrichment* (nonlocal, gradient, microstructural, multiphysics), and the *problem type* (plane waves, surface waves, reflection and transmission, transient response). A typical paper chooses one entry on each axis and computes dispersion curves or response histories. Table 2 lists representative works from our reference list by problem type; the sorting is editorial and many papers appear in more than one row.

2.3 Recurrent observations from the literature

Reading across these streams, we note patterns that shape the rest of this review.

- **The heat law and the elastic enrichment are treated as independent modules.** Most interface and surface-wave papers combine one of several heat laws with one of several enrichments, so the dispersion relations differ only through $G(s)$ and through the elastic operator. This is why a unified treatment is possible and why a limit-checking discipline matters.
- **Energy ratios and amplitude ratios dominate the reflection literature.** The interface papers cited above report energy ratios, whose sum must equal one in a lossless setting. A sum different from one is therefore a diagnostic of either dissipation, an error in the flux definition, or both; Section 21 discusses this check in detail.
- **Parameter values are rarely tied to measurements.** Lags and fractional orders are typically assigned convenient values, so many figures demonstrate a model rather than a material.
- **The thermodynamic status of the Taylor-expanded models is delicate.** The papers of Chiriţă and co-workers [44, 45] examine thermodynamic compatibility and continuous dependence of the time-differential lag models, and our own stability computation (Section 9) shows how strongly the truncation order matters.

3 Thermomechanical foundations

This section collects the balance laws, the thermodynamic structure of linear thermoelasticity, the reasons why Fourier's law is problematic for fast processes, and the heat-flux template that

Table 2: Representative literature by problem type and constitutive enrichment (selection from the reference list; classification by the titles and stated subjects of the works).

Problem type	Representative works
Plane waves, stability, dispersion	Prasad–Kumar [63]; Tiwari–Kumar (nonlocal TPL) [79]; van der Merwe–van Rensburg (modal comparison of heat laws) [46]; Sarkar et al. (nonlocal type II) [80].
Surface waves	Biswas (porous nonlocal, orthotropic; piezothermoelastic layer) [81, 82]; Tung (nonlocal piezoelastic Rayleigh) [83]; Gupta et al. (nonlocal piezo-semiconductor Rayleigh wave) [84]; Aldandani et al. [85]; Ivanov–Savova (energy flux of surface waves) [86].
Reflection at free or insulated boundaries	Das et al. [87]; Kalkal et al. (nonlocal micropolar with rotation) [88]; Singh–Guha (piezothermoelastic fibre-reinforced) [89]; Yadav et al. [90]; Gupta et al. [91]; Singh et al. (flexoelectric) [92]; Jiao et al. (flexomagnetic) [93].
Interfaces and energy partition	Barak et al. [94–97]; Gupta–Kumar [98, 99]; Kumar et al. [100–107]; Fu–Wei (micropolar imperfect interface) [108]; Liu et al. (graded piezoelectric) [109].
Transient response, rods, bodies	Singh–Pal [110, 111]; Biswas [112]; Sur et al. [113, 114]; Bayones et al. [115]; Li et al. (laser pulse, bilayer) [116]; Abouelregal [117–119]; Lata et al. [120].
Damping in resonators	Yang et al. (DPL damping in nonlocal nanoplates) [121]; Kumar–Devi (couple stress plates) [122].
Fractional and memory models	Povstenko [52, 54–56]; Sherief et al. [57]; Ezzat–El-Karamany [58]; Wang–Li [59, 60, 123]; Yu et al. [124]; Bachher–Sarkar [125].
Size effects	Eringen [15, 16]; Aifantis [17]; Askes–Aifantis [18, 50]; Maranganti–Sharma [126, 127]; Lazar–Agiasofitou [128]; Zang et al. [129]; Li et al. [130].
Variational and uniqueness theory	Marotti de Sciarra [131]; Srivastava–Mukhopadhyay [132]; Chandrasekharaiah [39]; Chiriţă et al. [44, 45].

organises the remainder of the review. It is self-contained so that the later sections can refer to equation numbers rather than rederive.

3.1 Balance laws for small disturbances

Let a body occupy a region Ω and let $\boldsymbol{u}(\boldsymbol{x},t)$ be the displacement from a reference state at uniform temperature T_0 and zero stress. For small displacement gradients the strain is $\varepsilon_{ij} = \tfrac{1}{2}(u_{i,j}+u_{j,i})$. The local forms of the balance of linear momentum and of energy are

$$\rho\,\ddot{u}_i = \sigma_{ij,j} + \rho\, f_i, \tag{1}$$

$$\rho\,\dot{e} = \sigma_{ij}\dot{\varepsilon}_{ij} - q_{i,i} + \rho\, r, \tag{2}$$

where ρ is the (reference) density, f_i the body force per unit mass, e the specific internal energy, q_i the heat flux and r the heat supply per unit mass. Balance of angular momentum implies a symmetric stress in the absence of couple stresses. The second law, in the Clausius–Duhem form used by Coleman and Noll [133],

$$\rho\,\dot{\eta} \;\geq\; \frac{\rho r}{T} - \left(\frac{q_i}{T}\right)_{,i}, \tag{3}$$

with specific entropy η and absolute temperature T, restricts the constitutive relations. Eliminating r between (2) and (3) and introducing the Helmholtz free energy $\psi = e - T\eta$ gives

$$-\rho\left(\dot{\psi} + \eta\dot{T}\right) + \sigma_{ij}\dot{\varepsilon}_{ij} - \frac{q_i T_{,i}}{T} \;\geq\; 0. \tag{4}$$

For a material with $\psi = \psi(\varepsilon_{ij}, T)$, (4) must hold for arbitrary $\dot{\varepsilon}_{ij}$ and $\dot{T}$; the coefficients of those rates must vanish,

$$\sigma_{ij} = \rho \frac{\partial \psi}{\partial \varepsilon_{ij}}, \qquad \eta = -\frac{\partial \psi}{\partial T}, \tag{5}$$

and the remaining term gives the *heat-conduction inequality*

$$q_i T_{,i} \;\le\; 0. \tag{6}$$

Fourier's law $q_i = -k_{ij} T_{,j}$ with positive-semidefinite k_{ij} satisfies (6) identically; so does any law for which the flux is, to leading order, opposed to the temperature gradient. This is the point at which lagged and relaxed laws require care (Section 3.3).

3.2 Linear thermoelasticity

Expanding $\rho\psi$ to second order about $(\varepsilon_{ij}, T) = (0, T_0)$, with $\theta = T - T_0$, gives

$$\rho\psi = \tfrac{1}{2} C_{ijkl}\varepsilon_{ij}\varepsilon_{kl} - \beta_{ij}\varepsilon_{ij}\theta - \frac{\rho c}{2T_0}\theta^2, \tag{7}$$

where C_{ijkl} are the isothermal elastic moduli, β_{ij} the thermal-stress tensor and c the specific heat at constant strain (we do not carry the constant and linear terms in θ, which only fix the reference values of the energy and entropy). Equations (5) give

$$\sigma_{ij} = C_{ijkl}\varepsilon_{kl} - \beta_{ij}\theta, \qquad \rho\eta = \beta_{ij}\varepsilon_{ij} + \frac{\rho c}{T_0}\theta. \tag{8}$$

The first of these is the Duhamel–Neumann relation. For an isotropic solid $C_{ijkl} = \lambda\delta_{ij}\delta_{kl} + \mu(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk})$ and $\beta_{ij} = \beta\delta_{ij}$ with $\beta = (3\lambda + 2\mu)\alpha_T$, since free thermal expansion $\varepsilon_{ij} = \alpha_T\theta\,\delta_{ij}$ must produce zero stress: $\sigma_{ij} = (3\lambda + 2\mu)\alpha_T\theta\,\delta_{ij} - \beta\theta\,\delta_{ij} = 0$. The energy balance $\rho T\dot{\eta} = -q_{i,i} + \rho r$ (entropy form of (2) when the stress power is accounted for by the free energy), linearised about T_0 using the second of (8), gives

$$\rho c\,\dot{\theta} + T_0\,\beta_{ij}\dot{\varepsilon}_{ij} = -q_{i,i} + \rho r. \tag{9}$$

Together with (1), (8) and a heat-flux law, (9) closes the system. The coefficients carry a clear physical reading: ρc is the heat capacity per unit volume at constant strain; $T_0\beta^2/(\rho c)$ is the difference between the adiabatic and isothermal longitudinal moduli, namely

$$\lambda_{\mathrm{ad}} + 2\mu_{\mathrm{ad}} = \lambda + 2\mu + \frac{T_0\beta^2}{\rho c}, \qquad \text{equivalently} \qquad c_{\mathrm{ad}}^2 = c_1^2(1+\varepsilon), \quad \varepsilon = \frac{\beta^2 T_0}{\rho^2 c\, c_1^2}. \tag{10}$$

Table 3 gives the coupling parameter ε and the characteristic frequency $\omega_* = \rho c c_1^2/k$ for representative solids. The coupling is of order 10^{-2} for common metals and much smaller for low-expansion materials; the characteristic frequency lies far above the ultrasonic range for bulk solids (10^{10}–10^{12} Hz) and drops into the megahertz–gigahertz range only when waves are confined to micro-scale structures (Section 27).

3.3 Beyond Fourier's law

Why Fourier's law is parabolic. Inserting $q_i = -kT_{,i}$ in (9) with constant coefficients and no coupling gives $\rho c\,\dot{\theta} = k\nabla^2\theta$. The fundamental solution is a Gaussian that is nonzero at all distances for every positive time, so the signal speed is unbounded. Because the Gaussian decays so rapidly, the infinite speed has no observable consequence for slow processes; for processes on the time scale of the carrier relaxation it has.

Table 3: Thermoelastic coupling ε and characteristic frequency $f_* = \omega_*/2\pi$ computed from rounded handbook-type values (300 K) of density, specific heat, conductivity, expansion coefficient, Young's modulus and Poisson ratio; the reader should replace them by data for the specimen at hand.

Material	c_1 (m/s)	ε	$f_* = \omega_*/2\pi$ (Hz)	$\alpha = k/\rho c$ (m^2/s)
Aluminium	6198	0.027	6.3×10^{10}	9.8×10^{-5}
Copper	4540	0.019	2.8×10^{10}	1.2×10^{-4}
Silicon	8854	0.00056	1.4×10^{11}	9.2×10^{-5}
Steel (carbon)	5778	0.0097	4.1×10^{11}	1.3×10^{-5}
Fused silica	5972	7.2e-06	6.6×10^{12}	8.6×10^{-7}

Relaxation of the flux. A natural modification, due to Cattaneo and Vernotte [25, 26], lets the flux relax to its Fourier value with relaxation time τ_0:

$$q_i + \tau_0 \dot{q}_i = -k\,\theta_{,i}. \tag{11}$$

With (9) and no coupling, $k\nabla^2\theta = \rho c(\dot\theta + \tau_0\ddot\theta)$, a damped wave equation with front speed $c_T = \sqrt{k/(\rho c \tau_0)}$. In kinetic terms τ_0 is the mean time between momentum-destroying collisions of the heat carriers and c_T is of the order of the carrier speed divided by $\sqrt{3}$ in three dimensions; in phonon hydrodynamics a different, "second sound", regime exists in which normal (momentum-conserving) collisions dominate [29–31].

Thermodynamic compatibility via extended irreversible thermodynamics. The Cattaneo law does not follow from (3) with a free energy that depends only on (ε_{ij}, T): the flux must enter the state space. In extended irreversible thermodynamics [134–136] the entropy depends also on q_i. In one dimension, without deformation, take

$$\eta = \eta_{\mathrm{eq}}(e) - \frac{\tau_0}{2\rho k T^2}\, q^2, \qquad \rho\dot e = -q_{,x} \tag{12}$$

and an entropy flux q/T. Then

$$\rho\dot\eta + \left(\frac{q}{T}\right)_{,x} = -\frac{q}{kT^2}\left(\tau_0\dot q + kT_{,x}\right), \tag{13}$$

and for the Cattaneo law the bracket equals $-q$, so the entropy production is $q^2/(kT^2) \geq 0$. This is the minimal sense in which the relaxed law is thermodynamically admissible: the entropy contains a quadratic correction in the flux. The correction is of second order in the disturbance, which is why the linear theory is not sensitive to it. In lagged models the corresponding entropy functional has to be constructed model by model, and for some Taylor truncations of the lag equations no admissible functional exists unless parameter restrictions hold [44, 45].

Thermal displacement and the Green–Naghdi route. Green and Naghdi [36, 38] take instead an entropy *balance* with an entropy flux and postulate constitutive relations in terms of the *thermal displacement* v, $\dot v = \theta$ (so $v = \int \theta\,\mathrm{d}t$). The flux q_i may depend on $\theta_{,i}$ and $v_{,i}$; type II ($q_i = -k^* v_{,i}$) conserves energy, type III adds Fourier dissipation. In wave problems v is the natural field since $\hat v = \hat\theta/s$.

Green–Lindsay: rates in the stress and entropy. Green and Lindsay [33] keep a single temperature but let the free energy and stress depend on $\dot\theta$ as well, introducing two relaxation times $\tau_0 \geq \tau_1 \geq 0$ in the stress and entropy relations; the heat law remains Fourier's. Solutions of the original theory can exhibit discontinuities at wave fronts that some authors regard as unphysical, and a modification with a strain-rate term has been proposed to remove them [137]. Section 8 returns to this theory.

3.4 The heat-flux template

Linear wave problems with constant coefficients depend on the heat-flux law only through its Laplace-domain relation between flux and temperature gradient. We therefore define the family

$$q_i + \tau_q \dot q_i = -k\left(\theta_{,i} + \tau_T \dot\theta_{,i}\right) - k^*\left(v_{,i} + \tau_v \dot v_{,i}\right), \qquad \dot v = \theta, \tag{14}$$

and note that after Laplace transformation (zero initial data)

$$\hat q_i = -G(s)\,\hat\theta_{,i}, \qquad G(s) = \frac{k\,(1+\tau_T s) + k^*(1+\tau_v s)/s}{1+\tau_q s}. \tag{15}$$

Table 4 lists the parameter choices that reproduce the principal models. For fractional and memory-dependent models $G(s)$ has the same structure with s replaced by non-integer powers or by kernel transforms (Section 11).

Table 4: Heat-flux theories as parameter choices in the template (14). All entries use the energy balance (9).

Theory	k	k^*	τ_q	τ_T	τ_v	Source
Classical (Fourier)	k	0	0	0	0	[1]
Lord–Shulman (Cattaneo)	k	0	τ_0	0	0	[25, 26, 32]
Dual-phase-lag (DPL), 1st order	k	0	τ_q	τ_T	0	[40, 41]
Green–Naghdi II	0	k^*	0	0	0	[38]
Green–Naghdi III	k	k^*	0	0	0	[36]
Three-phase-lag (TPL)	k	k^*	τ_q	τ_T	τ_v	[42]
MGT (flux form)	k	k^*	τ_q	0	0	[71]

Exact reductions and singular limits. The function $G(s)$ determines which reductions are regular. The following were verified symbolically (script `check_reductions.py`).

- **DPL $\to$ Lord–Shulman:** $\tau_T \to 0$ with $\tau_q = \tau_0$ fixed, $k^* = 0$. Letting $\tau_q \to 0$ with τ_T fixed gives instead $G = k(1+\tau_T s)$, a Fourier law with a temperature-rate term (Kelvin–Voigt-like conduction), not Cattaneo.
- **TPL $\to$ DPL:** $k^* \to 0$. For $k^* \neq 0$ the function $G(s) \sim k^*/s$ as $s \to 0$, so the low-frequency and long-time response is qualitatively different.
- **TPL $\to$ classical:** $k^* \to 0$ and all three lags $\to 0$.
- **GN-II:** $k = 0$, all lags zero, $G = k^*/s$; **GN-III:** all lags zero, $G = k + k^*/s$.
- **Large-s behaviour.** $G(\infty) = 0$ for Lord–Shulman (hyperbolic behaviour, finite thermal speed c_T); $G(\infty) = k\tau_T/\tau_q$ for DPL (diffusive again at high frequency with effective conductivity $k\tau_T/\tau_q$).
- **Fractional orders:** the endpoint $\alpha \to 2$ of the Caputo family is second order in time and is not Fourier's law (Section 11).

3.5 Energy balance and energy flux

For the Lord–Shulman model in the absence of body forces and heat supply, multiplying (1) by $\dot{u}_i$, (9) by θ/T_0 and using (11) gives

$$\frac{\partial \mathscr{E}}{\partial t} + P_{i,i} = -\frac{q_i q_i}{k T_0}, \qquad \mathscr{E} = \tfrac{1}{2}\rho \dot{u}_i \dot{u}_i + \tfrac{1}{2} C_{ijkl}\varepsilon_{ij}\varepsilon_{kl} + \frac{\rho c}{2T_0}\theta^2 + \frac{\tau_0}{2kT_0} q_i q_i, \tag{16}$$

$$P_i = -\sigma_{ij}\dot{u}_j + \frac{\theta\, q_i}{T_0}. \tag{17}$$

Here $\mathscr{E}$ is positive definite and P_i is the flux associated with it; the right-hand side is the dissipation. This quadratic (availability-type) energy and its flux P_i are what is meant by "energy flux" in linear wave theory; they differ from the flux $q_i - \sigma_{ij}\dot{u}_j$ of the *total* internal energy, whose heat-flux part is first order in the disturbance and averages to zero over a period. For the classical theory the same procedure gives (16) without the last term in $\mathscr{E}$. For the other members of the template an energy functional exists under parameter conditions that must be stated for each model: for Green–Naghdi type II the functional is conserved; for DPL and TPL (first-order truncations) the construction is more involved [43, 44]. We use (17) for interface energy partition in Section 21.

4 Kinetic origin of the hyperbolic and Guyer–Krumhansl laws

The lag models of Section 9 are usually justified by their formal structure. Their physical meaning comes from kinetic theory of phonons, and this section summarises how the two most-used laws arise there. We state standard results of phonon hydrodynamics at the level of structure; coefficient values depend on the dispersion model and should be taken from the original derivations.

4.1 Relaxation-time closure and the Cattaneo law

Write the linearised phonon Boltzmann equation for the deviation n' from equilibrium with a single relaxation time τ_R for momentum-destroying (resistive) scattering. Taking the first moment, with the energy flux q as the moment variable, and using the first-order closure that the second moment is proportional to the temperature gradient, gives

$$\tau_R \dot{q} + q = -k\nabla\theta, \qquad k = \tfrac{1}{3} c_v v^2 \tau_R, \tag{18}$$

where v is the phonon group speed. This is the Cattaneo law with $\tau_q = \tau_R$. It is valid when the closure holds, namely when the phonon mean free path is small compared with the wavelength, so that $\omega\tau_R \ll 1$ and $\ell_{\mathrm{mfp}} k \ll 1$. Beyond this the closure fails, and the finite speed $\sqrt{k/(c_v\tau_R)} = v/\sqrt{3}$ should not be expected to describe a ballistic front.

4.2 Normal scattering and the Guyer–Krumhansl law

If momentum-conserving (normal) collisions with time τ_N dominate over resistive ones, $\tau_N \ll \tau_R$, the expansion to the next order yields the Guyer–Krumhansl equation

$$\tau_R \dot{q} + q = -k\nabla\theta + \ell^2\left(\nabla^2 q + 2\nabla\nabla\cdot q\right), \tag{19}$$

with a length ℓ of order $v\sqrt{\tau_N\tau_R}$ up to a model-dependent numerical factor ($1/5$ in a common Debye-type derivation). In one dimension the bracket becomes $3q_{,xx}$. Together with $c_v\dot{\theta} = -q_{,x}$ the uncoupled dispersion relation is

$$k_w^2 = -\frac{c_v s\,(1+\tau_R s)}{k + 3\ell^2 c_v s}, \qquad s = -i\omega. \tag{20}$$

The law Eq. (19) is not of the form $\hat{q} = -G(s)\hat{\theta}_{,x}$ with G independent of k_w: the Laplacian term makes the response wavenumber dependent. It is therefore outside the template of Section 3 and is treated separately in Section 22. The practical consequence is that size effects and relaxation effects cannot be separated by choosing different $G(s)$.

4.3 Dispersion of the Guyer–Krumhansl law

Figure 2 (script `scripts/fig_gk.py`) compares the uncoupled thermal branch for the Fourier, Cattaneo and Guyer–Krumhansl laws, in units with $\chi = k/c_v = 1$ and $\tau_R = 1$. Reading the numbers at $\omega\tau_R = 0.1,\ 1,\ 10,\ 100$:

- Fourier: phase speed 0.45, 1.41, 4.48, 14.1 (in units of $\sqrt{\chi/\tau_R}$), growing as $\sqrt{\omega}$, and attenuation $\mathrm{Im}\,k_w = 0.22,\ 0.71,\ 2.2,\ 7.1$.
- Cattaneo ($\ell = 0$): 0.43, 0.91, 1.00, 1.00, i.e. the speed saturates at $\sqrt{\chi/\tau_R}$, and the attenuation saturates at $1/(2\sqrt{\chi\tau_R}) = 0.5$.
- Guyer–Krumhansl with $\ell^2 = 0.05$: 0.43, 0.95, 1.57, 5.3 and attenuation 0.22, 0.53, 3.9, 18; with $\ell^2 = 0.3$: 0.45, 1.34, 4.2, 13.4 and attenuation 0.22, 0.71, 2.3, 7.4.

At low frequency the three laws coincide with diffusion. At high frequency the Cattaneo law approaches a finite front, whereas the Guyer–Krumhansl law returns to diffusive behaviour: from Eq. (20), $k_w^2 \to i\omega\tau_R/(3\ell^2)$ for $\omega \to \infty$, so both $\mathrm{Re}\,k_w$ and $\mathrm{Im}\,k_w$ grow as $\sqrt{\omega}$ and the phase speed grows as $\sqrt{6\ell^2\omega/\tau_R}$ (confirmed numerically to 0.2%). The linear Guyer–Krumhansl law therefore has no finite signal speed in this uncoupled form. Its role is to represent non-local (Knudsen-type) corrections at intermediate frequencies, not to enforce a front.

4.4 What this means for model choice

(i) A lag law is a statement about relaxation of the flux; if the heat-carrier mean free path is comparable to the sample, the Guyer–Krumhansl structure or a full kinetic model is the physical starting point. (ii) Parameters τ_q, τ_T obtained by fitting a lag law to transient data are effective and need not equal a microscopic relaxation time. (iii) A lag law with τ_q of the order of 10^{-12} s describes phonon relaxation only if a closure of the type above is valid at the wavelengths and frequencies of interest.

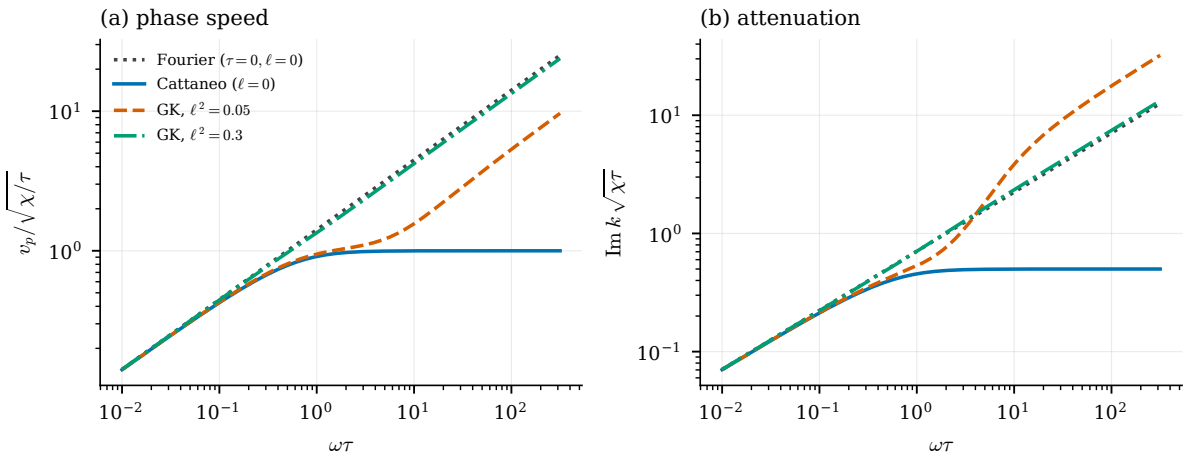


Figure 2: Thermal branch for Fourier, Cattaneo and Guyer–Krumhansl laws ($\chi = \tau_R = 1$): (a) phase speed, (b) attenuation. The dispersion relation is Eq. (20).

5 Mathematical preliminaries

The later sections use a small set of tools repeatedly. They are collected here with the conventions of this review, so that every statement about branches, damping and dispersion has a single definition.

5.1 Fourier–Laplace conventions

Fields depend on $\exp[i(kx-\omega t)]$ and the Laplace variable is $s=-i\omega$, so that $\partial_t \to s$ and $\partial_x \to ik$. For a causal response, i.e. one that vanishes before the excitation, the transfer functions $G(s)$ and the resulting $k(\omega)$ are analytic in the upper half of the complex ω-plane ($\operatorname{Re} s > 0$). Two consequences are used below: the wave number is determined by $k^2 = -\rho c\, s/G(s)$ for the uncoupled thermal branch, and the appropriate root is the one whose spatial amplitude does not grow, $\operatorname{Im} k \geq 0$, with $\operatorname{Re} k > 0$ for propagation in $+x$.

5.2 Branches and the choice of root

The dispersion relation is a polynomial in $K = k^2$ with coefficients depending on s, so each root $K_j(s)$ gives two wave numbers $\pm\sqrt{K_j}$. The rules adopted are: (i) for each K_j, take the square root with $\operatorname{Im} k \geq 0$; (ii) a "branch" is a $k(\omega)$ followed continuously in ω, because the labels "elastic-type" and "thermal-type" are meaningful only away from hybridisation, where roots approach each other; (iii) near hybridisation, which in the Lord–Shulman model occurs for $\tau\omega_* \approx 1$, tracking by continuity with a sufficiently fine frequency grid is necessary, and selecting the root nearest an expected value fails. Both failures were met in producing the figures of this review.

5.3 Spatial and temporal damping

Two measures of loss appear. For real ω and complex k one defines $Q_x^{-1} = 2\operatorname{Im} k/\operatorname{Re} k$; for real k and complex ω one defines $Q_t^{-1} = 2|\operatorname{Im}\omega|/|\operatorname{Re}\omega|$. For weak damping the two are related by the

group-to-phase velocity ratio, $Q_t^{-1} \simeq (v_g/v_p)\, Q_x^{-1}$. In the clamped-bar example of Section 20 the classical peak $Q_t^{-1} = 0.0488$ at $\varepsilon = 0.1$ compares with the Debye value $\varepsilon/2 = 0.05$ of the spatial measure; the difference is a few percent and is of the order ε expected from that relation. Resonator experiments measure Q_t, wave-propagation experiments Q_x, and the two should not be compared without the conversion.

5.4 Causality and the sum rule

A causal response obeys the Kramers–Kronig relations, which tie dispersion to loss: a medium with a loss peak must show a change of speed. For weak damping the relation takes the form of a sum rule,

$$\int_0^\infty Q_x^{-1}\, d\ln\omega = \pi\, \frac{v_0 - v_\infty}{\bar{v}}, \tag{21}$$

where v_0 and v_∞ are the phase speeds at zero and infinite frequency and $\bar{v}$ is their mean. For the Debye form $Q^{-1} = \varepsilon x/(1+x^2)$ the left side is $\pi\varepsilon/2$ and the right side is $\pi(\sqrt{1+\varepsilon}-1) \approx \pi\varepsilon/2$, which is consistent with the adiabatic and isothermal speeds of Section 6. We checked Eq. (21) numerically on the exact coupled classical dispersion relation (script `scripts/check_kk.py`): at $\varepsilon = 0.01$ the integral is 1.562994×10^{-2} against 1.562991×10^{-2} on the right, and at $\varepsilon = 0.001$ the two sides are 1.570009×10^{-3} and 1.570011×10^{-3}. The practical use is as a consistency test on any computed or measured loss curve: if a model predicts a loss peak but no corresponding speed change, or the reverse, it violates causality or has been truncated inconsistently.

5.5 Support of the solution and finite signal speed

The Laplace transform of a solution supported in $x \le ct$ decays as $e^{-sx/c}$ for $\mathrm{Re}\, s \to \infty$. Conversely, a transfer function with $k(s) \sim s/c$ at large s implies a front travelling at c, which is the origin of the criterion used in Section 8: finite signal speed requires $sG(s)$ to tend to a finite, non-zero limit. Models for which $k_w \sim \sqrt{s}$ at large s (classical, DPL, TPL, GN-III, and the Guyer–Krumhansl law of Section 4) have no front, and a transient problem for them has a solution that is nonzero everywhere immediately after the excitation, however small.

5.6 Complex roots of the delay law

When G contains an exponential, as in the exact delay law of Section 10, $k^2 = -\rho c\, s/G(s)$ is an entire function times s and has infinitely many branches. Truncations replace this by a rational function with finitely many roots. Roots of the delay law that the truncation does not contain, and spurious roots that it introduces, are a likely reason why truncated and exact laws differ at $\omega\tau \gtrsim 1$, although this has to be checked for each truncation.

6 Classical coupled thermoelasticity

6.1 Field equations

For an isotropic solid with Fourier conduction $q_i = -k\theta_{,i}$, (1), (8) and (9) give the Navier-type system

$$\rho\ddot{\boldsymbol{u}} = \mu\nabla^2\boldsymbol{u} + (\lambda+\mu)\nabla(\nabla\cdot\boldsymbol{u}) - \beta\nabla\theta + \rho\boldsymbol{f}, \tag{22}$$

$$\rho c\,\dot{\theta} = k\nabla^2\theta - T_0\beta\,\nabla\cdot\dot{\boldsymbol{u}} + \rho r. \tag{23}$$

The thermodynamic content is fully encoded in the two coupling terms: the thermal body force $-\beta\nabla\theta$ in (22) and the dilatation heating $-T_0\beta\nabla\cdot\dot{\boldsymbol{u}}$ in (23). Their product, $T_0\beta^2$, is what appears in the single coupling constant ε of (10); the reciprocal structure (the same β in both) is a consequence of the existence of the free energy (7). A sign or factor error in either term breaks the reciprocity and is a common source of unphysical results; in our dispersion checks (Appendix A) this is exactly what a symbolic comparison with the field equations detects.

Helmholtz decomposition. Writing $\boldsymbol{u} = \nabla\phi + \nabla\times\boldsymbol{\psi}$ with $\nabla\cdot\boldsymbol{\psi} = 0$ decouples the problem in an unbounded isotropic body into

$$\rho\ddot{\phi} = (\lambda+2\mu)\nabla^2\phi - \beta\theta, \qquad \rho c\dot{\theta} = k\nabla^2\theta - T_0\beta\nabla^2\dot{\phi}, \qquad \rho\ddot{\boldsymbol{\psi}} = \mu\nabla^2\boldsymbol{\psi}. \tag{24}$$

The shear potential $\boldsymbol{\psi}$ obeys an uncoupled wave equation with speed $c_2 = \sqrt{\mu/\rho}$: transverse waves are neither attenuated nor dispersed by thermoelastic coupling in an unbounded isotropic medium. Only the dilatational potential couples to temperature. At boundaries and interfaces, shear and dilatational motions are coupled again by boundary conditions, which is why surface waves and reflected shear waves are affected by thermal coupling (Sections 18 and 21).

6.2 Uniqueness and energy

Multiplying (22) by $\dot{u}_i$ and (23) by θ/T_0 and integrating gives the energy identity (16) without the flux-relaxation term, with dissipation $\int k|\nabla\theta|^2/T_0\,\mathrm{d}V \geq 0$. From it follow uniqueness of solutions of the initial–boundary value problem under standard boundary conditions (Dirichlet, Neumann or Robin on each of the mechanical and thermal parts) and continuous dependence on data. Variational formulations and reciprocity theorems are available [22, 23, 131], and for the generalized theories these identities are the starting point of well-posedness proofs [39, 71].

6.3 Longitudinal plane waves

For one-dimensional dilatational motion, $\boldsymbol{u} = u(x,t)\boldsymbol{e}_x$, (22)–(23) reduce to

$$\rho\ddot{u} = (\lambda+2\mu)u_{,xx} - \beta\theta_{,x}, \qquad \rho c\,\dot{\theta} + T_0\beta\,\dot{u}_{,x} = -q_{,x}. \tag{25}$$

Introducing $\hat{q} = -G(s)\hat{\theta}_{,x}$ (so $G = k$ for the classical theory) and seeking solutions $\propto \exp[\mathrm{i}(kx - \omega t)]$, we obtain the characteristic equation, which is quadratic in $K = k^2$:

$$\left(\rho c_1^2 K - \rho\omega^2\right)\left(s\rho c + G(s)K\right) + \beta^2 T_0\, s\, K = 0, \qquad s = -\mathrm{i}\omega. \tag{26}$$

Equation (26) holds for every member of the template of Section 3.4; a derivation and the symbolic check against the field equations are in Appendix A. Two wavenumber branches result: an *elastic-type* branch with $k \approx \omega/c_1$ for weak coupling, and a *thermal-type* branch with $k^2 \approx -s\rho c/G(s)$. We call them types and not "waves" because the classical thermal branch is a diffusive wave with $\mathrm{Re}\,k = \mathrm{Im}\,k$, which attenuates by a factor $e^{-2\pi}$ per wavelength.

Weak-coupling expansion. Let $x = \omega/\omega_*$ with $\omega_* = \rho c c_1^2/k$. For $\varepsilon \ll 1$ and $G = k$, set $K = \omega^2/c_1^2 + \delta K$ in (26). To first order in ε,

$$c_1^2 K = \omega^2\Big[1 + \varepsilon\,\frac{-1+\mathrm{i}x}{1+x^2}\Big] \;\Longrightarrow\; v_p \simeq c_1\Big[1 + \frac{\varepsilon}{2(1+x^2)}\Big], \qquad Q^{-1} \equiv \frac{2\,\mathrm{Im}\,k}{\mathrm{Re}\,k} \simeq \varepsilon\,\frac{x}{1+x^2}. \tag{27}$$

The phase speed interpolates between the adiabatic value $c_1(1+\varepsilon/2) \simeq c_1\sqrt{1+\varepsilon}$ at low frequency and the isothermal speed c_1 at high frequency, and the inverse quality factor Q^{-1} has the Debye form with a maximum $\varepsilon/2$ at $\omega = \omega_*$. In the language of anelasticity the relaxation strength is ε and the relaxation time is $1/\omega_*$. This is precisely Zener's thermoelastic relaxation [24], here derived for a plane compressional wave rather than for a vibrating reed. The ordering of the limits is opposite to what one may expect at first: the classical theory is *adiabatic* at low frequency and *isothermal* at high frequency, because conduction over a wavelength is efficient when the wavelength is short ($\alpha k^2 \gg \omega$) and negligible when it is long. We confirmed (27) against the exact roots of (26): for $\varepsilon = 10^{-3}$ the exact value of Q^{-1}/ε at $\omega = \omega_*$ is 0.49999984 against the predicted 0.5 (script `fig_zener_q.py`).

6.4 Thermoelastic damping

Figure 4(a) compares the exact classical Q^{-1} with the Debye form for two couplings; the weak-coupling formula is accurate to within a few percent already at $\varepsilon = 0.02$ and the curve shape is independent of ε to leading order. Panel (b) shows the effect of the heat-flux law at the larger coupling $\varepsilon = 0.1$. All three models share the low-frequency rise $Q^{-1} \propto \omega$ and the maximum near ω_*. The generalized laws increase the maximum and raise the high-frequency tail, by a factor that depends on $\omega\tau_q$ only when that product approaches order one. The relevant parameter is $\omega_*\tau_q$. For the metals of Table 3, $\omega_* \sim 10^{11}$–10^{12} rad/s, so with a lag time of order a picosecond the product is of order one near the *bulk* damping peak, and the heat law then matters; but that peak lies at tens of gigahertz and is reached only in very high-frequency experiments. For micro- and nano-resonators the relevant relaxation frequency is set by the structure size rather than by ω_* (see below) and lies far below $1/\tau_q$, where Zener's classical formula is adequate; the lagged corrections become important for ultra-high-frequency nanostructures (Section 27).

Relation to flexural damping. In a thin beam of thickness h vibrating in flexure, the temperature field varies across the thickness with a thermal relaxation time $\tau_Z \simeq h^2/(\pi^2\alpha)$ (Zener), and the damping has the same Debye form with ω_* replaced by $1/\tau_Z$ and the relaxation strength $\Delta = E\alpha_T^2 T_0/(\rho c)$ (with Young's modulus E) [24]. Lifshitz and Roukes [138] derived the corresponding exact expression for beams and discussed its use for micro- and nano-mechanical resonators. For a silicon beam of thickness one micrometre, $\alpha \approx 9\times10^{-5}\,\mathrm{m}^2/\mathrm{s}$ gives a relaxation frequency of order 10^8 Hz, which is in the range of present-day resonators. Generalized and nonlocal corrections to this classical result have been studied for plates and beams [121, 122].

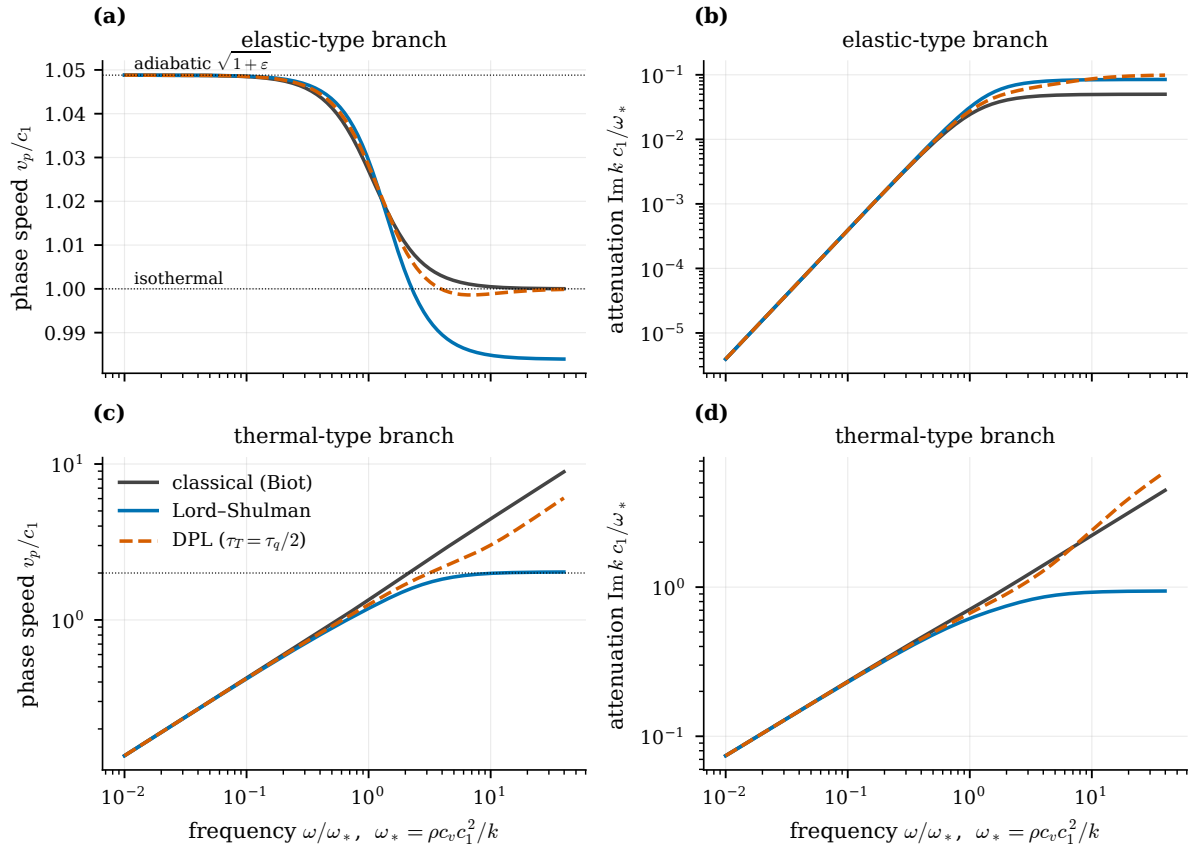


Figure 3: Phase speed (left) and attenuation (right) of the elastic-type (top) and thermal-type (bottom) branches for the classical, Lord–Shulman and dual-phase-lag models. Parameters: $\varepsilon = 0.1$ (larger than for metals so that dispersion is visible), $\tau_0 = \tau_q = 0.25/\omega_*$ giving $c_T = 2c_1$, and $\tau_T = \tau_q/2$ for DPL (a stable choice for the first-order model, see Section 9). Dotted lines: isothermal speed c_1, adiabatic speed $c_1\sqrt{1+\varepsilon}$, thermal speed c_T. Branches are followed by continuity in frequency.

6.5 Anisotropy and other classical extensions

For anisotropic media the field equations keep their form with C_{ijkl}, β_{ij} and k_{ij}; Dhaliwal and Sherief extended the generalized theory to anisotropic media [139]. The coupled problem no longer decouples into shear and dilatational parts, but quasi-longitudinal, quasi-shear and thermal branches exist, and for symmetric directions the one-dimensional analysis above applies with the appropriate effective moduli. For orthotropic, transversely isotropic and piezoelectric crystals, the Stroh formalism [140, 141] gives a systematic route to surface waves.

6.6 What the classical theory cannot describe

Three predictions of the classical theory conflict with finite-speed heat transport: the instantaneous thermal signal, the unbounded thermal phase speed (Figure 3c, black curve), and the monotonic growth of the thermal attenuation with frequency. The frequency range in which these matter depends on the time scale τ_q of heat carriers and on the dimension of the structure; Section 31 gives the criteria.

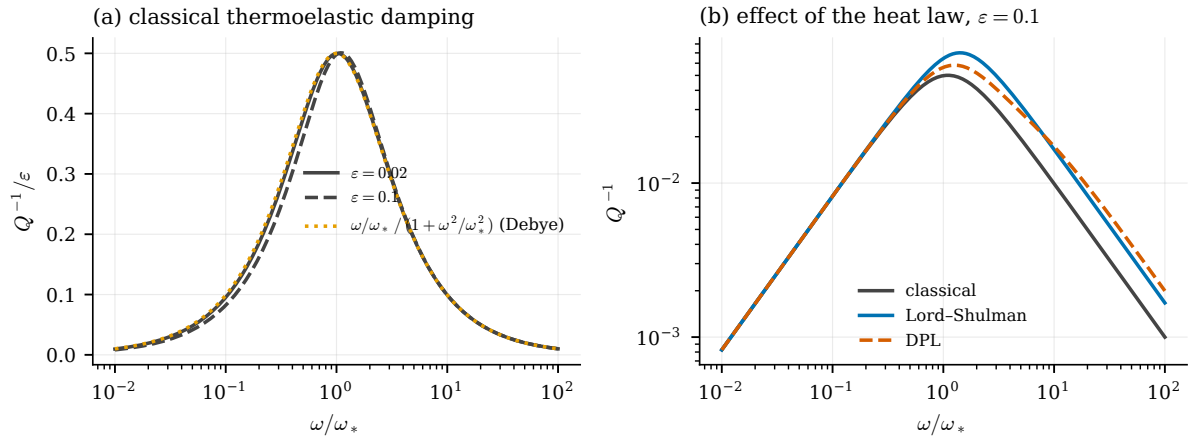


Figure 4: Thermoelastic damping of the elastic-type branch, $Q^{-1} = 2\,\mathrm{Im}\,k/\,\mathrm{Re}\,k$ against frequency. (a) Classical theory for two couplings, normalised by ε, compared with the weak-coupling Debye form (27). (b) Effect of the heat law at $\varepsilon = 0.1$, with $\tau_q = 0.25/\omega_*$ and DPL $\tau_T = \tau_q/2$.

7 Thermoelastic damping in flexure: Zener and exact Fourier solutions

Longitudinal waves (Section 6) are only one of the settings in which the thermoelastic loss appears. In micro-resonators the dominant case is flexure of a thin beam, where the temperature gradient across the thickness h relaxes by heat conduction in the transverse direction.

7.1 Two classical results

Zener's single-relaxation treatment gives

$$Q^{-1} = \Delta\,\frac{\omega\tau_Z}{1+(\omega\tau_Z)^2}, \qquad \tau_Z = \frac{h^2}{\pi^2\chi}, \qquad \Delta = \frac{E\alpha_T^2 T_0}{c_v}, \tag{28}$$

with χ the thermal diffusivity, α_T the linear expansion coefficient and c_v the volumetric heat capacity. The exact solution of the Fourier problem for a thin rectangular beam in Euler–Bernoulli flexure, due to Lifshitz and Roukes, is

$$Q^{-1} = \Delta\left[\frac{6}{\xi^2} - \frac{6}{\xi^3}\,\frac{\sinh\xi + \sin\xi}{\cosh\xi + \cos\xi}\right], \qquad \xi = h\sqrt{\frac{\omega}{2\chi}}. \tag{29}$$

Both reduce to $Q^{-1} \simeq \Delta\,\omega\tau_Z$ for $\omega\tau_Z \ll 1$.

7.2 Quantitative comparison

Figure 5 compares Eqs. (28) and (29) (script `scripts/fig_flexTED.py`). With $x = \omega\tau_Z$ the exact solution peaks at $Q^{-1}/\Delta = 0.494$ near $x = 1.008$, against 0.5 at $x = 1$ for the Zener form. The relative error of the Zener form is $+1.3\%$ at $x = 0.01$, 0.1 and 1, -5% at $x = 10$ and -14% at $x = 100$. In practice the single-relaxation form is therefore accurate to a few percent through the peak and on its low-frequency side, and underestimates the loss at $x \gtrsim 10$, where the exact solution decays as $x^{-1/2}$ whereas the Zener form decays as x^{-1}.

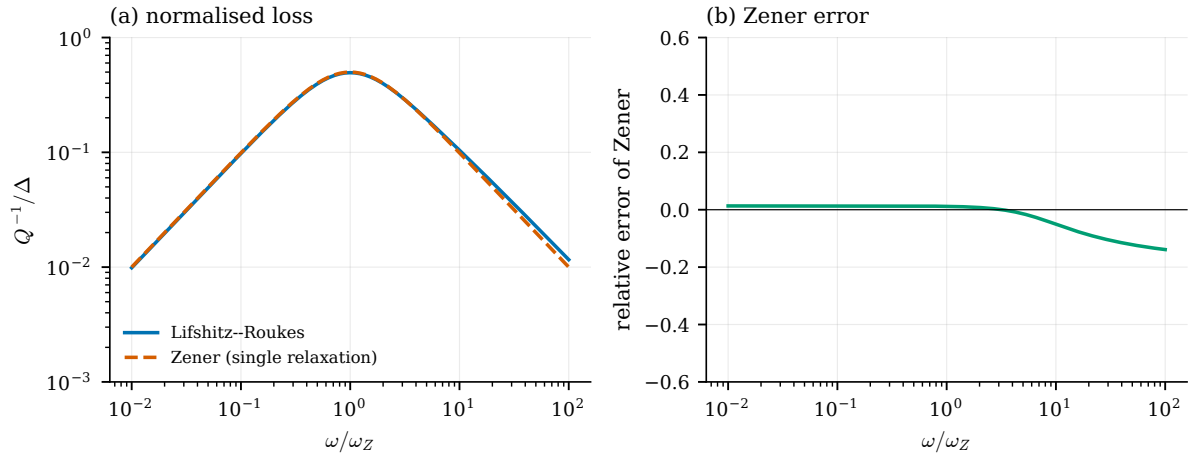


Figure 5: (a) Thermoelastic loss of a thin beam in flexure: exact Fourier solution Eq. (29) and the Zener form Eq. (28). (b) Relative error of the Zener form.

7.3 Where hyperbolic corrections enter

Equation (29) assumes parabolic heat conduction across the thickness. Replacing the Fourier law by a lagging law (Section 9) changes the result only when the transverse relaxation rate $\pi^2\chi/h^2$ approaches the inverse lag. For $\tau_q \sim 10^{-12}$ s and $h \sim 100$ nm, with $\chi \sim 10^{-4}$ m^2/s, the product $\omega\tau_q$ at the Zener peak is of order 10^{-2}, so the parabolic result is adequate; corrections matter for h well below ~ 10 nm or when ballistic transport is relevant, in which case the Guyer–Krumhansl form (Section 4) is the appropriate next step rather than a pure lag law.

8 Hyperbolic theories: LS, GL and GN

8.1 A common dispersion equation

The three classical generalized theories can be written, for plane longitudinal motion, in one equation. Let τ_0 denote the flux relaxation time (Lord–Shulman) or the thermal relaxation time (Green–Lindsay), and let $\tau^\dagger$ be the lag that multiplies the thermal-stress and dilatation coupling. Elimination of $\hat{u}$ and $\hat{\theta}$ gives

$$\left(\rho c_1^2 K + \rho s^2\right)\left(s\rho c\,(1+\tau_0 s) + kK\right) + \beta^2 T_0\, s K\,(1+\tau^\dagger s) = 0, \tag{30}$$

with $\tau^\dagger = 0$ and $\tau_0 = 0$ for the classical theory, $\tau^\dagger = \tau_0$ for Lord–Shulman (this is (26) multiplied by $1+\tau_0 s$), and $\tau^\dagger = \tau_1$, the second relaxation time, for Green–Lindsay. The unified form of Dhaliwal and Sherief [139] makes the same point at the level of the field equations: the theories differ in which of the two coupling terms carries a time-derivative correction.

8.2 Lord–Shulman theory

The Lord–Shulman (LS) theory introduces Cattaneo's law (11) with the energy balance (9). Eliminating the flux gives the temperature equation

$$k\nabla^2\theta = (1+\tau_0\partial_t)\left(\rho c\,\dot{\theta} + T_0\beta\,\dot{\varepsilon}_{kk}\right) - (1+\tau_0\partial_t)\rho r, \tag{31}$$

so that the lagged derivative acts on *both* the heat-capacity term and the dilatation-coupling term. The latter is a modelling consequence of applying the Cattaneo law to the total heat flux, and it is the source of the difference from the Green–Lindsay equations.

Wave fronts. Take $\omega \to \infty$ in (30). The leading terms in s give, with $c_T^2 = k/(\rho c \tau_0)$ and $\varepsilon = \beta^2 T_0/(\rho^2 c c_1^2)$,

$$v^4 - v^2\left[c_1^2(1+\varepsilon\,\tau^\dagger/\tau_0) + c_T^2\right] + c_1^2 c_T^2 = 0, \tag{32}$$

whose two roots are the speeds of the two wave fronts (an elastic-type and a thermal-type front), $v_\mathrm{e} < v_\mathrm{t}$ when $c_T > c_1$ (and the labels exchange for $c_T < c_1$). For LS, $\tau^\dagger = \tau_0$ and the equation is the standard high-frequency front equation of the LS theory; for GL it is (32) with the factor τ_1/τ_0. We verified (32) numerically against the roots of (30) at $\omega/\omega_* = 2\times 10^3$ for both theories (script `check_fronts.py`): with $\varepsilon = 0.1$, $\tau_0 = 0.25/\omega_*$ ($c_T = 2c_1$), the LS fronts are $0.9839\,c_1$ and $2.0327\,c_1$, and with $\tau_1 = 0.1/\omega_*$ the GL fronts are $0.9934\,c_1$ and $2.0132\,c_1$, in agreement with (32) to four digits. Two features are worth recording.

1. The elastic front speed is *not* the isothermal speed c_1, nor the adiabatic speed: it is displaced from c_1 by the interaction with the thermal front, and the two speeds repel one another (their product is $c_1 c_T$, their sum of squares $c_1^2(1+\varepsilon)+c_T^2$). When $c_T = c_1$ the degeneracy is lifted by an amount proportional to $\sqrt{\varepsilon}$: $v^2 = c_1^2[2+\varepsilon \pm \sqrt{\varepsilon(4+\varepsilon)}]/2 \approx c_1^2(1\pm\sqrt{\varepsilon})$.
2. The leading front speeds depend only on the high-frequency terms, so they are independent of k except through c_T.

Discontinuities and well-posedness. Because the system is hyperbolic, initial data with a jump produce solutions with jumps carried along the fronts, and these jumps decay exponentially along the propagation path; this is visible in the heat-pulse problem of Section 16. The system is well posed for $\tau_0 > 0$ and energy decays in bounded bodies under standard boundary conditions, from the energy identity (16).

Interpretation. The principal modelling questions are the physical size of τ_0 and whether Cattaneo's law, with a single exponential relaxation, is an adequate description in the frequency range of interest. Joseph and Preziosi [34, 35] review evidence for and against; large relaxation times have been reported for nonhomogeneous materials [142, 143], although the interpretation of such measurements has been debated. For solids the strongest evidence of wave-like heat propagation is second sound at cryogenic temperatures [27, 28], where the governing regime is the Guyer–Krumhansl hydrodynamic one [30], not the simple Cattaneo law.

8.3 Green–Lindsay theory

The Green–Lindsay (GL) theory [33] has the constitutive relations

$$\sigma_{ij} = C_{ijkl}\varepsilon_{kl} - \beta_{ij}\big(\theta + \tau_1\dot\theta\big), \qquad k\theta_{,ii} = \rho c\big(\dot\theta + \tau_0\ddot\theta\big) + T_0\beta_{ij}\dot\varepsilon_{ij}, \tag{33}$$

with $\tau_0 \geq \tau_1 \geq 0$. The heat conduction term is Fourier's, so the theory does not fit the flux template (14); the finite thermal speed instead comes from the inertia term $\tau_0\ddot\theta$. Substituting in plane waves gives (30) with $\tau^\dagger = \tau_1$. Three differences from LS are worth noting. First, the thermal equation is not of Cattaneo type, so the relation to kinetic arguments is less direct. Second, the entropy is a function of temperature rate, so it is the thermal displacement-like quantity

that appears in the constitutive structure, and the interpretation of τ_0 and τ_1 as independent material constants requires care. Third, solutions of the original theory can contain discontinuities that some authors consider unphysical; a modification with a strain-rate term was proposed to remove them [137]. Wave fronts follow (32) with $\tau^\dagger = \tau_1$. A distinguishing consequence: in GL the coupling is weaker at the wave fronts than in LS by the factor $\tau_1/\tau_0 \le 1$.

8.4 Green–Naghdi types I, II and III

The Green–Naghdi (GN) theories [36–38] postulate an entropy balance with the thermal displacement v ($\dot v = \theta$) as a primary field. In the template the three types are

$$\text{I: } q_i = -k\theta_{,i}; \qquad \text{II: } q_i = -k^* v_{,i}; \qquad \text{III: } q_i = -k\theta_{,i} - k^* v_{,i}, \tag{34}$$

with $G = k$, k^*/s and $k + k^*/s$, respectively. Their uncoupled thermal equations are, with $\rho c = 1$ for brevity,

$$\text{II: } \ddot\theta = c_*^2 \nabla^2 \theta, \quad c_*^2 = k^*/\rho c; \qquad \text{III: } \ddot\theta = c_*^2 \nabla^2 \theta + \alpha \nabla^2 \dot\theta. \tag{35}$$

Type II. Thermal disturbances propagate as undamped waves of speed c_*, and with coupling the two fronts satisfy (32) with $c_T \to c_*$, $\tau^\dagger/\tau_0 \to 1$. In this case, however, the relation is *exact for all frequencies*, not only asymptotically: the roots of the quadratic (26) with $G = k^*/s$ are non-dispersive, with $v^4 - v^2[c_1^2(1+\varepsilon) + c_*^2] + c_1^2 c_*^2 = 0$ at every ω. We confirmed this numerically for $\varepsilon = 0.1$ and $c_* = c_1$: the exact speeds $0.854309 c_1$ and $1.170537 c_1$ are the same at $\omega/\omega_* = 0.5$, 2 and 10, and agree with the formula. The theory is energy-conserving, so there is no thermoelastic damping at all; it is the only member of the family with this property, and consequently the benchmark for energy-based checks.

Type III. The type III equation is a damped wave equation with a Kelvin–Voigt-type damping term $\alpha \nabla^2 \dot\theta$. Two consequences deserve emphasis. At *low* frequency $G \simeq k^*/s$ and the thermal branch has the type-II speed c_*, whereas at *high* frequency $G \to k$ and the branch becomes diffusive: its phase speed is unbounded (Figure 7, magenta curve), exactly as in the classical theory. Thus, in contrast to LS, type III is wave-like at low frequencies and diffusive at high frequencies, and its thermal signals do not have a finite propagation speed in the strict sense (the strongly damped wave equation has a parabolic smoothing effect). The practical interpretation is that type III describes weakly damped thermal waves at moderate frequencies; it is not an "improved Cattaneo law". Statements in the literature that type III possesses finite thermal wave speed should be read as referring to the undamped limit.

Uniqueness and stability. Chandrasekharaiah proved uniqueness for type II [39]; complete solutions for type II were given by [144]; further work on the theory without energy dissipation includes its general formulation [145], spatial behaviour [146], micropolar media [147] and materials with microstructure [148]; the effect of thermomechanical coupling and relaxation times on the wave spectrum is studied in [149]. The type II theory has attracted attention as a conservative reference model; combined with nonlocal elasticity it has been analysed for wave dispersion [80].

9 Lagging and third-order models: DPL, TPL and MGT

9.1 The lag idea and its Taylor truncations

The dual-phase-lag (DPL) model [40, 41] states that the temperature gradient at time $t+\tau_T$ drives the heat flux at time $t+\tau_q$:

$$q_i(\boldsymbol{x},t+\tau_q) = -k\,\theta_{,i}(\boldsymbol{x},t+\tau_T). \tag{36}$$

To obtain a differential equation, both sides are expanded in Taylor series. The truncation order is a modelling decision, not a detail, and three variants are in use:

$$\text{(i) first order in both:}\quad q_i+\tau_q\dot{q}_i = -k\big(\theta_{,i}+\tau_T\dot{\theta}_{,i}\big), \tag{37}$$

$$\text{(ii) second order in } \tau_q\text{, first in } \tau_T:\quad q_i+\tau_q\dot{q}_i+\tfrac{1}{2}\tau_q^2\ddot{q}_i = -k\big(\theta_{,i}+\tau_T\dot{\theta}_{,i}\big), \tag{38}$$

$$\text{(iii) second order in both:}\quad q_i+\tau_q\dot{q}_i+\tfrac{1}{2}\tau_q^2\ddot{q}_i = -k\big(\theta_{,i}+\tau_T\dot{\theta}_{,i}+\tfrac{1}{2}\tau_T^2\ddot{\theta}_{,i}\big). \tag{39}$$

Version (i) is the member of the template of Section 3.4. Versions (ii) and (iii) have higher-order $G(s)$: for (ii), $G(s)=k(1+\tau_T s)/(1+\tau_q s+\tau_q^2 s^2/2)$.

Why the truncation order matters for stability. The stability condition $\tau_T \geq \tau_q/2$, often quoted as *the* stability condition of DPL thermoelasticity, was derived by Quintanilla [43] for the version in which the flux is expanded to second order in τ_q and the gradient to first order in τ_T, i.e. variant (ii). Related stability results for the same model appear in [150–153]. We checked this by computing, for the coupled one-dimensional problem with real wavenumber, the largest real part of the temporal roots s of the characteristic polynomial and maximising it over k. For the first-order version (i) the largest growth rate is negative for every $\tau_T>0$ tested (τ_T/τ_q from 0.05 to 1.2, wavenumbers up to $10^3 c_1^{-1}\tau_q^{-1}$, $\varepsilon=0.1$); for the second-order version (ii) it is positive when $\tau_T<\tau_q/2$, vanishes at $\tau_T=\tau_q/2$ and is negative above (Figure 6; script `fig_stability.py`). The high-wavenumber growth rate saturates, e.g. at $\tau_q\,\mathrm{Re}\,s = 0.571$ for $\tau_T/\tau_q=0.3$, so the instability is a *mode-independent exponential growth at short wavelengths* — an ill-posedness of the model, not a slow instability of a few modes.

The consequence for the literature is practical. Papers that use the first-order version and state that they "impose" $\tau_T\geq\tau_q/2$ for stability are applying a condition that belongs to another model; conversely, papers that use the second-order version with $\tau_T<\tau_q/2$ are computing solutions of an ill-posed problem, and any numerical result is then dominated by the growing high-wavenumber modes. The thermodynamic and continuous-dependence analyses of the time-differential lag models [44, 45] and modal comparisons with Fourier and Cattaneo laws [46] address the same issue from a different angle. Note also that first-order truncation is valid only when τ_q and τ_T are small compared with the time scale of the phenomenon ($\omega\tau\ll 1$), a condition that is not satisfied in the frequency range where the lag models are claimed to differ from Fourier's law; we return to this in Section 31.

9.2 Dispersion and the shape of the lag response

For version (i), $G(s)=k(1+\tau_T s)/(1+\tau_q s)$ has two limits, $G(0)=k$ and $G(\infty)=k\tau_T/\tau_q$. The thermal branch is therefore diffusive at both ends, with effective diffusivities α at low and $\alpha\tau_T/\tau_q$ at high frequency, and wave-like only at intermediate frequency if $\tau_T<\tau_q$ (Figure 7).

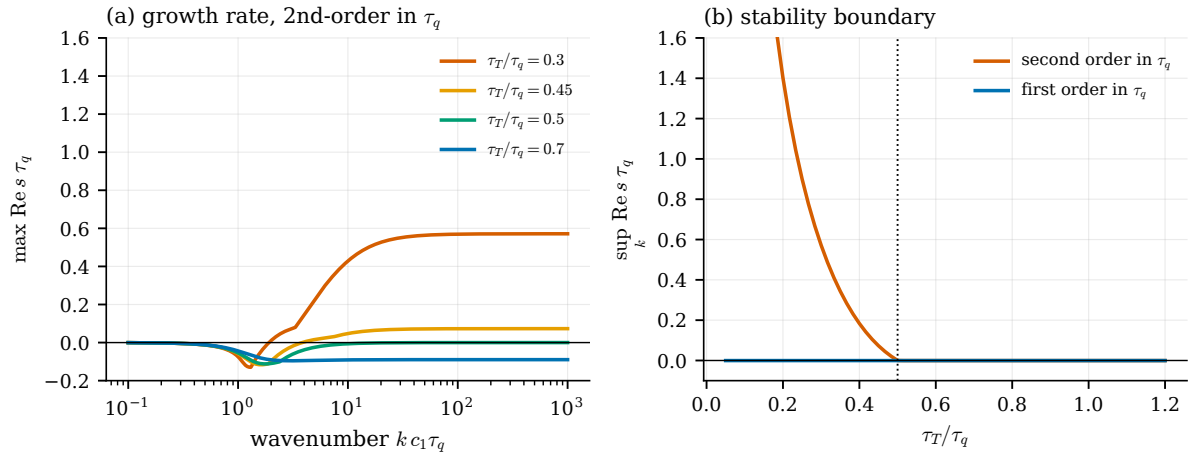


Figure 6: Temporal growth rate of the coupled one-dimensional DPL problem ($\varepsilon = 0.1$, $\tau_q = 1$ in units where $c_1 = \omega_* = 1$). (a) Maximum real part of the temporal roots against wavenumber for the second-order-in-τ_q truncation (38): for $\tau_T/\tau_q < 1/2$ the growth rate tends to a positive constant at large k; for $\tau_T/\tau_q \geq 1/2$ it is non-positive. (b) The supremum over k against τ_T/τ_q for the two truncations; the first-order version (37) (blue) has no instability.

The statement that "$\tau_T < \tau_q$ means wave-like conduction and $\tau_T > \tau_q$ means diffusive conduction" is therefore correct only in the sense of an intermediate regime. When $\tau_T = \tau_q$, $G(s) \equiv k$ and DPL collapses to Fourier's law, whatever the common value of the lags. When $\tau_T > \tau_q$ the effective conductivity increases with frequency.

9.3 Three-phase-lag model

The three-phase-lag (TPL) model of Roy Choudhuri [42] adds a thermal-displacement term to the DPL law,

$$q_i + \tau_q \dot{q}_i = -\left[k\left(\theta_{,i} + \tau_T \dot{\theta}_{,i}\right) + k^*\left(v_{,i} + \tau_v \dot{v}_{,i}\right)\right], \qquad \dot{v} = \theta, \tag{40}$$

which is the template. Truncations of the lag equations to higher order, introducing more constants, are common in applications [118, 119]. Because TPL includes GN-III as the special case $\tau_q = \tau_T = \tau_v = 0$, it inherits the type-III behaviour at low frequency ($G \sim k^*/s$) unless $k^* = 0$. For $\tau_q > 0$, $G(\infty) = k\tau_T/\tau_q$ (if $\tau_T > 0$) so the high-frequency limit is diffusive as in DPL. The thermodynamic compatibility and continuous dependence of the TPL model are studied in [44]. For plane waves the nonlocal TPL dispersion was studied in [79], combinations with memory-dependent derivatives in [110–112] and with magnetic fields and nonlocal elasticity in [115].

9.4 Moore–Gibson–Thompson thermoelasticity

The Moore–Gibson–Thompson (MGT) equation, a third-order-in-time linear model arising in nonlinear acoustics [66], has been studied as an abstract evolution equation with memory and in the critical regime [67–70]. Quintanilla [71] introduced a thermoelastic theory with the flux law

$$\tau \dot{q}_i + q_i = -k\theta_{,i} - k^* v_{,i}, \tag{41}$$

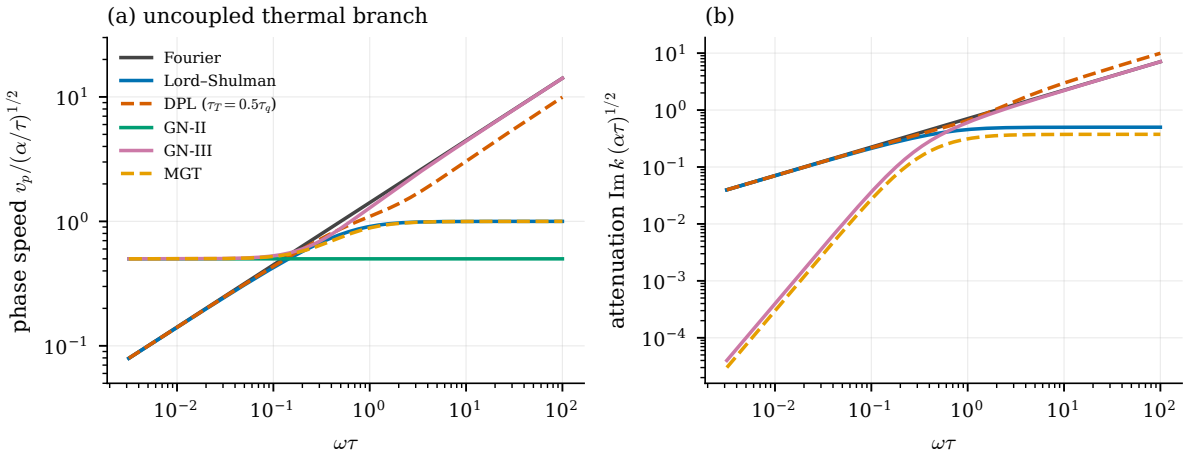


Figure 7: Uncoupled thermal branch for the models of the template with $\rho c = 1$, $k = 1$, $\tau = 1$ (and $k^* = 0.25$ where relevant): (a) phase speed, (b) attenuation. Fourier and GN-III become diffusive at high frequency (unbounded speed); LS saturates at c_T; DPL is diffusive at both ends; MGT is wave-like at both ends, with speed $c_* = \sqrt{k^*/\rho c}$ at low frequency and $c_T = \sqrt{k/\rho c\tau}$ at high frequency; GN-II is non-dispersive.

which is the Green–Naghdi III law with a relaxation of the flux, i.e. the template with $\tau_q = \tau$, $\tau_T = \tau_v = 0$. The uncoupled thermal equation is third order in time,

$$\tau\,\dddot{\theta} + \ddot{\theta} = c_*^2\nabla^2\theta + \alpha\nabla^2\dot{\theta} \quad (\rho c = 1)\,. \tag{42}$$

Its structure explains the dispersion shown in Figure 7: $G(s) = (k + k^*/s)/(1 + \tau s)$ gives $G \simeq k^*/s$ at low frequency (wave speed c_*) and $G \simeq k/(\tau s)$ at high frequency (wave speed $c_T = \sqrt{k/\rho c\tau}$). The model thus has finite signal speed at high frequency, in contrast to GN-III; and a low-frequency regime with undamped-type thermal waves. The decay of radial solutions of MGT thermoelasticity is analysed in [72]. The wave-dispersion questions that remain open include the sign conditions that guarantee decay for the full coupled system with all parameters positive and the behaviour of the thermal front for finite τ; applications to piezo-semiconductors with fractional MGT have appeared [84, 85].

9.5 Hyperbolic two-temperature models

Two-temperature theories [61, 62] distinguish the conductive temperature φ from the dynamic temperature θ, linked by $\theta = \varphi - a\nabla^2\varphi$ with $a \geq 0$ a length-squared constant, and hyperbolic versions combine this with LS or GN heat laws [63–65]. In plane waves the two temperatures differ by the factor $1 + ak^2$, so a wavenumber dependence enters the dispersion relation through a; this plays a role comparable to a nonlocal length scale as regards high-wavenumber behaviour. Stability and characterization of plane waves in the hyperbolic two-temperature theory are in [63].

9.6 Summary of the template models

Table 5 compiles the qualitative properties that were verified in Sections 8–9. Table 5 gives a more discriminating picture than the common statement “all generalized theories have finite

Table 5: Properties of the uncoupled thermal branch in the template models (verified from $G(s)$ and by the scripts listed in Appendix C).

Model	Low-ω thermal speed	High-ω thermal speed	Finite signal speed	Dissipative
Classical	$\propto \sqrt{\omega}$	$\propto \sqrt{\omega}$	no	yes
Lord–Shulman	$\propto \sqrt{\omega}$	$\to c_T$	yes	yes
DPL (1st order)	$\propto \sqrt{\omega}$	$\propto \sqrt{\omega}$ $(\alpha\tau_T/\tau_q)$	no (strict)	yes
GN II	c_*	c_*	yes	no
GN III	c_*	$\propto \sqrt{\omega}$	no (strict)	yes
TPL ($k^* \neq 0$)	c_*	$\propto \sqrt{\omega}$ if $\tau_T > 0$	no (strict)	yes
MGT	c_*	$\to c_T$	yes	yes

thermal speed". A thermal front with finite speed requires that $sG(s)$ tend to a finite nonzero limit as $s \to \infty$ (then $K \simeq -s\rho c/G \propto \omega^2$); this holds for LS, MGT and GN-II and fails for the classical law, DPL, TPL and GN-III in the first-order template.

10 Fidelity of Taylor-truncated lag models

Section 9 treats the dual-phase-lag (DPL) law through Taylor truncations of the shifted fluxes. This is a modelling choice, not an identity, and its consequences deserve a quantitative look. Here we compare the truncations with the *exact* delay law on the uncoupled thermal branch.

10.1 Setting

Take $\tau_T = 0$ in the DPL law, $q(t+\tau_q) = -k\,\theta_{,x}(t)$. In the Laplace/Fourier representation the exact delay gives $G(s) = k\,e^{-\tau_q s}$, the first-order truncation gives $G = k/(1+\tau_q s)$ (Cattaneo), and the second-order truncation gives $G = k/(1+\tau_q s+\tau_q^2 s^2/2)$. With $s = -i\omega$ the thermal branch satisfies $k_w^2 = -\rho c\, s/G(s)$, where k_w is the wavenumber. All three differ only through G, so they coincide as $\omega\tau_q \to 0$.

10.2 Numerical comparison

Figure 8 shows phase speed and attenuation for the three laws, computed by `scripts/fig_lag_truncation.py`. The relative error in k_w against the exact delay is

$\omega\tau_q$	0.05	0.1	0.3	1.0
first order	6×10^{-4}	2.5×10^{-3}	2.2×10^{-2}	0.22
second order	1×10^{-5}	8×10^{-5}	2.2×10^{-3}	0.08

Both truncations are accurate for $\omega\tau_q \lesssim 0.3$ and lose accuracy at $\omega\tau_q \sim 1$, which is exactly where lagging effects are claimed to matter. Conclusions drawn at $\omega\tau_q \gtrsim 1$ from a truncated law therefore describe the truncated model, not the delay law.

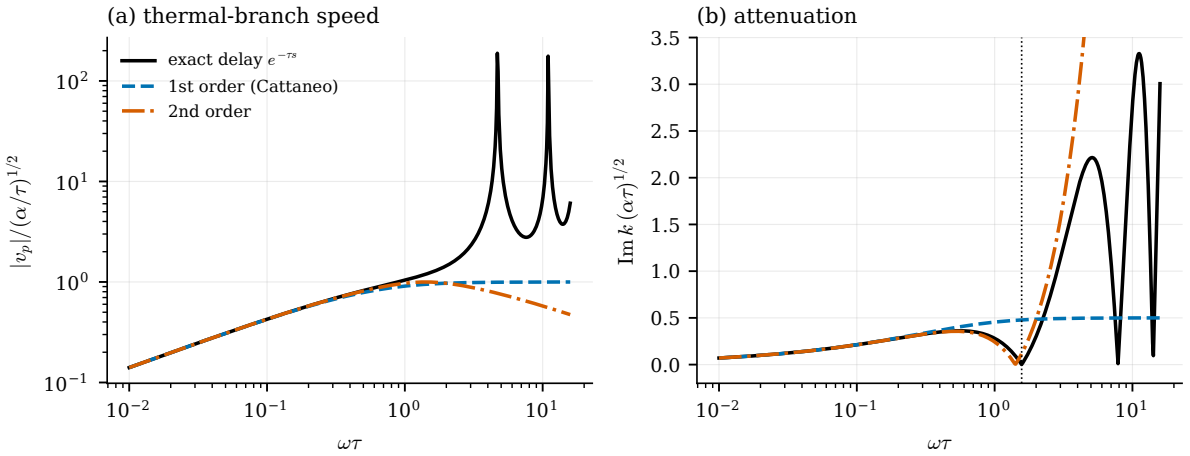


Figure 8: Thermal branch for the exact delay $e^{-\tau s}$, first-order and second-order truncations: (a) speed, (b) attenuation. Roots are taken with $\mathrm{Im}\,k \geq 0$, so the real part of k of the delay law can be negative at higher $\omega\tau$ and the spikes of $|v_p|$ mark zeros of $\mathrm{Re}\,k$. The dotted line marks $\omega\tau = \pi/2$, where the attenuation of the principal root of the delay law changes sign.

10.3 Sign change of attenuation in the exact delay law

For the exact delay the principal root is $k_w = \sqrt{\rho c\,\omega/k}\; e^{i(\pi/4-\omega\tau_q/2)}$, so $\mathrm{Im}\,k_w < 0$ once $\omega\tau_q > \pi/2$ (numerically checked: $\mathrm{Im}\,k_w > 0$ at $\omega\tau_q = 1$, $= 0$ at $\pi/2$, < 0 at 2). Choosing $\mathrm{Re}\,k_w > 0$ then gives spatial growth, and choosing decay gives a negative phase speed. The truncations avoid this behaviour, but only because they discard the high-frequency part of the delay. This is a reason for caution when a delay law is used beyond $\omega\tau_q \approx \pi/2$ and a reason to state, in any application, which truncation is used.

10.4 Practical rule

(i) Report the truncation order together with τ_q, τ_T. (ii) Check $\omega\tau_q \lesssim 0.3$ before trusting a truncated law as an approximation of the delay law. (iii) Do not read agreement between two truncated laws at $\omega\tau_q \sim 1$ as confirmation of the delay model.

11 Fractional and memory-dependent models

11.1 Motivation

Fractional and memory-dependent derivatives replace the local-in-time structure of Cattaneo's and Tzou's laws by an operator that weights the past, so that the response at time t depends on the history of the gradient or flux. Three motivations are cited: anomalous (sub- or super-diffusive) heat transport in heterogeneous and porous media, relaxation spectra that are not single-exponential, and the wish to interpolate between diffusive and wave-like conduction with a single parameter.

11.2 Caputo-fractional heat conduction

The Caputo derivative of order $\alpha > 0$ with $m-1 < \alpha \le m$, m integer, is [51, 154]

$$ {}^{C}D^{\alpha} f(t) = \frac{1}{\Gamma(m-\alpha)} \int_0^t (t-\xi)^{m-\alpha-1} f^{(m)}(\xi)\,\mathrm{d}\xi, \qquad \mathscr{L}\{{}^{C}D^{\alpha} f\} = s^{\alpha}\hat{f} - \sum_{j=0}^{m-1} s^{\alpha-1-j} f^{(j)}(0). \tag{43} $$

The Laplace rule in (43) is the reason for the Caputo definition: initial conditions have their usual physical meaning. Two families of fractional heat-conduction models are in common use.

1. *Time-fractional heat equation* [52–54]: $\rho c\, {}^{C}D^{\alpha}\theta = k\nabla^2\theta$ with $0 < \alpha \le 2$, interpolating between the diffusion ($\alpha = 1$) and a wave-like ($\alpha = 2$) equation, with subdiffusion for $\alpha < 1$ and diffusion–wave behaviour for $1 < \alpha < 2$.
2. *Fractional Cattaneo-type law* [55, 57, 58]: $q_i + \tau^{\alpha}\, {}^{C}D^{\alpha} q_i = -k\theta_{,i}$, with $G(s) = k/(1+(\tau s)^{\alpha})$ in the Laplace domain.

Further variants (fractional DPL and TPL, with fractional orders on one or both sides of the lag equation) are numerous; reviews and critical discussions of the logic of fractional thermoelasticity are given in [52–54, 56, 124, 155]; fractional-order generalised models with two relaxation times, with two temperatures and with the new Caputo definition are treated in [156–158].

The diffusion–wave equation. For the first family and prescribed boundary temperature θ_0 on the half-line $x > 0$, with zero initial data, the Laplace-domain solution is $\hat{\theta} = (\theta_0/s)\exp(-x s^{\alpha/2}/\sqrt{a})$, $a = k/\rho c$ (with a fractional time scale absorbed in a). For $\alpha = 1$ this inverts to $\theta_0 \operatorname{erfc}(x/2\sqrt{at})$, which we reproduced numerically with Talbot inversion to eight digits (0.4795001 at $x/\sqrt{at} = 1$). For $\alpha = 2$ it gives the step $\theta_0 H(t - x/\sqrt{a})$, a pure wave with no dispersion or damping; for $\alpha < 1$ the solution is a stretched Mittag-Leffler-type decay with heavy tails [159, 160]. Figure 9(a) displays the family. Two cautions: (i) the endpoint $\alpha = 2$ is *not* Fourier's law, but a undamped wave equation, and (ii) for $1 < \alpha < 2$ the equation requires two initial conditions, one of which ($\dot{\theta}(0) = 0$) is chosen by convention.

11.3 Memory-dependent derivative

The memory-dependent derivative (MDD) of Wang and Li [59] is defined for a delay (memory) window $\tau > 0$ by

$$ D_{\tau} f(t) = \frac{1}{\tau} \int_{t-\tau}^{t} K(t-\xi)\, f'(\xi)\,\mathrm{d}\xi, \qquad K(u) = 1 - \frac{2bu}{\tau} + \frac{a u^2}{\tau^2}, \tag{44} $$

with a, b constants; the choices $a = b = 1$ ($K = (1-u/\tau)^2$), $a = 0, b = \tfrac{1}{2}$ ($K = 1 - u/\tau$) and $a = b = 0$ ($K = 1$) are popular (Figure 9b). The operator differs from the fractional derivative in having a finite window, a bounded kernel, and a clear time-domain meaning as a weighted average of the rate over the last τ time units; Wang and Li's comparison with the fractional derivative [60, 123] discusses the relation of the two. MDD versions of lag models replace the first-order derivatives in (37) or (40) by D_{τ} [110–114, 117, 161, 162].

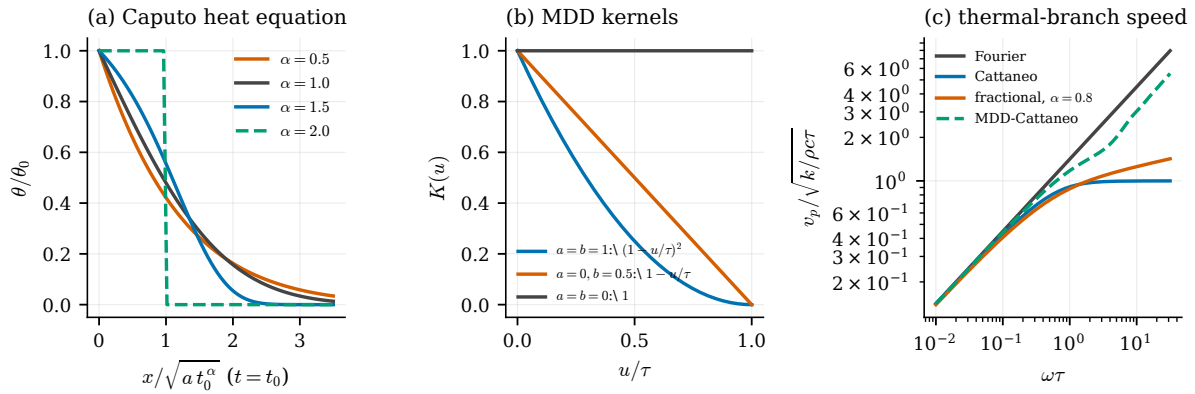


Figure 9: (a) Caputo time-fractional heat equation, boundary temperature step, profiles at $t = t_0$ for $\alpha = 0.5, 1, 1.5, 2$ (Talbot inversion; the case $\alpha = 2$ is the exact step front). (b) Kernels of the memory-dependent derivative (44). (c) Phase speed of the uncoupled thermal branch for Fourier, Cattaneo, fractional Cattaneo with $\alpha = 0.8$ and MDD-Cattaneo ($a = b = 1$), normalised by the Cattaneo wave speed.

Laplace transform of the MDD. Because f' vanishes for negative arguments, (44) is a convolution of f' with $K(u)\mathbf{1}_{[0,\tau]}(u)/\tau$, and

$$\mathscr{L}\{D_\tau f\} = s\,\kappa(s)\,\hat{f}, \qquad \kappa(s) = \frac{1}{\tau}\int_0^\tau K(u)\,e^{-su}\,\mathrm{d}u. \tag{45}$$

For the kernel $K = (1-u/\tau)^2$, $\kappa(0) = \frac{1}{3}$ and $s\kappa(s) \to K(0)/\tau = 1/\tau$ as $s \to \infty$. An MDD version of Cattaneo's law, $q_i + \tau D_\tau q_i = -k\theta_{,i}$, therefore has $G(s) = k/\big(1+\tau s\kappa(s)\big)$, which tends to $k/2$ (not to zero!) at high frequency: the thermal branch of this particular MDD-Cattaneo law becomes *diffusive* at high frequency with half the low-frequency diffusivity, in contrast to the ordinary Cattaneo law, for which $G \to 0$ and the speed saturates (Figure 9c, green dashed curve versus blue). The reason is that a bounded kernel with $K(0) = 1$ gives $\tau D_\tau \to$ a time-local multiple of the identity at high frequency, which acts as a constant added to the denominator rather than a derivative. This is a model-specific property: a kernel with $K(0) = 0$ would give $\tau s\kappa(s) \to 0$ and hence $G \to k$ (Fourier behaviour) at high frequency, and other MDD formulations of the lag laws behave differently, so the consequence for wave fronts must be checked for the variant at hand.

11.4 Position of these models in the template

Because $G(s)$ for fractional and MDD models is not rational, the dispersion relation (26) is still quadratic in K with $G(s)$ evaluated at complex $s = -\mathrm{i}\omega$ (with principal branches), so that the analysis of Section 6.3 carries over verbatim; only the closed-form front-speed analysis and the transform inversion change. The energy functionals, uniqueness and thermodynamic admissibility of these models are less developed than for the template models, and their thermodynamic admissibility has been discussed with differing conclusions. A practical concern is that the extra order or memory parameter can be tuned to fit a single dataset without independent confirmation. The reinterpretation of fractional thermoelasticity in terms of anomalous heat conduction [124] and the thermodynamic consistency analyses of time-differential lag models [44, 45] provide the context in which one should choose among them.

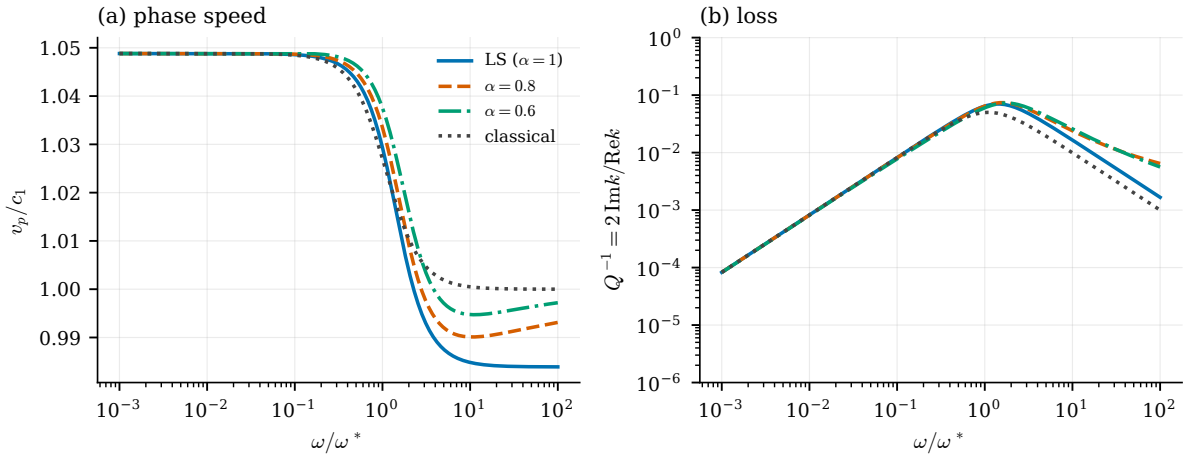


Figure 10: Coupled longitudinal wave with a fractional Cattaneo law: (a) phase speed, (b) loss Q^{-1} for $\alpha = 1, 0.8, 0.6$ and the classical model ($\varepsilon = 0.1$, $\tau\omega^* = 0.25$).

12 Coupled fractional dispersion

The memory models of Section 11 are usually analysed on the heat equation alone. Here we put a fractional Cattaneo law, $q + \tau^\alpha D^\alpha q = -k\,\theta_{,x}$ with a Caputo derivative D^α and $0 < \alpha \leq 1$, into the coupled longitudinal problem of Section 6. In the Laplace representation $G(s) = k/[1 + (\tau s)^\alpha]$, and the polynomial of Eq. (30) is used unchanged, with the principal branch $(-i\omega)^\alpha = \omega^\alpha e^{-i\pi\alpha/2}$.

Figure 10 (script `scripts/fig_frac_coupled.py`) shows phase speed and loss for the elastic-like branch, tracked by continuity in ω, for $\varepsilon = 0.1$, $\tau\omega^* = 0.25$ (so that the thermal front speed is twice c_1 and the branches do not hybridise degenerately). The numbers that can be read from it are:

- At low frequency all models give the adiabatic speed $c_1\sqrt{1+\varepsilon} = 1.0488\,c_1$.
- The classical Debye peak is $Q^{-1} = 0.0501 \approx \varepsilon/2$ at $\omega/\omega^* \approx 1.1$.
- The lagging and fractional models shift and raise the peak: 0.0701 at $\omega/\omega^* = 1.42$ for $\alpha = 1$, 0.0741 at 1.58 for $\alpha = 0.8$ and 0.0728 at 1.75 for $\alpha = 0.6$.
- At the top of the plotted range $v_p/c_1 = 0.984, 0.993, 0.997$ for $\alpha = 1, 0.8, 0.6$, and 1.000 for the classical model.

The qualitative message is that the fractional order moves the position and height of the loss peak by amounts comparable to the Cattaneo–classical difference, so it cannot be treated as a small correction. These numbers belong to this parameter set; a different $\tau\omega^*$ changes them, and the figure should be reproduced for the material at hand.

13 Wave propagation in three dimensions

13.1 Plane waves in an isotropic generalized thermoelastic solid

In an unbounded isotropic solid with any heat-flux law of the template, (24) generalises to

$$\rho\ddot{\phi} = (\lambda + 2\mu)\nabla^2\phi - \beta\theta, \quad s\rho c\,\hat{\theta} + T_0\beta s\nabla^2\hat{\phi} = G(s)\nabla^2\hat{\theta}, \quad \rho\ddot{\boldsymbol{\psi}} = \mu\nabla^2\boldsymbol{\psi}, \tag{46}$$

and a plane wave in direction $\boldsymbol{n}$, $\propto \exp[\mathrm{i}(k\,\boldsymbol{n}\cdot\boldsymbol{x} - \omega t)]$, has the same characteristic equation (26) for the dilatational–thermal system. The dispersion analysis of Section 6.3 therefore applies without change to waves in any direction of an isotropic body. The shear wave has speed c_2 and is not affected by the heat-flux law, for any member of the template and for fractional or memory models as well. For general anisotropy the quasi-longitudinal, quasi-shear and thermal branches couple through β_{ij} and k_{ij}, and the characteristic equation becomes a polynomial of higher degree; Dhaliwal and Sherief's formulation [139] is the standard reference, and for numerical dispersion the Stroh formalism [140, 141] is convenient.

Counting branches. The number of branches in a bulk plane-wave problem is 2 (dilatational–thermal) $+2$ (shear polarisations) in an isotropic medium, with G rational of low degree; the degree of the polynomial in K equals 2 only if G is rational of the kind in the template. For the second-order lag version (ii) (38), or higher-order models, the number of thermal-type branches increases. Several additional branches appear in published dispersion curves of such models, and it is advisable to separate genuine physical branches from spurious roots created by Taylor truncation.

13.2 Reflection at a free surface: structure of the problem

A plane dilatational wave incident on a plane free surface in an isotropic thermoelastic half-space generates reflected waves of three kinds: a dilatational (elastic-type) wave, a thermal-type wave (evanescent or strongly damped), and a shear (SV) wave. Boundary conditions are the traction-free conditions together with a thermal condition, either *insulated* (zero normal flux, $\partial_n\theta = 0$ for any G) or *isothermal* ($\theta = 0$). The amplitude ratios are obtained from a 3×3 linear system and have been computed for many models; the comparison of insulated and isothermal boundaries for generalized theories is a classical exercise [87], and extensions with nonlocal, micropolar, piezoelectric and rotational effects are numerous [88–91]. We will not tabulate their results; instead, Section 18 gives the structure of the free-surface problem used for Rayleigh waves, which is the same determinant, and Section 21 treats the interface problem with a verified energy check.

13.3 Waves in plates, rods and layered structures

In bounded waveguides, the thermal and elastic fields are coupled by lateral boundary conditions and a family of guided modes arises (Lamb modes in plates [163], Pochhammer–Chree modes in rods). The thermoelastic version has one additional family of thermal modes per thermal condition, and the dispersion curves of the elastic modes are perturbed by the thermal coupling in the same way as in Figure 3: small speed corrections and a damping that peaks where the thermal relaxation frequency of the cross-section matches the wave frequency. For thin plates and beams the flexural damping reduces to Zener's classical result (Section 6.4). Texts on the elastic waveguide problem are [164, 165]; for the thermoelastic waveguide with generalized laws the literature consists mainly of analyses of specific configurations [82, 117].

14 Anisotropy: thermal coupling of quasi-longitudinal waves in cubic crystals

Most of the theory above is isotropic. Real devices use single crystals, and the question is how the coupling strength ε depends on the propagation direction and which modes are affected.

14.1 Christoffel problem with thermal coupling

For a cubic crystal the thermal expansion is isotropic, $\alpha_{ij} = \alpha_T \delta_{ij}$, so the thermal-stress tensor is $\beta_{ij} = \beta_0 \delta_{ij}$ with $\beta_0 = (c_{11} + 2c_{12})\alpha_T$, and heat conduction is scalar. For a plane wave with direction $\boldsymbol{n}$ the field equations give, in the low-frequency (adiabatic) limit $\omega \ll \omega_*$, the Christoffel equation

$$\left[\Gamma_{ik}(\boldsymbol{n}) + \frac{\beta_0^2 T_0}{c_v} n_i n_k - \rho v^2 \delta_{ik}\right] p_k = 0, \qquad \Gamma_{ik} = C_{ijkl} n_j n_l, \tag{47}$$

where c_v is the volumetric heat capacity, and in the high-frequency (isothermal) limit the same equation without the thermal term. The thermal term is a rank-one matrix along $\boldsymbol{n}$, so it couples only to the polarisation component along the propagation direction, and it vanishes for purely transverse modes. Along the symmetry directions of a cubic crystal the coupling is therefore governed by

$$\varepsilon_{\boldsymbol{n}} = \frac{\beta_0^2 T_0}{c_v M_{\boldsymbol{n}}}, \qquad M_{[100]} = c_{11},\; M_{[110]} = \tfrac{1}{2}(c_{11} + c_{12} + 2c_{44}),\; M_{[111]} = \tfrac{1}{3}(c_{11} + 2c_{12} + 4c_{44}), \tag{48}$$

with $M_{\boldsymbol{n}}$ the longitudinal modulus along $\boldsymbol{n}$, and the quasi-longitudinal speed rises from the isothermal to the adiabatic value by the factor $\sqrt{1 + \varepsilon_{\boldsymbol{n}}}$. Both transverse modes along these directions are unaffected.

14.2 Numerical values

Table 6 gives the result for three cubic metals and semiconductors, computed with the script `scripts/fig_aniso.py` from rounded handbook values of the elastic constants, expansion coefficient, density and specific heat at $T_0 = 300\,\mathrm{K}$; these inputs should be replaced by the measurements for a specific sample. The speed shifts obtained from the full eigenvalue problem agree with $\varepsilon_{\boldsymbol{n}}/2$ to the digits shown.

Table 6: Direction dependence of the thermoelastic coupling of quasi-longitudinal waves ($T_0 = 300\,\mathrm{K}$, rounded handbook inputs). $\varepsilon_{\boldsymbol{n}}$ from Eq. (48); shift is $100(v_{\mathrm{ad}}/v_{\mathrm{iso}} - 1)$.

	[100]		[111]		ratio
Crystal	ε	shift (%)	ε	shift (%)	$\varepsilon_{[100]}/\varepsilon_{[111]}$
Cu	0.0238	1.18	0.0169	0.84	1.41
Al	0.0325	1.61	0.0306	1.52	1.06
Si	6.5×10^{-4}	0.032	5.3×10^{-4}	0.026	1.23

For copper the coupling differs by 41% between [100] and [111], so the thermoelastic loss peak height (proportional to $\varepsilon_{\boldsymbol{n}}$ in the weak-coupling Debye form) and the adiabatic–isothermal speed gap are direction dependent by a comparable amount. For aluminium the anisotropy of

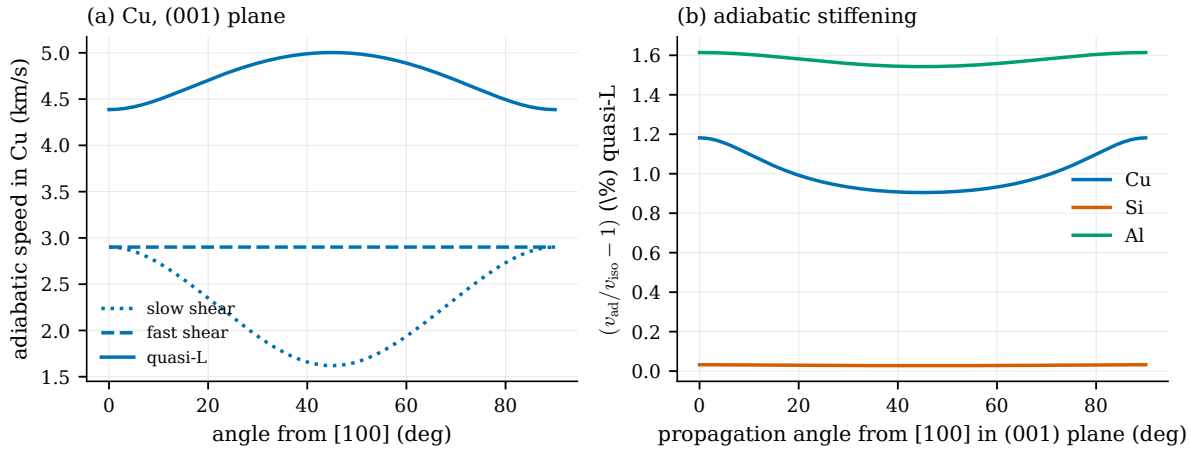


Figure 11: (a) Adiabatic phase speeds of the three modes of copper in the (001) plane as functions of the propagation angle from [100]. (b) Relative stiffening $v_{\mathrm{ad}}/v_{\mathrm{iso}} - 1$ of the quasi-longitudinal mode for Cu, Si and Al.

the coupling is only 6% because the elastic anisotropy is weak, and for silicon the coupling is tiny at all angles, so the longitudinal-wave thermoelastic loss is very small in that material.

14.3 What changes for the heat-flux law

Anisotropic heat conduction (non-cubic crystals) makes G a tensor, so the template of Section 3 generalises to $\hat{q}_i = -G_{ij}(s)\hat{\theta}_{,j}$ and the lag times may differ along different crystal axes. For the cubic case used here, conduction is isotropic and only the elastic and coupling anisotropy enter, so every scalar result of the preceding sections carries over with $c_1^2 \to M_{\boldsymbol{n}}/\rho$ and $\varepsilon \to \varepsilon_{\boldsymbol{n}}$ for propagation along a high-symmetry direction. Away from those directions the quasi-longitudinal polarisation is not along $\boldsymbol{n}$, the coupling is reduced by $(\boldsymbol{n}\cdot\boldsymbol{p})^2$, and a quasi-transverse mode acquires a small thermal component.

15 Reflection at a thermoelastic free surface: numerical results

Section 13 described the structure of the free-surface problem. Here we give numbers, and a validation, for a plane P wave incident on an insulated traction-free surface of an isotropic thermoelastic half-space.

15.1 Method

The half-space is $y > 0$ with plane strain and the same non-dimensionalisation as in Section 18 ($\rho = c_1 = c_v = \omega_*^{-1} = 1$, $\nu = 0.3$, $\varepsilon = 0.1$). The tangential wavenumber $k_x = \mathrm{Re}(k_P)\sin\theta$ is real and conserved. The incident elastic-type wave has amplitude one; the reflected field consists of a P wave, a thermal wave that decays with distance from the surface, and an SV wave, and the three amplitudes follow from the 3×3 system of two traction conditions and the insulation condition. The columns of the system are those of the Rayleigh determinant of

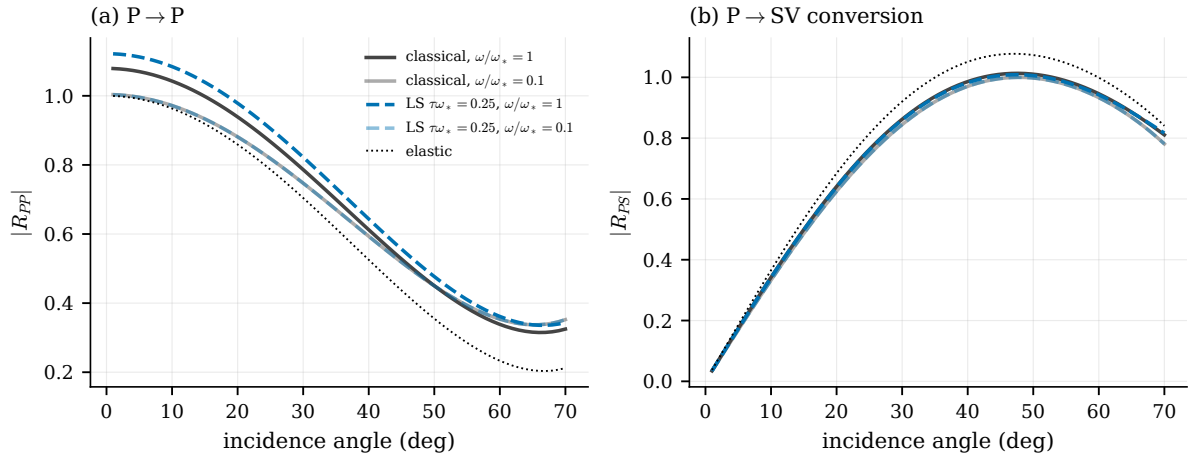


Figure 12: Displacement-amplitude reflection coefficients at an insulated free surface for classical and Lord–Shulman heat flux at $\omega/\omega_* = 1$ (dark) and 0.1 (light), $\varepsilon = 0.1$; the dotted line is the purely elastic result.

Section 18, with the sign of the transverse wavenumber reversed for the incident wave (script `scripts/fig_reflection.py`).

15.2 Validation

For $\varepsilon = 10^{-10}$ and $\omega/\omega_* = 0.5$ the system reduces to the elastic problem. The computed displacement-amplitude coefficients agree with the classical closed forms for a free surface,

$$R_{PP} = \frac{\sin 2\theta \sin 2\phi - \kappa^2 \cos^2 2\phi}{\sin 2\theta \sin 2\phi + \kappa^2 \cos^2 2\phi}, \qquad R_{PS} = \frac{2\kappa \sin 2\theta \cos 2\phi}{\sin 2\theta \sin 2\phi + \kappa^2 \cos^2 2\phi}, \tag{49}$$

with $\kappa = c_1/c_2$ and $\sin\phi = \kappa^{-1}\sin\theta$, to 1.5×10^{-10} in $|R_{PP}|$ and 8×10^{-11} in $|R_{PS}|$ over $1^\circ \le \theta \le 70^\circ$ (for example 0.6970 and 0.9283 at 30°).

15.3 Results

Figure 12 compares classical and Lord–Shulman ($\tau\omega_* = 0.25$) heat flux at $\omega/\omega_* = 0.1$ and 1.

- At $\omega/\omega_* = 0.1$ the two heat laws are indistinguishable to 10^{-4}: $|R_{PP}| = 0.878, 0.509, 0.349$ at 20°, 45°, 60° for both. The thermal coupling changes the elastic values only at the level of ε.
- At $\omega/\omega_* = 1$ the classical law gives $|R_{PP}| = 0.935, 0.517, 0.333$ at the same angles and the Lord–Shulman law gives 0.975, 0.546, 0.355, i.e. differences of 4%, 5.5% and 7% between the two heat laws. The mode-converted amplitude changes by less, $|R_{PS}| = 0.648$ versus 0.647 at 20° and 1.012 versus 1.007 at 45°.

The displacement amplitude $|R_{PS}|$ exceeds one at 45°; this reflects the difference between displacement and energy normalisation of P and SV waves, not a gain, and conclusions about energy must be drawn from the energy flux (Section 21).

15.4 Reading the result

The effect of the heat law on reflection is of the order of several percent only near $\omega \sim \omega_*$, which for metals lies in the tens of GHz,. For frequencies in the MHz range the reflection coefficients of a thermoelastic solid are those of an elastic solid with the adiabatic moduli, whatever the heat law. This supports the model-selection argument of Section 31 and, more practically, means that discrepancies between reflection measurements and elastic predictions in the MHz range cannot be attributed to the heat-flux law.

16 Transient problems: heat pulses and thermal shock

Steady-state dispersion analysis tells us how a monochromatic wave propagates. Transient problems test the *fronts*: finite speed, jumps and their decay. Two benchmark problems are used here because their Laplace-domain solutions are closed-form and they can be inverted numerically to any accuracy; they also serve as test cases for numerical codes (Section 25).

16.1 Uncoupled heat pulse in a half-space

Consider the semi-infinite rod $x > 0$, initially at $\theta = 0$, with prescribed boundary flux $q(0,t) = q_0 H(t)$ and no coupling. For $G(s) = k(1+\tau_T s)/(1+\tau_q s)$ with $\tau_T = 0$ (Lord–Shulman), $\tau_q = \tau$, the solution is

$$\hat{\theta}(x,s) = \frac{q_0\,(1+\tau s)}{k\,s\,m(s)}\,e^{-x m(s)}, \qquad m(s) = \sqrt{\frac{s(1+\tau s)}{\alpha}}, \tag{50}$$

and $\tau = 0$ gives Fourier's solution $\theta = (q_0/k)\big[2\sqrt{\alpha t/\pi}\,e^{-x^2/4\alpha t} - x\operatorname{erfc}(x/2\sqrt{\alpha t})\big]$. Large-$s$ asymptotics, $m(s) \simeq s/c_T + 1/(2\ell)$ with $c_T = \sqrt{\alpha/\tau}$ and $\ell = \sqrt{\alpha\tau} = c_T\tau$, give the front structure: the signal is exactly zero ahead of $x = c_T t$; at the front it jumps by

$$\big[\theta\big]_{x=c_T t} = \frac{q_0}{\rho c\, c_T}\,e^{-x/2\ell}, \tag{51}$$

decaying exponentially with distance, and at the boundary the temperature jumps from zero to $q_0/\rho c c_T$ at $t = 0^+$, instead of growing as $\sqrt{t}$. The Cattaneo curves of Figure 13 were obtained by numerical Talbot inversion [166] of (50) (with the exact zero set ahead of the front); the ripples near the front are the Gibbs-type error of the inversion at a discontinuity. At late times the two solutions coincide (the ratio at $t = 40\tau$ differs from one by less than one percent), which is the sense in which the Cattaneo model is a high-frequency correction to Fourier's law.

Boundary conditions decide the sign. Statements such as "second sound lowers the surface temperature" or "raises" it are meaningless without the boundary condition. With a prescribed flux, the hyperbolic model gives a larger surface temperature than Fourier's law at early times in this normalisation (Figure 13); with a prescribed surface temperature, it is the flux that jumps, with an overshoot relative to Fourier's. Laser-heating models therefore require a careful statement of how the heat input is represented [116].

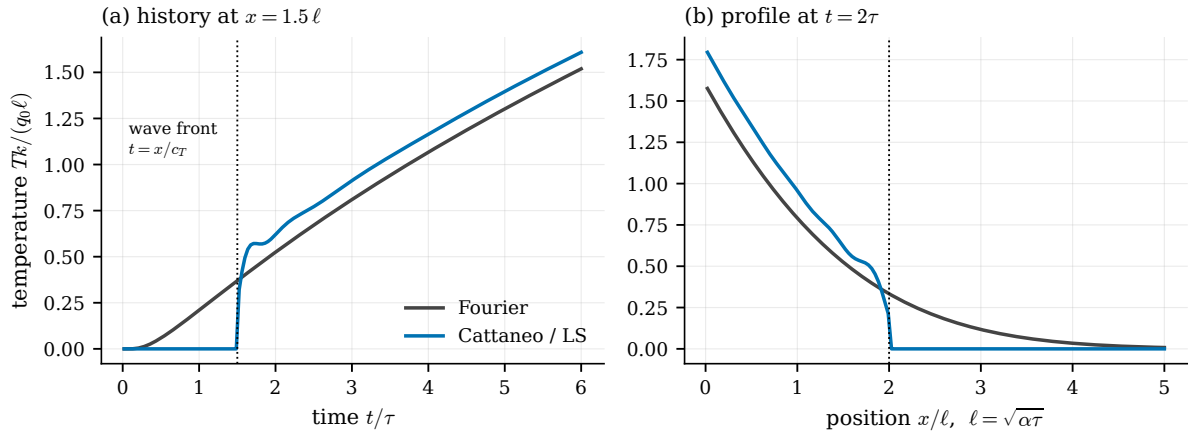


Figure 13: Uncoupled heat pulse from a prescribed boundary flux. Left: temperature history at $x = 1.5\ell$. Right: profile at $t = 2\tau$. The Lord–Shulman curve is exactly zero beyond the front (dotted); the classical solution has no front. At early times the Lord–Shulman surface temperature exceeds the classical one for this boundary condition.

16.2 Thermal shock of a coupled half-space

Next consider the half-space $x > 0$ with the boundary at temperature $\theta_0 H(t)$ and traction-free, coupled thermoelastic one-dimensional problem. In the Laplace domain, with fields $\propto e^{-mx}$ the characteristic equation is (26) with $K = -m^2$, giving two decay rates $m_1(s)$, $m_2(s)$ ($\mathrm{Re}\, m > 0$). In units $\rho = c = c_1 = k = T_0 = 1$ (so that lengths are in c_1/ω_* and time in $1/\omega_*$), with $\beta = \sqrt{\varepsilon}$, the displacement and stress amplitudes associated with a temperature amplitude A_j are

$$U_j = \frac{\beta\, m_j}{s^2 - m_j^2} A_j, \qquad \sigma_j = -m_j U_j - \beta A_j \tag{52}$$

and the boundary conditions $\sum_j A_j = \theta_0/s$, $\sum_j \sigma_j(0) = 0$ determine A_1, A_2; the second condition reduces to $\sum_j A_j\, s^2/(s^2 - m_j^2) = 0$. We verified that the resulting transform satisfies $s\hat{\theta}(0,s) = 1$ and $\hat{\sigma}(0,s) = 0$ to machine precision, and invert numerically. Figure 14 shows temperature and stress at a point $x = 2c_1/\omega_*$ for the classical, Lord–Shulman and DPL laws ($\varepsilon = 0.1$, $\tau_q = 0.25/\omega_*$, DPL $\tau_T = \tau_q/2$).

The following qualitative features are robust. (i) The thermal part of the response is almost identical in all three models after the first elastic transit, because the differences between heat laws are confined to the first moments after loading; the long-time evolution is governed by the classical diffusion. (ii) The stress is largest, and most sensitive to the heat law, at the elastic front, where the Lord–Shulman model gives about 20 % larger compressive peak than the classical one in this example (peak values -0.30 against -0.25, in units of ρc_1^2). (iii) Before the elastic front arrives the classical solution already carries stress, because the thermal body force acts instantly everywhere; in Lord–Shulman the stress is essentially zero until the thermal front arrives at $t = 1$ and then rises more steeply. The numerical amplitudes depend on the dimensionless parameters and should not be extrapolated to real materials without recomputation.

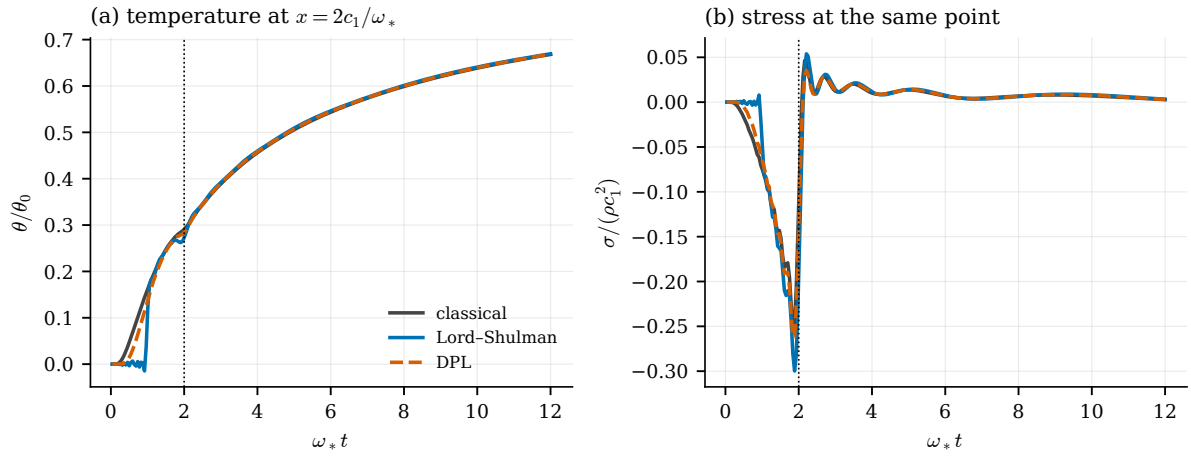


Figure 14: Thermal shock of a coupled half-space (boundary temperature step, traction-free surface), recorded at $x = 2c_1/\omega_*$. Dotted line: arrival of the elastic wave front ($x = c_1 t$). The Lord–Shulman thermal front reaches the point at $t = x/c_T = 1$; the stress develops a negative (compressive) pulse and jumps at the elastic front. Small oscillations are inversion error at discontinuities.

16.3 Lessons for laser heating and for numerical codes

Short-pulse laser heating in metals is the main application driving hyperbolic models; analyses of bilayers and films under pulsed heating combine fractional and memory-dependent laws [116]. For numerical codes, the benchmark problems above provide a stringent check, because a code that smears the front also misses the stress peak. The classical statement of the thermal shock problem is due to Danilovskaya [3], and thermal-shock solutions of the coupled problem are a standard test in texts [8, 21, 22].

17 Thermal-wave resonance in a finite rod

The uncoupled heat law offers a small, exactly solvable example of how a finite size reveals the hyperbolic character: the normal modes of a rod with insulated ends.

17.1 Modes of the Cattaneo equation

For a rod $0 < x < L$ with $q = 0$ at both ends, eliminating q from the Cattaneo system gives $\tau\theta_{tt} + \theta_t = \chi\theta_{xx}$, with $\theta \propto \cos(k_n x)e^{st}$, $k_n = n\pi/L$, and

$$\tau s^2 + s + \chi k_n^2 = 0 \quad \Rightarrow \quad s_\pm = \frac{-1 \pm \sqrt{1 - 4\tau\chi k_n^2}}{2\tau}. \tag{53}$$

For the Fourier law ($\tau = 0$) every mode decays at the real rate χk_n^2. For $\tau > 0$ the modes with $k_n < k_c = 1/(2\sqrt{\chi\tau}) = 1/(2\ell)$, where $\ell = c_T\tau$ is the relaxation length, are overdamped with two real rates, and the modes with $k_n > k_c$ are oscillatory with frequency $\omega_n = \sqrt{\chi k_n^2/\tau - 1/(4\tau^2)}$, decaying at the common rate $1/(2\tau)$. Thus the thermal wave appears as a damped oscillation only in the modes of wavelength shorter than $4\pi\ell$, and then it always decays at the rate $1/2\tau$ independently of mode number.

17.2 Size at which oscillation is possible

The lowest mode ($n = 1$) is oscillatory if $L < 2\pi\ell$. For copper, with $\chi = 1.2 \times 10^{-4}\,\mathrm{m^2/s}$ (Table 3), a lag $\tau = 1\,\mathrm{ps}$ gives $c_T = 1.1 \times 10^{4}\,\mathrm{m/s}$ and $\ell = 1.1 \times 10^{-8}\,\mathrm{m}$, so a rod shorter than 68 nm would have an oscillatory first mode; for $\tau = 10\,\mathrm{ps}$ it is $\ell = 3.4 \times 10^{-8}\,\mathrm{m}$ and $L < 210\,\mathrm{nm}$. In both cases the size is comparable to or below the carrier mean free paths for which the continuum conduction law itself is questionable, which agrees with the conclusion of Section 30: the oscillatory regime of the Cattaneo law is reached only where the underlying assumptions of the law fail.

17.3 Relation to the coupled problem

In the coupled problem of Section 20 the elastic-like mode inherits damping from the thermal modes through the factor ε, but the heat-conduction modes themselves are of the type above and are damped; there is no regime in which heat-driven oscillations persist. This is the energy-dissipative nature of the Cattaneo law ($\mathrm{d}\mathscr{E}/\mathrm{d}t = -\int q^2/kT_0 \le 0$, Eq. (16)), in contrast to the conservative Green–Naghdi type II law, whose uncoupled thermal modes are undamped oscillations $\omega_n = c_T k_n$.

18 Surface waves: thermoelastic Rayleigh waves

18.1 Formulation

Consider the half-space $y > 0$ in plane strain, with waves propagating along x, $\propto e^{\mathrm{i}kx}$ and decaying with depth. With potentials $\boldsymbol{u} = \nabla\phi + \nabla \times (\psi \boldsymbol{e}_z)$, each coupled dilatational–thermal mode has $\phi_j, \theta_j \propto e^{-m_j y}$ with

$$m_j^2 = k^2 + \kappa_j, \qquad G\kappa^2 - \big(Gs^2 + s + \varepsilon s\big)\kappa + s^3 = 0 \quad (\rho = c = c_1 = T_0 = 1), \tag{54}$$

which is (26) with $K = -\kappa$, and the shear potential decays with $m_s^2 = k^2 + s^2/c_2^2$. The ratio of temperature to potential amplitude in each coupled mode is $\theta_j = (\kappa_j - s^2)\phi_j/\beta$. The boundary conditions on $y = 0$ are traction-free, $\sigma_{yy} = \sigma_{xy} = 0$, and either an insulated ($\partial_y\theta = 0$, which is $q_y = 0$ for any G) or an isothermal ($\theta = 0$) thermal condition. With

$$\sigma_{yy} = \sum_{j=1}^{2} \big[\lambda\,\kappa_j + 2\mu m_j^2 - \beta\,\theta_j/\phi_j\big]\phi_j + 2\mu\,\mathrm{i}k m_s\,\psi, \tag{55}$$

$$\sigma_{xy} = \mu\Big[-2\mathrm{i}k \sum_{j=1}^{2} m_j\phi_j + (m_s^2 + k^2)\psi\Big], \tag{56}$$

the three conditions form a homogeneous 3×3 linear system for (ϕ_1, ϕ_2, ψ) whose determinant $\mathscr{D}(s,k)$ must vanish. The root $s(k)$ with $\mathrm{Re}\, m_j > 0$ gives the complex Rayleigh frequency $\omega = \mathrm{i}s$ for real k (temporal damping); the same determinant at real ω and complex k gives spatial damping. Differences between formulations in the literature (which wave attenuation measure is plotted, which branch of the square roots is used, how the thermal boundary condition is chosen) account for many apparent disagreements among published Rayleigh dispersion curves [81, 83–85, 167–171].

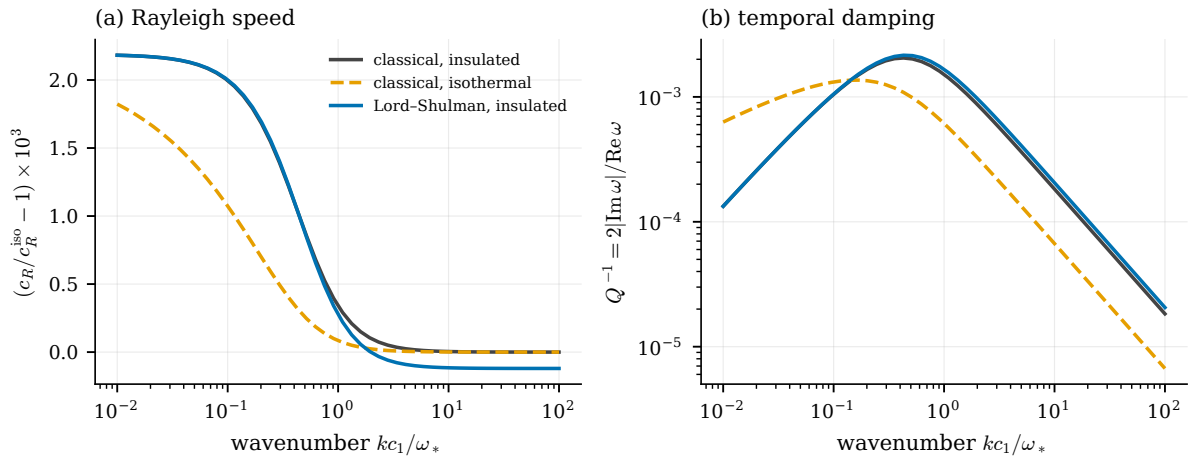


Figure 15: Thermoelastic Rayleigh wave in a half-space ($\nu = 0.3$, $\varepsilon = 0.05$). (a) Speed relative to the purely elastic Rayleigh speed. (b) Temporal damping $Q^{-1} = 2|\mathrm{Im}\,\omega|/\mathrm{Re}\,\omega$ at real wavenumber k in units of ω_*/c_1. Curves: classical theory with insulated and with isothermal surface, and Lord–Shulman with $\tau_q = 0.25/\omega_*$ and insulated surface.

Check of the limit. For $\varepsilon \to 0$ the determinant factorises and the root reduces to the classical Rayleigh equation $(2 - c^2/c_2^2)^2 = 4\sqrt{1 - c^2/c_1^2}\sqrt{1 - c^2/c_2^2}$ [172]. For $\nu = 0.3$ (so $c_2^2/c_1^2 = (1-2\nu)/(2-2\nu) = 2/7$) the root is $c_R = 0.4957\,c_1 = 0.9274\,c_2$; the numerical determinant root at $\varepsilon = 10^{-4}$ reproduces this value to the digits displayed, and the thermoelastic root approaches it at large k where the response becomes isothermal.

18.2 Results

Figure 15 shows the speed change and damping of the thermoelastic Rayleigh wave for $\varepsilon = 0.05$, $\nu = 0.3$. The speed increases by about 2×10^{-3} at low wavenumber relative to the purely elastic Rayleigh speed (the adiabatic limit) and tends to the elastic value at large k (the isothermal limit), in the same way as the bulk wave (Figure 3a), but the effect is about an order of magnitude smaller than for the compressional wave because only part of the surface motion is dilatational. The damping peaks at $kc_1/\omega_* \approx 0.4$ with a quality-factor reciprocal of about 2×10^{-3}. The thermal boundary condition matters at low wavenumber: an isothermal surface shifts the damping peak to smaller k and lowers the speed change, since the surface temperature fluctuation is suppressed. The Lord–Shulman law ($\tau_q = 0.25/\omega_*$, insulated surface) changes the results little at low wavenumber but increasingly at large k: relative to the classical insulated surface the damping differs by less than 1 % for $kc_1/\omega_* \le 0.1$, by 5 % at the damping peak, by 10 % at $kc_1/\omega_* = 1$ and saturates at about 12 % for $kc_1/\omega_* \ge 10$; the speed offset is -1.2×10^{-4} in relative terms at large k. This is the pattern expected from the bulk wave: the heat law matters once the product of the frequency and the lag is of order one.

18.3 Anisotropic, piezoelectric and nonlocal surface waves

For anisotropic media the Stroh formalism [140, 141] gives the displacement–traction state vector as a first-order system in y, and the secular equation becomes the vanishing of a 3×3 (or larger, when piezoelectric and thermal fields are added) determinant built from the eigenvectors

with decaying exponents; this is the route taken for piezothermoelastic layers and half-spaces [82]. For piezoelectric materials, new surface-wave types arise: the Bleustein–Gulyaev shear-horizontal wave exists for transversely isotropic piezoelectric substrates even though a pure SH wave cannot exist at an ordinary elastic free surface; Rayleigh waves in nonlocal piezoelastic half-spaces and piezo-thermoelectric semiconductors have been treated with Eringen-type nonlocality and with fractional or Klein–Gordon-type extensions [83–85, 128]. Energy aspects of surface thermoelastic waves are discussed in [86]. In each of these extensions the structure of the problem is identical to the determinant above, with additional exponents and boundary conditions; the physical effects are the shift of the Rayleigh root by electromechanical stiffening (by K^2, see Section 23), by nonlocal dispersion (Section 22), and by thermal damping as shown here.

18.4 Practical remarks on secular equations

Three technical points recur. (i) The decaying branch of each square root must be chosen with $\operatorname{Re} m > 0$, and in the complex plane this introduces branch cuts across which the determinant is not analytic; root finding that ignores this can jump to spurious roots. (ii) The determinant has trivial zeros ($s = 0$ for instance) that must be excluded when tracking roots; we found that a naive secant search from a poor initial guess converged to such a trivial root. (iii) Continuation in k from a reliable starting point (here the isothermal Rayleigh root) is more robust than independent root finding at each k.

19 Thermoelastic Lamb waves in a plate

Surface waves on a half-space (Section 18) have a counterpart in plates, where the two free faces couple the dilatational, shear and thermal fields. The plate problem is where the flexural thermoelastic damping of Section 7 is recovered from the full field equations, which gives a stringent test of the formulation.

19.1 Formulation

Consider a plate $-h < y < h$ in plane strain, with waves $\exp[i(kx - \omega t)]$ along x and real k. The fields are the two dilatational–thermal modes $j = 1, 2$, which are the two roots κ_j of the coupled quadratic of Section 6 with transverse wavenumbers $m_j = \sqrt{\kappa_j + k^2}$, and the shear mode with $m_s = \sqrt{s^2/c_2^2 + k^2}$ with $s = -i\omega$. Each mode appears as $e^{\pm m y}$, so there are six amplitudes. The boundary conditions on $y = \pm h$ are $\sigma_{yy} = \sigma_{xy} = 0$ and either an insulated ($\theta_{,y} = 0$) or an isothermal ($\theta = 0$) face, which gives a 6×6 determinant $D(s; k, h)$ whose roots are the plate modes. For a given real k the roots s are complex, and $Q^{-1} = 2|\operatorname{Im}\omega|/\operatorname{Re}\omega$ measures the temporal damping. The heat-flux model enters only through $G(s)$ in the quadratic, so lagging and fractional laws are handled by substitution. The implementation is in `scripts/lamb_te.py`.

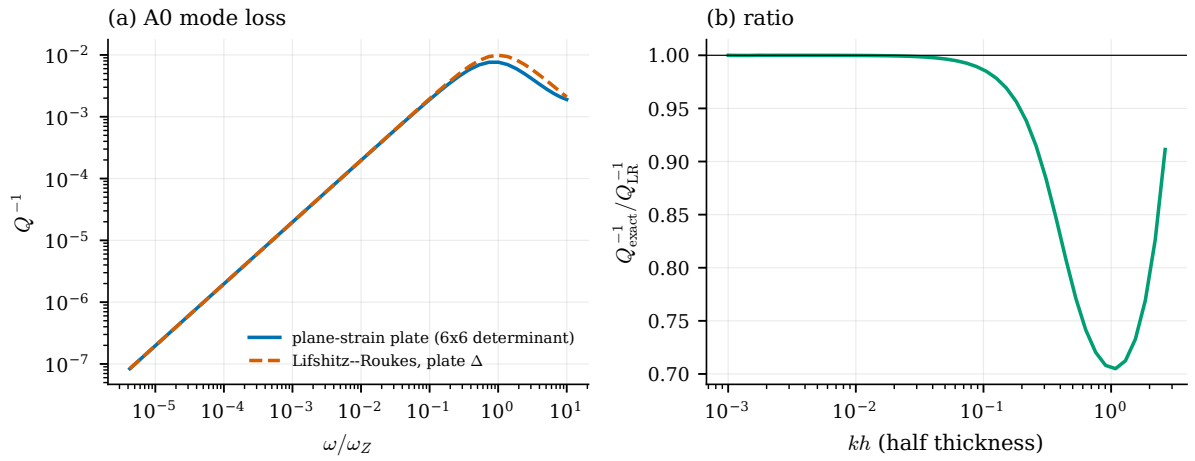


Figure 16: Lowest antisymmetric (flexural) mode of a thermoelastic plate, plane strain, $\varepsilon = 0.05$, $\nu = 0.3$, $h = 20\,c_1/\omega_*$, insulated faces. (a) Loss from the 6×6 determinant and from Eq. (29) with the plane-strain Δ. (b) Ratio of the two against kh.

19.2 Validation

Two checks were made.

1. *Elastic limit.* With $\varepsilon = 10^{-8}$ and $kh = 0.3$ the antisymmetric root is $\omega = 0.04426280008$ (units $c_1 = 1$, $h = 1$), against 0.04426280008 from the Rayleigh–Lamb equation $\tan(qh)\,4k^2pq + (q^2-k^2)^2\tan(ph) = 0$ solved separately; they agree to about 10^{-10}.
2. *Thin-plate thermoelastic damping.* For $\varepsilon = 0.05$, $\nu = 0.3$ and a plate with $h = 20\,c_1/\omega_*$ the loss of the lowest antisymmetric mode was computed over $10^{-3} \lesssim kh \lesssim 1.3$. For $\omega \ll \omega_Z = \pi^2\chi/(2h)^2$ the loss follows Eq. (29) with the plane-strain value $\Delta = E\alpha_T^2 T_0(1+\nu)/[(1-\nu)c_v]$, with ratio 1.000 for $kh \lesssim 0.015$, 0.998 at $kh = 0.036$, and 0.989 at $kh = 0.088$.

The factor $(1+\nu)/(1-\nu)$ rather than 1 is what plane strain requires (stiffness $E/(1-\nu^2)$ and expansion $(1+\nu)\alpha_T$), and the numerics confirm it. A beam formula quoted with the uniaxial Δ would underestimate the plane-strain damping by a factor 1.86 at $\nu = 0.3$.

19.3 Beyond the thin-plate limit

Figure 16 shows the loss and its ratio to Eq. (29). The ratio falls to 0.94 at $kh = 0.22$, to 0.77 at $kh = 0.53$ (where $\omega = 1.01\,\omega_Z$) and to 0.71 at $kh = 1.3$. The peak loss is $Q^{-1} = 7.6\times10^{-3}$ at $\omega/\omega_Z = 1.01$, against 9.9×10^{-3} for the thin-plate formula. Thus the thin-plate formula overestimates the loss by about 30% at the peak once the wavelength is only a few thicknesses, presumably because shear and rotary-inertia effects, absent from the Euler–Bernoulli kinematics behind Eq. (29), become important; this computation does not isolate that cause. Resonator designs with $kh \gtrsim 0.3$ therefore need the full determinant or a Mindlin-type thermoelastic model.

19.4 Role of the heat-flux law

In Eq. (29) heat flows across the thickness, so the relevant dimensionless number is the product of the lag with the transverse relaxation rate $\omega_Z = \pi^2\chi/(2h)^2$. For thicknesses of several tens of nanometres in metals, $\tau_q\omega_Z \ll 1$ and the parabolic result is adequate (Section 7); for sub-ten-

nanometre films, or when the phonon mean free path is comparable to the thickness, a kinetic or Guyer–Krumhansl treatment (Section 4) is the more appropriate refinement. The determinant above does not by itself settle which regime applies; it only provides the correct reference result for each choice of $G(s)$.

20 Resonant modes of a clamped bar

The loss measured in a resonator is a temporal quantity: a standing mode with real wavenumber $k_n = n\pi/L$ decays in time with a complex frequency ω_n. Here the plane-wave polynomial of Eq. (30) is used with real k and complex $s = -i\omega$.

20.1 Setting

Take a bar of length L with both ends clamped ($u = 0$) and insulated ($\theta_{,x} = 0$). Each eigenfunction has the form $u \propto \sin k_n x$, $\theta \propto \cos k_n x$ with $k_n = n\pi/L$, so the boundary conditions are satisfied exactly by each coupled mode and the modal frequencies are the roots of the polynomial of Eq. (30) in s at $K = k_n^2$. For Lord–Shulman heat flux the polynomial is quartic in s, for the classical law cubic. The mode tracked is the elastic-like root, followed by continuity in k from the adiabatic starting value $c_1\sqrt{1+\varepsilon}\,k$. In units $c_1 = \omega_* = 1$ and with $\varepsilon = 0.1$ the results of the script `scripts/fig_bar_modes.py` are as follows.

20.2 Results

- At small k all models give the same loss (numerically 8.7×10^{-4} at $k = 0.01$), about 13% below the first-order Debye value $\varepsilon k/(1+k^2)$ because of the $O(\varepsilon)$ difference between temporal and spatial loss measures.
- The classical law gives a Debye-type peak $Q^{-1} = 0.0488$ at $k c_1/\omega_* = 1.07$, and the modal speed falls from the adiabatic value $1.049\,c_1$ to c_1 at large k.
- For Lord–Shulman with $\tau\omega_* = 0.25$ the peak rises to 0.0676 at $k = 1.39$, and the high-k modal speed settles at $0.984\,c_1$.
- For $\tau\omega_* = 1$, where the thermal front speed equals c_1, the peak is 0.146 at $k = 1.84$, three times the classical value, and the high-k modal speed is $1.17\,c_1$. This is the degenerate case of equal front speeds, where the two fronts split by coupling to $c^2 = c_1^2[1 + \varepsilon/2 \pm \sqrt{\varepsilon + \varepsilon^2/4}]$, giving $1.17\,c_1$ for the upper root.

The curves are shown in Fig. 17. Near $\tau\omega_* \approx 1$ the thermal and elastic branches hybridise, so the label "elastic-like" ceases to be sharp and results must be read together with the branch-tracking procedure.

20.3 Consequence for resonator design

A bar of fixed length and a sequence of overtones traverse the curve of Fig. 17 from left to right. The classical model predicts the worst loss when $k_n c_1 \approx \omega_*$, and the lag raises it only if $\tau\omega_*$ is of order one, where the lag law is also least trustworthy. For all materials of Table 3 ω_* exceeds 10^{11} rad/s, so the position of the peak lies at wavelengths of order $c_1/\omega_* \sim 10$ nm, and bar resonators of micron size operate on the left, linear-in-k flank where the models coincide.

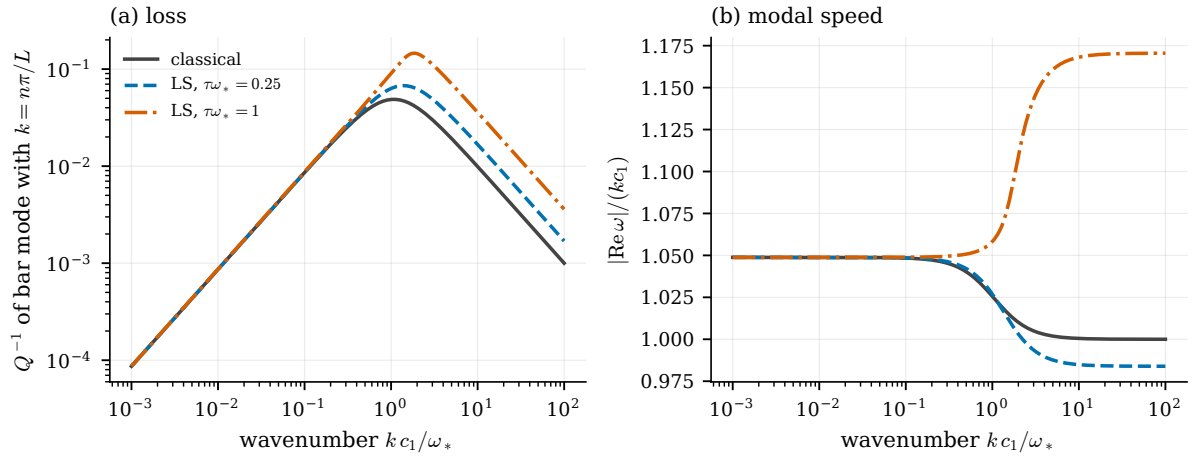


Figure 17: Loss $Q^{-1} = 2|\operatorname{Im}\omega|/|\operatorname{Re}\omega|$ and modal speed of the elastic-like mode of a clamped insulated bar ($\varepsilon = 0.1$) against $k = n\pi/L$ for classical and Lord–Shulman heat flux.

21 Interfaces, reflection and energy partition

Reflection and transmission at interfaces is the most heavily studied wave problem in generalized thermoelasticity and multiphysics media; Table 2 lists many of the works. Most of them report *energy ratios* and use the requirement that the ratios sum to unity as a validation. This section builds the problem from the elastic baseline, then adds thermal coupling and shows when and why the unit-sum check is satisfied or violated, since this is a point on which the literature is not uniform.

21.1 Welded elastic interface at normal incidence

Two elastic half-spaces with impedances $Z_j = \rho_j c_{1j}$ are perfectly bonded at $x = 0$. For a wave incident from medium 1, continuity of displacement and of traction gives the displacement amplitude coefficients

$$r = \frac{Z_1 - Z_2}{Z_1 + Z_2}, \qquad t = \frac{2Z_1}{Z_1 + Z_2}, \tag{57}$$

and the time-averaged energy fractions $\mathscr{R} = r^2$, $\mathscr{T} = 4Z_1Z_2/(Z_1+Z_2)^2$ satisfy $\mathscr{R} + \mathscr{T} = 1$ (Figure 18). The displacement transmission coefficient t can reach two for a soft second medium without violating conservation, since only the energy fractions are bounded.

21.2 Oblique incidence on an elastic interface

For oblique incidence of a P wave, Snell's law conserves the horizontal slowness $p = \sin\theta_1/c_{1,P} = \sin\phi_1/c_{1,S} = \sin\theta_2/c_{2,P} = \sin\phi_2/c_{2,S}$ and four waves are generated (reflected P and SV, transmitted P and SV). Continuity of $u_x, u_z, \sigma_{zz}, \sigma_{xz}$ gives four linear equations for the amplitudes (the Zoeppritz system). We solved it for a pair of media ($\rho_1, c_P, c_S = 2.5, 5.0, 2.8$ and $\rho_2, c_P, c_S = 3.0, 8.0, 4.5$ in arbitrary consistent units), with energy flux per wave $P_z = -\frac{1}{2}\operatorname{Re}(\sigma_{zj}\overline{\dot{u}_j})$. The sum of the four energy fractions equals one to rounding error ($|\mathrm{sum} - 1| < 2 \times 10^{-15}$ over 0.5°–89.5°), and at normal incidence the P–P reflection fraction (0.099268) equals the impedance formula (0.099268). Figure 19(a) shows

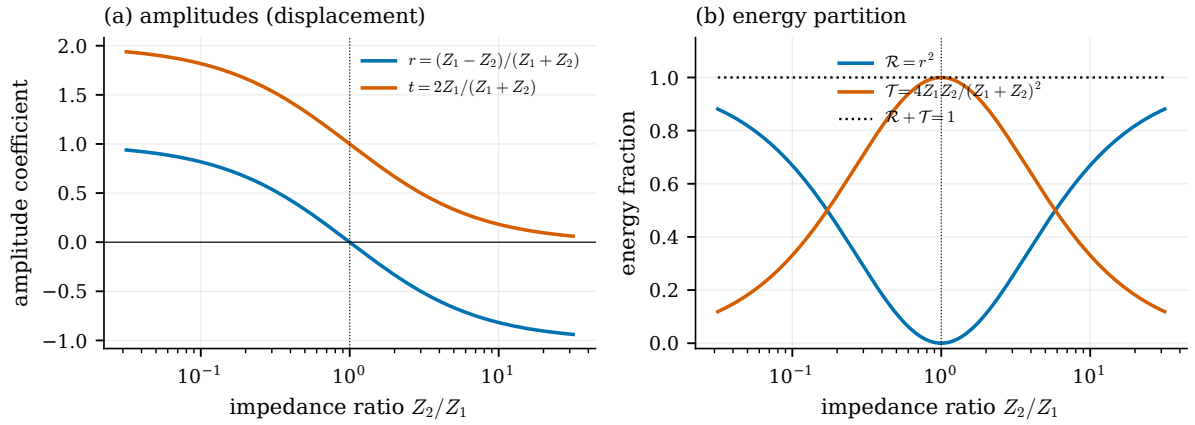


Figure 18: Normal incidence on a welded elastic interface. (a) Displacement reflection and transmission coefficients. (b) Energy fractions; $\mathscr{R}+\mathscr{T}=1$ for every impedance ratio.

the characteristic features: a critical angle (38.7° here), beyond which the transmitted P wave is evanescent and carries no energy; a rise of the reflected P fraction to total reflection of P energy in the P–P channel only at grazing incidence, with mode conversion in between.

21.3 Thermoelastic interfaces

When both media are thermoelastic, two further wave types appear in each medium and the interface conditions include thermal continuity. For perfect thermal contact, θ and the normal flux q_n are continuous together with displacement and traction. In normal incidence there are four amplitudes (reflected elastic and thermal waves, transmitted elastic and thermal waves) and four equations. We implemented this for two media with different densities, wave speeds, heat capacities, conductivities, coupling constants and lag times, with Lord–Shulman conduction; incident wave of elastic type. The amplitude ratios and the flux of each wave follow from (17). The tests we performed are: (i) as $\varepsilon \to 0$ (here $\beta = 10^{-6}$) the reflected elastic energy fraction (0.169550173) equals the impedance formula $(Z_1-Z_2)^2/(Z_1+Z_2)^2$ to twelve digits; (ii) the *net* flux, computed from the total fields on either side of the interface, is continuous to 10^{-15} for all frequencies tested.

When does “energy fractions sum to one” hold? The exact statement is item (ii): the net flux of (17) into the interface equals the net flux out of it, because the continuity conditions imply continuity of $\sigma\dot{u}$ and of θq. This is not the same as saying that the *sum of the fluxes carried by each wave separately* equals the incident flux. For homogeneous waves in lossless media the two statements coincide, because the cross terms between distinct modes vanish in the time average (this is what the elastic checks above confirm numerically). In a dissipative medium, the modes have complex wavenumbers and the cross terms between, for example, the elastic-type and the thermal-type wave at the interface do not vanish, so

$$\sum_{n} \mathscr{F}_n \neq 1 \quad \text{in general, while} \quad \mathscr{F}_{\text{net}} \equiv 1. \tag{58}$$

Figure 19(b,c) shows an example: the sum of the four mode fractions is within 1.5 % of unity for $\omega \lesssim 2\omega_*$ but reaches 1.34 near $\omega \approx 8\omega_*$ and individual fractions can exceed one (the transmitted elastic fraction reaches 1.1). This is not a violation of energy conservation: the cross terms

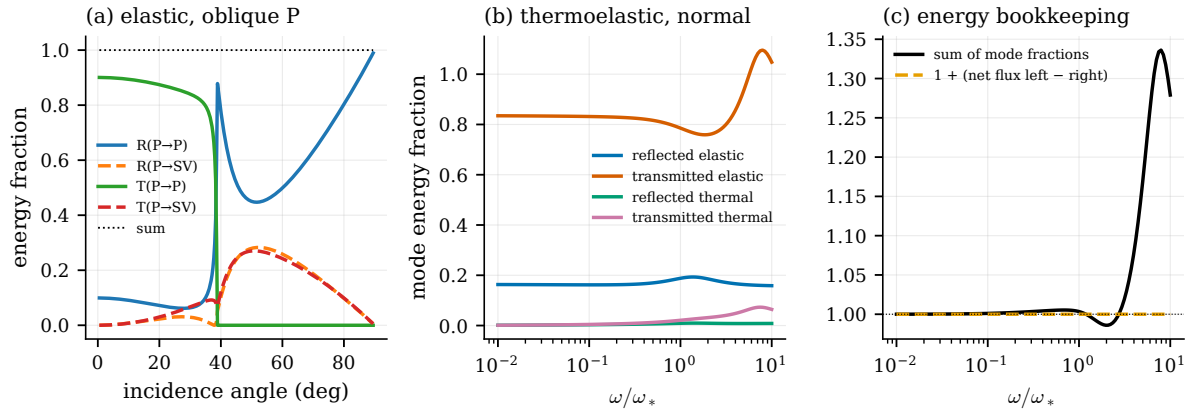


Figure 19: Energy partition at welded interfaces. (a) Oblique P incidence on an elastic–elastic interface: reflected and transmitted P and SV energy fractions; their sum is unity at every angle. (b) Normal incidence on a thermoelastic–thermoelastic interface with Lord–Shulman conduction ($\varepsilon_1 = 0.1$, $\varepsilon_2 = 0.3$, $\tau_{q1} = 0.25/\omega_*$, $\tau_{q2} = 0.15/\omega_*$): energy fractions of the four generated waves against frequency. (c) The sum of mode fractions deviates from unity at high frequency because of interference terms, while the net flux of (17) is continuous across the interface (dashed).

account for the difference. The consequence for published results is practical. An energy-ratio sum that deviates from unity can be caused by (a) a wrong coupling coefficient or boundary condition, (b) a flux definition inconsistent with the model, or (c) genuine interference of modes in a dissipative medium; the first two are errors, the third is not, and a correct check must use the net flux, or include the interference terms, rather than the sum of mode fluxes. Studies that report energy ratios at thermoelastic and piezothermoelastic interfaces [94–97, 99–104] should be read with this point in mind: when the media are lossy, a unit sum of mode ratios is not a necessary condition and a failure of the sum to equal one is not by itself evidence of error.

21.4 Imperfect interfaces

Real interfaces are rarely perfect. The common idealisation replaces continuity by spring conditions: the traction is continuous and proportional to the jump in displacement, $\sigma = K_n[u]$ (normal) and $K_t[u_t]$ (tangential); thermal imperfections add a thermal contact resistance, $[\theta] = R_\theta q$ (the Kapitza-type jump), and for micropolar or piezoelectric media analogous electric and rotational conditions. In the limit $K \to \infty$ one recovers the welded interface and for $K \to 0$ a free surface. With thermal resistance the net flux of (17) is no longer continuous: the jump $[\theta]q/T_0 = R_\theta q^2/T_0$ is the entropy production at the interface, and the energy balance must include it. Imperfect-interface wave problems have been treated for micropolar solids [108], piezoelectric composites [173, 174], and a fluid–double-porous thermoelastic interface [105].

21.5 Fluid–solid and multiphysics interfaces

When one medium is an ideal fluid the shear conditions change (no shear traction, tangential displacement discontinuous), and the energy ratios have been computed for fluid–piezothermoelastic and fluid–double-porous interfaces [98, 101, 105]. In piezoelectric media electric boundary conditions (continuity of potential and of normal electric displacement, or an electrode condition) add two equations per interface; the reflection and refraction at

elastic–magneto-thermoelastic diffusion solids with voids [106], micropolar piezothermoelastic half-spaces [107], functionally graded piezoelectric microstructures [109] and flexoelectric or flexomagnetic microstructures [92, 93] extend the same procedure. All of these share the structure of Sections 18 and the present one: a linear system for amplitudes whose entries are fixed by the dispersion roots of each medium, and an energy flux that must be defined consistently with the model.

21.6 A checklist for interface calculations

The following checks, which cost little, catch most errors in interface calculations:

1. Reduce the coupling to zero and compare with the elastic impedance formulas (normal incidence) or the Zoeppritz equations.
2. Verify the Snell invariants (horizontal slowness) for all generated waves, including evanescent ones.
3. Verify continuity of the net flux (17) (not of the sum of mode fluxes) across a perfect interface; for imperfect interfaces include the interface dissipation.
4. Test the boundary conditions independently: substitute the computed amplitudes back into each boundary equation.
5. For lagged and memory models, state the flux definition used for the energy ratio and whether interference terms are included.

22 Size effects: nonlocal and gradient models

22.1 Why local elasticity fails at small wavelengths

Local elasticity predicts non-dispersive waves, $\omega = c\,k$, for all wavelengths. A crystal is a lattice, and its acoustic branch bends over and reaches zero group velocity at the zone boundary. For a monatomic chain with nearest-neighbour spring constant κ_s, mass m and spacing a, the exact relation is $\omega = 2\sqrt{\kappa_s/m}\,|\sin(ka/2)| = c_0 k\,\mathrm{sinc}(ka/2)$ with $c_0 = a\sqrt{\kappa_s/m}$ [175], which at small k gives $\omega \simeq c_0 k\,[1-(ka)^2/24]$. Nonlocal and gradient models add one length scale to capture the leading deviation from local behaviour in a continuum framework. Maranganti and Sharma [127] estimate the scales at which classical elasticity breaks down for various materials, and show that size effects can also generate electromechanical coupling in non-piezoelectric solids [126]. Micropolar thermoelasticity without energy dissipation was formulated by Ciarletta [147] and Quintanilla [148]; strain gradient and higher-order nonlocal formulations appear in [176, 177], and size-dependent piezoelectric nanoshells are treated in [178].

22.2 Eringen's nonlocal elasticity

In the integral model the stress at $\boldsymbol{x}$ is a weighted average of the local stress over a neighbourhood, $\sigma_{ij}(\boldsymbol{x}) = \int \alpha(|\boldsymbol{x}-\boldsymbol{x}'|)\,C_{ijkl}\varepsilon_{kl}(\boldsymbol{x}')\,\mathrm{d}V'$, with a kernel α normalised to unity. For the exponential kernel in one dimension, $\alpha(|x|) = e^{-|x|/\ell}/(2\ell)$, the integral equation is equivalent to the differential equation

$$\left(1-\xi^2\partial_x^2\right)\sigma = c\,\varepsilon, \qquad \xi = \ell, \tag{59}$$

(because α is the Green's function of $1-\ell^2\partial_x^2$), and in three dimensions with the Yukawa-type kernel $e^{-r/\ell}/(4\pi\ell^2 r)$ the equation $(1-\xi^2\nabla^2)\sigma_{ij} = C_{ijkl}\varepsilon_{kl}$ [15, 16, 47]. Substituting in the momentum balance gives

$$\omega^2 = \frac{c_0^2 k^2}{1+\xi^2 k^2}, \qquad v_g = \frac{c_0}{(1+\xi^2 k^2)^{3/2}}, \tag{60}$$

where $\xi = e_0 a$ in Eringen's notation (e_0 a dimensionless material constant and a an internal characteristic length). Equation (60) is the rational analogue of the lattice relation: ω is bounded by c_0/ξ and the group velocity decays to zero at large k.

Calibration against the lattice. Matching (60) to the chain at the zone boundary $k = \pi/a$, where $\omega = 2c_0/a$, requires $(\xi/a)^2 = \frac{1}{4} - 1/\pi^2$, i.e. $\xi/a = 0.3856$; this reproduces the value $e_0 \approx 0.39$ quoted by Eringen [16] and is the origin of the often-used choice. Matching instead the long-wavelength curvature requires $\xi^2 = a^2/12$ ($\xi = 0.289a$). The two calibrations differ by about one third in ξ and thus by a factor of about 1.8 in ξ^2, so any quantitative prediction that depends on ξ depends on the calibration target (Figure 20). The group velocity of the Eringen model at the zone boundary is $0.26c_0$, whereas the chain has $v_g = 0$ there.

22.3 Strain-gradient elasticity

Aifantis-type gradient elasticity adds the gradient of the strain [17, 49],

$$\sigma = c\,(\varepsilon - \ell^2\varepsilon_{,xx}), \qquad \omega^2 = c_0^2 k^2(1+\ell^2 k^2), \qquad v_g = c_0\frac{1+2\ell^2 k^2}{\sqrt{1+\ell^2 k^2}}. \tag{61}$$

Both phase and group velocities *increase* with k for $\ell^2 > 0$, the opposite of lattice dispersion. To reproduce the lattice curvature one needs $\ell^2 = -a^2/12 < 0$, which gives $\omega^2 = c_0^2 k^2(1 - a^2k^2/12)$ and an instability ($\omega^2 < 0$) beyond $k = \sqrt{12}/a$ — outside the Brillouin zone, but a warning that the pure gradient model with the lattice sign is not well posed at high wavenumber. The dispersive-stabilised versions, with a gradient of inertia as well as of strain [18, 50], interpolate between (60) and (61): with strain-gradient length ℓ_s and inertia-gradient length ℓ_d, $\omega^2 = c_0^2 k^2(1+\ell_s^2 k^2)/(1+\ell_d^2 k^2)$, which remains stable and reproduces the lattice shape for $\ell_d > \ell_s$ (the Eringen form is the case $\ell_s = 0$).

22.4 Other enrichments

Couple-stress theory [179], micropolar theory [13, 73], flexoelectricity and surface elasticity add length scales through different mechanisms. The Klein–Gordon-type nonlocal elasticity [128] (a Helmholtz-type operator acting on the displacement rather than the stress) has been used with piezo-thermoelastic media [180]. Nonlocal variational principles are discussed in [181]. The physical ingredients differ, but at the level of plane-wave dispersion most enrichments reduce to a rational function of k^2 multiplying the local stiffness or the inertia, and the choice among them is largely a matter of which feature of the underlying lattice or microstructure one wishes to represent.

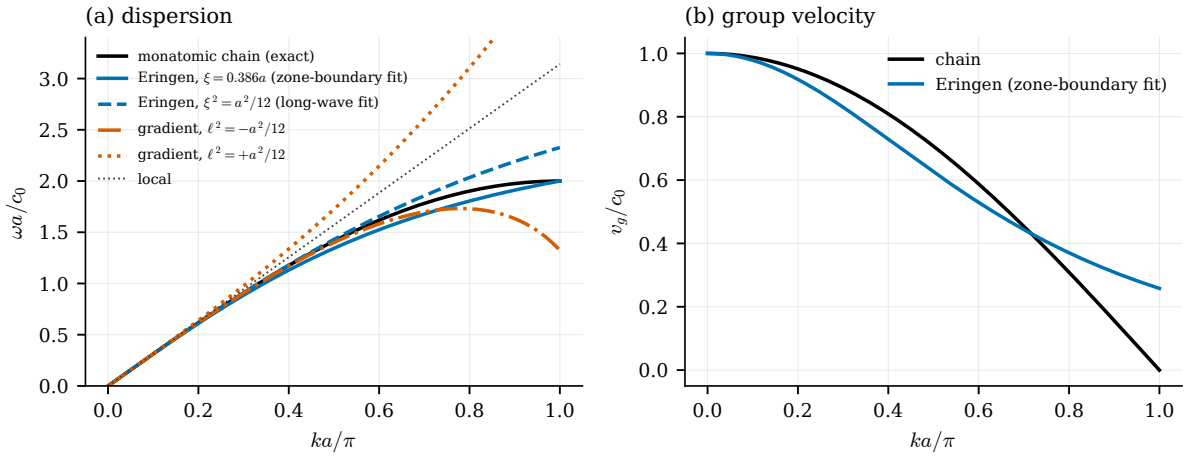


Figure 20: Nonlocal and gradient dispersion compared with the exact monatomic chain. (a) Dispersion: Eringen with two calibrations (zone-boundary and long-wavelength), gradient with positive and negative length-squared, and the local model. (b) Group velocity of the chain and of the Eringen model with the zone-boundary calibration.

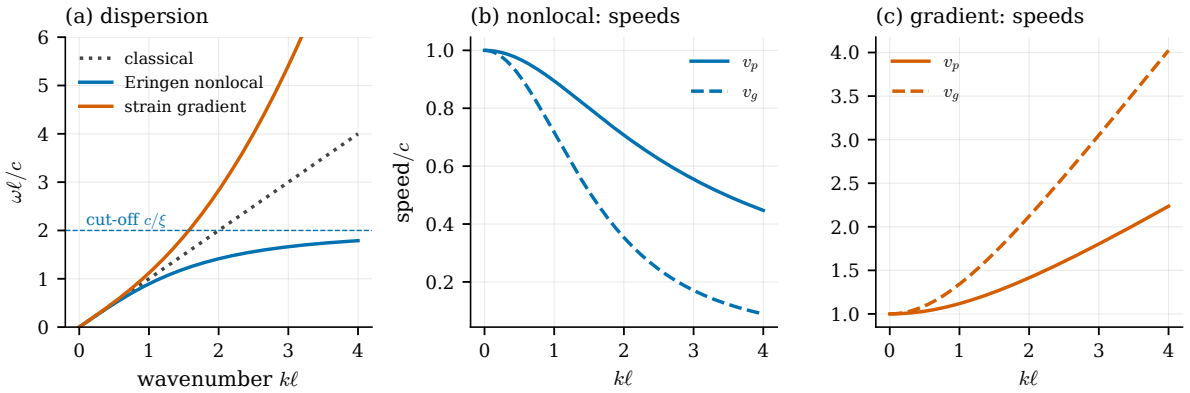


Figure 21: Generic size-effect dispersion for $\xi = \ell = 0.5$ in units of the characteristic length. (a) Frequency against wavenumber: the Eringen model saturates at c_0/ξ, the gradient model rises faster than linearly. (b,c) Phase and group speeds.

22.5 Nonlocal thermoelasticity

Eringen [48] extended nonlocality to thermoelasticity. Two forms of the constitutive relation are in use in recent papers:

$$\text{(A)} \quad (1-\xi^2\partial_x^2)\,\sigma = c\,\varepsilon - \beta\,\theta, \tag{62}$$

$$\text{(B)} \quad (1-\xi^2\partial_x^2)\,(\sigma+\beta\,\theta) = c\,\varepsilon, \tag{63}$$

that is, whether the thermal stress is smoothed with the elastic stress (B) or not (A). The two forms give different dispersion polynomials: with $K = k^2$, (A) is $\left[\rho(c_1^2+\xi^2 s^2)K + \rho s^2\right](s\rho c + GK) + \beta^2 T_0 s K = 0$, and (B) adds the factor $(1+\xi^2 K)$ to the last term. Both reduce to the local relation (26) as $\xi \to 0$ and to the nonlocal elastic relation (60) as $\varepsilon \to 0$; we confirmed both limits numerically. Figure 22 compares the elastic-type branch. In the classical theory with $\varepsilon = 0.1$ and $\xi = 0.3c_1/\omega_*$ (elastic cut-off at $\omega = 3.3\omega_*$), the main effect on the phase speed is the nonlocal dispersion itself: the speed falls steadily with frequency, to about $0.3c_1$ at 95 %

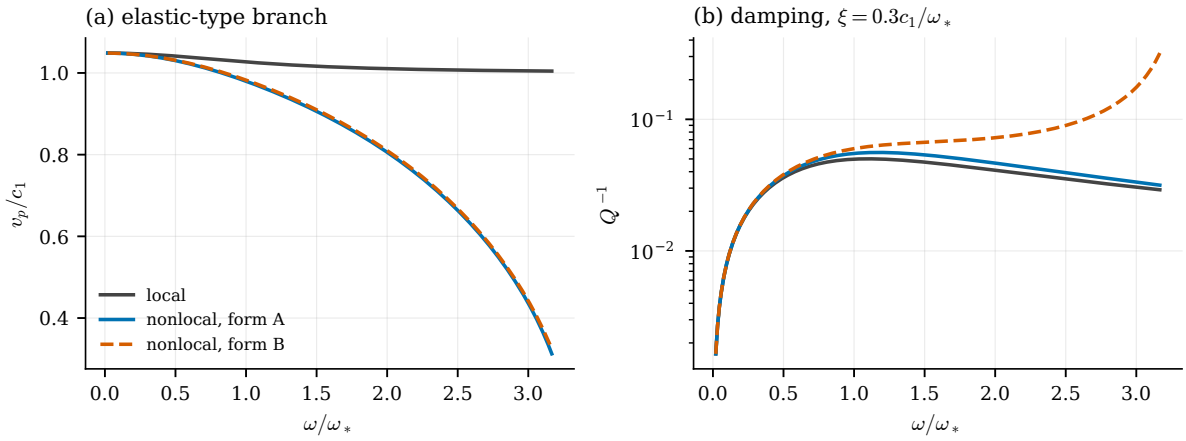


Figure 22: Elastic-type branch of the nonlocal classical thermoelastic problem ($\varepsilon = 0.1$, $\xi = 0.3c_1/\omega_*$, frequency up to 95% of the elastic cut-off c_1/ξ): phase speed (a) and damping (b) for the local model and the two nonlocal forms.

of the cut-off, and the two forms give almost the same speed. The damping differs: form (A) stays close to the local curve (its maximum is about 15 % larger and slightly shifted) whereas form (B), whose thermal coupling is multiplied by $1+\xi^2 K$ and thus grows with wavenumber, shows a damping that increases steeply on approaching the cut-off, reaching $Q^{-1} \approx 0.3$. The difference arises entirely from the high-wavenumber coupling. A paper that reports nonlocal thermoelastic damping or dispersion should therefore state which form is used; the literature on nonlocal thermoelastic waves [79, 80, 88, 120, 125, 130, 182] uses various forms.

22.6 Nonlocality in the heat flux

The Guyer–Krumhansl equation [29, 30] adds spatial derivatives of the flux,

$$q_i + \tau_R \dot{q}_i = -k\theta_{,i} + \ell_{GK}^2 \left(\nabla^2 q_i + 2\, q_{j,ji} \right), \tag{64}$$

and describes heat in the hydrodynamic phonon regime, where momentum-conserving collisions dominate. In one dimension the extra term is $3\ell_{GK}^2 q_{,xx}$ and $G(s,k) = k/(1+\tau_R s + 3\ell_{GK}^2 k^2)$ depends on the wavenumber, so the dispersion polynomial is no longer quadratic in K but retains its structure. For low-temperature crystals, ballistic and size-dependent conduction in nanostructures are modelled with this and related equations; the combination with elastic nonlocality remains little explored (Section 33).

22.7 Caveats

The differential nonlocal model produces well-known inconsistencies for bounded bodies (for example in some bending problems with particular boundary conditions) that do not appear in the infinite-medium plane-wave analysis above. The length scales are phenomenological; whether the lattice, the zone-boundary or some experimental dispersion is the calibration target has a leading effect on the quantitative predictions.

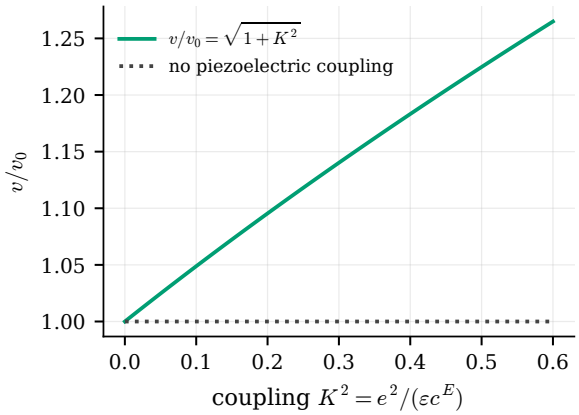


Figure 23: Piezoelectric stiffening of the longitudinal wave speed, (66).

23 Coupling to further fields

Thermoelastic waves are one member of a larger class: whenever a further field is energetically coupled to deformation and temperature, the dispersion relation acquires additional branches or renormalised constants, and the heat-flux law acts as a module that can be exchanged without altering the coupling structure. This section treats the piezoelectric coupling in detail, because it admits an exact reduction, and then surveys the principal other couplings at the level of structure.

23.1 Piezoelectric stiffening

Consider a one-dimensional piezoelectric bar in the quasi-electrostatic approximation, with $E = -\varphi_{,x}$ and the magnetic field neglected [179, 183]. The constitutive relations are

$$\sigma = c\,u_{,x} - e\,E, \qquad D = \varepsilon_e E + e\,u_{,x}, \tag{65}$$

with elastic constant c at constant field, piezoelectric constant e and permittivity ε_e at constant strain. Gauss's law $D_{,x} = 0$ (no free charge) gives $E = -(e/\varepsilon_e)u_{,x}$ up to a uniform field, so

$$\sigma = \Big(c + \frac{e^2}{\varepsilon_e}\Big)u_{,x}, \qquad v = \sqrt{\frac{c + e^2/\varepsilon_e}{\rho}} = v_0\sqrt{1+K^2}, \quad K^2 = \frac{e^2}{\varepsilon_e c}. \tag{66}$$

The electromechanical coupling constant K^2 ranges from a fraction of a percent in weakly piezoelectric crystals to several tens of percent in strongly piezoelectric ceramics, so the stiffening, shown in Figure 23, is large enough to matter in resonator design.

23.2 Thermopiezoelectric waves: exact renormalisation

With a pyroelectric coefficient p the relations extend to

$$\sigma = c u_{,x} - eE - \beta\theta, \qquad D = e u_{,x} + \varepsilon_e E + p\theta, \qquad \rho\eta = \beta u_{,x} + pE + \frac{\rho c_E}{T_0}\theta, \tag{67}$$

the last being the entropy (the pyroelectric and thermal-stress coefficients enter symmetrically, as required by the existence of a thermodynamic potential [11, 12, 78]). Eliminating E through

$D_{,x} = 0$ gives a purely thermoelastic system with the renormalised constants

$$\tilde{c} = c + \frac{e^2}{\varepsilon_e}, \qquad \tilde{\beta} = \beta - \frac{ep}{\varepsilon_e}, \qquad \rho\tilde{c}_v = \rho c_E - \frac{T_0 p^2}{\varepsilon_e}, \tag{68}$$

and therefore all results of Sections 6–9 hold with $c_1^2 \to \tilde{c}/\rho$, $\beta \to \tilde{\beta}$, $c \to \tilde{c}_v$ in the definitions of ε and ω_*. We verified this reduction symbolically by computing the 3×3 determinant of the coupled system in (u, θ, φ) and comparing it with the renormalised thermoelastic polynomial: the ratio is a constant factor independent of all material parameters except the trivial scaling of the Gauss equation (script `check_thermopiezo.py`). Three observations follow. (i) The piezoelectric stiffening raises the isothermal speed by $\sqrt{1+K^2}$ and therefore changes ω_* and ε. (ii) The product ep enters $\tilde{\beta}$ with a sign: depending on the orientation and sign conventions, the pyroelectric effect can *reduce* or *increase* the thermal stress, and the sign conventions for e and p vary across the literature, so a reported effect of the pyroelectric coefficient should be accompanied by a statement of conventions [183]. (iii) The heat capacity under zero electric displacement ($D = 0$) is smaller than under zero field, by $T_0 p^2/\varepsilon_e$, the electric analogue of the difference between adiabatic and isothermal elasticity.

Limits of the quasi-electrostatic model. The model has no electromagnetic branch and is valid when elastic wave speeds are small compared with the speed of light and the wavelength is much smaller than the electromagnetic wavelength at the same frequency, which holds for acoustic waves in all practical piezoelectric solids but excludes polaritons and fully dynamic electromagnetic coupling.

Extensions in the literature. Piezothermoelastic waves with memory and fractional heat laws, semiconductor effects (photo-excited carriers and plasma waves), nonlocal piezoelasticity and surface effects, flexoelectric microstructure, and functionally graded or viscous variants have been treated in a large body of work [91, 92, 103, 104, 109, 117, 129, 130, 132, 180, 184, 185]. The structural point of (68) applies to all of those that retain a one-dimensional or plane-wave setting: the additional coupling to the electric field in the absence of free charges is an exact renormalisation, and only when carriers (semiconductors) or higher electric moments (flexoelectricity) are added does a genuinely new branch appear.

23.3 Magneto-thermoelasticity

For a perfectly conducting solid in a uniform magnetic field $\boldsymbol{H}_0$, the Lorentz force $\mu_0 \boldsymbol{J} \times \boldsymbol{H}_0$ acts as an additional body force. For a longitudinal wave along x with $\boldsymbol{H}_0$ perpendicular to the propagation direction and the displacement along the propagation direction, $h = -H_0 u_{,x}$, $J = -h_{,x}$ and the force is $\mu_0 H_0^2 u_{,xx}$, so that $\rho\ddot{u} = (\lambda + 2\mu + \mu_0 H_0^2) u_{,xx} - \beta\theta_{,x}$. The magnetic field thus adds the Alfvén term to the stiffness, $c_1^2 \to c_1^2 + v_A^2$ with $v_A^2 = \mu_0 H_0^2/\rho$, and the dispersion polynomial (26) holds with that substitution. Finite electrical conductivity and displacement current add a magnetic diffusion branch and a further coupling that depends on the conductivity [58, 106, 112, 115, 186, 187]; the fractional-order heat-conduction law in magneto-thermoelasticity of [58] is a standard reference. Magneto-electro-thermo-elastic solutions for transversely isotropic media are given in [188]; a modified fractional-order piezoelectric thermoelasticity model is given in [189], and a memory-dependent-derivative piezo-thermoelastic model in [190].

Table 7: Added fields and their effect on the plane-wave problem (structure only; Δn is the increase in the number of branches in one-dimensional dilatational motion).

Added field	Principal effect on waves	Δn	Remarks
Electric (quasi-static, no free charge)	stiffness $c \to c + e^2/\varepsilon_e$; β, c_v renormalised	0	Exact reduction, Eq. (68).
Electric with carriers (semiconductor)	plasma / diffusive branch	$+1$	Photo-thermo-elastic and piezo-semiconductor models.
Magnetic (perfect conductor)	stiffness $c_1^2 \to c_1^2 + v_A^2$	0	Finite conductivity adds a diffusion branch.
Concentration (diffusion)	third diffusive branch	$+1$	Coupling to temperature through the chemical potential.
Voids / porosity	extra dilatational wave(s)	$+1$ (single) $+2$ (double)	Slow waves, strong damping.
Microrotation (micropolar)	rotational waves with cut-off	transverse	Longitudinal speed modified.
Two temperatures	wavenumber-dependent θ vs. φ	0	Acts like a length scale.

23.4 Thermoelastic diffusion and chemical coupling

When a diffusing species is present, the concentration C and chemical potential add a third field: the stress contains a term proportional to C, the energy balance a term proportional to $\dot{C}$ and the species balance a diffusive flux coupled to temperature. The dispersion polynomial becomes cubic in K and a third, diffusive branch appears, with decay governed by the mass diffusivity. Higher-order and multi-lag diffusion models have been proposed [119], and reflection at interfaces with magneto-thermoelastic diffusion solids with voids has been computed [106].

23.5 Microstructure: micropolar, voids and porosity

Micropolar elasticity [13, 73, 74] adds microrotation as an independent field. In the isotropic case the dilatational wave is accompanied by a longitudinal wave of modified speed, the transverse waves couple to microrotation waves, and the rotational waves have a cut-off frequency. Materials with voids [75–77] add an independent volume-fraction field and one extra dilatational wave; thermal coupling is introduced in [77]; poro-thermoelasticity and double porosity add slow dilatational waves associated with pore fluid pressure [102, 105, 191]. Fractional-order thermoelasticity with voids, micropolar fractional models and two-temperature versions have been studied [81, 88, 107, 125, 192, 193]. Each microstructural field adds branches that appear as new roots of the characteristic equation; whether they are physically observable depends on the microinertia and the coupling coefficients, which are usually assigned values with limited experimental support.

23.6 Table of couplings

24 Piezoelectric shear-horizontal surface waves

The thermopiezoelectric coupling of Section 23 changes bulk stiffness, but piezoelectric media also support surface waves that have no elastic counterpart. The simplest is the Bleustein–Gulyaev (BG) wave, a shear-horizontal (SH) wave localised at the surface of a hexagonal (6mm) crystal. It is included here because it is the prototype of electromechanical surface localisation and because its closed form allows a clean statement of how the electrical boundary condition enters.

24.1 Formulation

Let the half-space $y > 0$ be a 6mm crystal with the polar axis along z and motion $u_z(x,y,t)$ polarised along z, with wave number k along x. The relevant constants are the shear stiffness c_{44}, piezoelectric constant e_{15} and permittivity ε, combined in $K^2 = e_{15}^2/(c_{44}\varepsilon)$ and the stiffened modulus $\bar{c} = c_{44}(1+K^2)$ with bulk speed $\bar{v} = \sqrt{\bar{c}/\rho}$. In the quasi-static approximation the fields are $u = A\,e^{-\alpha y}$ and $\varphi = (e_{15}/\varepsilon)\,u + B\,e^{-ky}$, with $\alpha = k\sqrt{1 - v^2/\bar{v}^2}$, and in the vacuum $y < 0$ the potential is $\varphi = C\,e^{ky}$. The boundary conditions at $y = 0$ are a free surface $\sigma_{yz} = c_{44}u_{,y} + e_{15}\varphi_{,y} = 0$ together with either of the following:

- *metallised surface*, $\varphi = 0$;
- *free (unmetallised) surface*, continuity of φ and of D_y across the interface with vacuum permittivity ε_0.

24.2 Results

Eliminating the amplitudes gives the closed forms

$$\frac{\alpha}{k} = \frac{K^2}{1+K^2} \quad \text{(metallised)}, \qquad \frac{\alpha}{k} = \frac{\varepsilon_0}{\varepsilon+\varepsilon_0}\,\frac{K^2}{1+K^2} \quad \text{(free)}, \tag{69}$$

so that the surface wave speed is $v = \bar{v}\sqrt{1-(\alpha_*/k)^2}$ with α_*/k from Eq. (69). These were checked against the numerical roots of the 3×3 secular determinant (script `scripts/check_bg.py`), which agree to the digits quoted. For $K^2 = 0.2$, $\bar{v} = 1.0954\,v_t$ (with $v_t = \sqrt{c_{44}/\rho}$), the metallised speed is $1.0801\,v_t$, i.e. 1.4% below $\bar{v}$, while for the free surface $v = 1.0954$, 1.0953 and $1.0916\,v_t$ for $\varepsilon_0/\varepsilon = 0.01$, 0.1 and 1. For $K^2 = 0.5$ the metallised value is $1.1547\,v_t$ against $\bar{v} = 1.2247\,v_t$.

Two consequences follow. First, the shift between free and metallised speeds, the usual electromechanical coupling measure of a surface wave, is of order K^4 and therefore small when K^2 is small. Second, the decay length $1/\alpha$ is very sensitive to the electrical condition: for $K^2 = 0.2$ it is about $1/k \cdot 6$ (roughly one wavelength) when metallised, but about $600/k$ (roughly 100 wavelengths) for a free surface with $\varepsilon_0/\varepsilon = 0.01$. A plate of a few wavelengths thickness therefore does not support the free-surface BG wave as a surface wave in a strict sense.

24.3 Thermal coupling

For this SH polarisation with uniform fields in z the shear strains $\varepsilon_{xz}, \varepsilon_{yz}$ do not couple to the (diagonal) thermal-stress tensor of a hexagonal crystal, and the pyroelectric coupling acts along the polar axis z, which is perpendicular to the in-plane electric field. The BG wave is therefore not coupled to temperature in this geometry, and thermal effects enter only through the temperature dependence of c_{44}, e_{15} and ε, which shifts the speed without introducing a loss. Coupled thermo-piezoelectric surface waves exist for other cuts and polarisations, where the sagittal-plane motion has a dilatational component, and for those the Rayleigh-type determinant of Section 18 has to be extended by the electrical conditions.

25 Solution methods and verification

25.1 Analytical methods

For linear problems with constant coefficients, the classical analytical tools carry over from elasticity and heat conduction with only the algebra changed by the coupling.

- **Plane-wave and normal-mode analysis.** For unbounded or periodic media, (26) gives complete frequency-domain information. Real frequency with complex wavenumber describes spatial decay under steady excitation; real wavenumber with complex frequency describes temporal decay of a free disturbance. The two attenuation measures coincide only for weak damping ($Q^{-1} \ll 1$), and the relation involves the group velocity: $\operatorname{Im}\omega \approx -v_g \operatorname{Im} k$. Using the wrong measure when comparing with an experiment is a common source of apparent inconsistency.
- **Integral transforms.** For half-spaces, layers and cylinders, Laplace transformation in time and Fourier (or Hankel) transformation in space reduce the problem to ordinary differential equations with exponential solutions, as in Sections 16 and 18. Carrying the heat-flux law as the function $G(s)$ means that one derivation covers the whole template.
- **Potential methods and eigenfunction expansion.** Helmholtz potentials decouple shear from dilatation in the bulk, and eigenfunction expansion in the transform domain is often used for finite domains [111].
- **Wavefront expansions.** Discontinuities propagate along characteristics determined by the high-frequency limit; (32) is an example. Jump amplitudes follow transport equations along the front and decay exponentially in LS-type theories.

25.2 Numerical inversion of Laplace transforms

Closed forms in the transform domain are typically inverted numerically. The commonly used algorithms are the Stehfest method [194], Durbin's Fourier-series method [195] (and the closely related method of Dubner and Abate [196]), the Honig–Hirdes method [197], and contour-deformation methods such as Talbot's [166]; Abate and Whitt [198] review the common framework. Because applications to thermoelastic problems often report numbers from one of these methods without a check, we benchmarked four implementations (Talbot, Stehfest, de Hoog et al. [199], and Cohen acceleration [200], as implemented in a standard multiprecision library with default parameters at 15 digits) on four transforms with known inverses, chosen to represent the features of thermoelastic responses: a smooth decay, an oscillation, a discontinuous

step with a delay, and a diffusive response (Table 8). Four points are worth recording. (i) For

Table 8: Maximum absolute error of four numerical Laplace inversion methods on test transforms (evaluated at $t = 0.5, 2, 5, 10$; for the step at $t = 0.5, 0.95, 1.05, 2, 5$). Default parameters, 15 digits. "4×10^{193}" indicates a numerical blow-up.

Test function	Talbot	Stehfest	de Hoog	Cohen
smooth: $1/(s+1)^2$	6.9e−18	7.1e−13	9.8e−18	6.9e−18
oscillatory: $1/\sqrt{s^2+1}$	3.9e−2	6.6e−4	3.6e−16	6.1e−16
step: e^{-s}/s	3.8e193	1.8e−1	1.4e−6	1.8e−1
diffusive: $e^{-\sqrt{s}}/s$	1.1e−16	5.5e−14	1.1e−16	1.1e−16

smooth and diffusive responses all four methods are essentially exact ($< 10^{-12}$), which is why most published diffusion-dominated results are insensitive to the method. (ii) For the step with a delay, Talbot's method *fails catastrophically* at times before the delay ($t = 0.95 < 1$): its contour, which extends into the left half-plane, is multiplied by e^{-s} and grows without bound. This is exactly the situation of a Cattaneo front, whose transform contains $e^{-x/c_T \cdot s}$. The remedy used in Section 16 is to set the solution to zero ahead of the known front and invert only for $t > x/c_T$. (iii) Stehfest and Cohen give errors of order 0.2 near the discontinuity (they smear the front and ring), while the de Hoog method yields 1.4×10^{-6} at the tested points. (iv) For oscillatory functions the default Talbot setting is inadequate at $t = 10$ (error 0.04), while de Hoog and Cohen are accurate. The methods can be tuned (number of terms, contour parameter), and the table is not a ranking; its message is that the choice and the checks matter. We recommend: invert with two independent methods, compare against a known limit, and treat results near discontinuities as unreliable beyond the known front position.

25.3 Time-domain numerics

For problems that do not admit transform solutions (heterogeneous, nonlinear, complex geometry) finite-element and finite-difference methods in space and time are used. Two issues are specific to hyperbolic thermal models. First, schemes designed for parabolic heat equations (implicit, with numerical dissipation) smear the thermal front and then cannot represent the finite-speed effect that motivates the model; explicit or high-resolution schemes with controlled dissipation, or space–time discontinuous Galerkin formulations, are appropriate. Second, the time step must resolve the relaxation time ($\Delta t \ll \tau_q$) and satisfy a CFL condition based on the *largest* wave speed, which is $\max(c_1, c_T)$ and not c_1 alone: when $c_T \gg c_1$ (as in many metals) the thermal front dominates the stability bound. Memory-dependent and fractional operators require storage of history and, in an unoptimised implementation, an $O(N^2)$ cost in the number of time steps; short-memory approximations and fast convolution reduce this, and should be accompanied by an error estimate. Finite-element implementations with memory-dependent heat transfer in fibre-reinforced cylinders [113] and rotating nano-beams [114] illustrate the approach.

25.4 A verification discipline

We find that the following checks catch most errors in new thermoelastic wave calculations, and we recommend that papers state which they performed.

1. **Reduce to a known limit**: set the coupling to zero (elastic and thermal problems separately), and set the lags to zero (Fourier).
2. **Check the dispersion relation symbolically** against the field equations, as in Appendix A; sign and factor errors in the coupling terms are detected immediately.
3. **Check energy**: for conservative members of the family (GN-II) the energy is constant; for the others, the energy balance (16) must hold with the stated dissipation; for interfaces, the net flux must be continuous (Section 21).
4. **Check high- and low-frequency asymptotics**: adiabatic and isothermal speeds, front speeds (e.g. (32)), classical diffusive scaling.
5. **Check branches**: track roots by continuity and ensure that Taylor truncation has not introduced spurious roots.
6. **Report dimensionless groups**: ε, ω/ω_*, $\omega\tau$, c_T/c_1, $\xi\omega/c_1$, so that results can be compared across studies.

26 A finite-difference benchmark for hyperbolic heat conduction

Time-domain numerics for hyperbolic heat laws are commonly validated against a smooth test problem, which hides the main difficulty: a front that the scheme must carry without spurious oscillation or excessive smearing. This section gives a small benchmark with an exact solution and reports what three standard schemes achieve.

26.1 Problem

The uncoupled Cattaneo system in first-order form,

$$\rho c\,\theta_t + q_x = 0, \qquad \tau\, q_t + q = -k\,\theta_x, \tag{70}$$

on $x > 0$ with $\theta = q = 0$ initially and $q(0,t) = q_0 H(t)$, is solved in units $\rho c = k = \tau = q_0 = 1$, so $c_T = 1$ and the front is at $x = t$. The exact solution is the inverse Laplace transform of Eq. (50), taken as zero ahead of the front. At $t = 5$ it gives $\theta = 1.7615,\ 0.5923,\ 0.0959$ at $x = 1, 3, 4.9$ and the jump at the front is $q_0 e^{-x/2\ell}/(\rho c c_T) = e^{-2.5} = 0.082$.

26.2 Schemes

The first-order system is advanced by Strang splitting: the source $-q/\tau$ is integrated exactly over half steps, and the hyperbolic part (characteristic speeds $\pm c_T$) is advanced by (i) Lax–Friedrichs, (ii) first-order upwind applied to the characteristic variables, or (iii) the second-order Lax–Wendroff scheme, all at Courant number 0.9. The boundary value $q = q_0$ is imposed through a ghost cell. The code is in `scripts/bench_schemes.py`.

26.3 Results

Errors are L^1 norms over the sampled interval at $t = 5$. Over $0.05 \le x \le 4.95$, which includes the front, the observed convergence orders between $N = 800$ and 1600 cells are 0.85 (Lax–Friedrichs), 0.94 (upwind) and 0.58 (Lax–Wendroff); the errors at $N = 1600$ are 2.7×10^{-3},

1.3×10^{-3} and 8.6×10^{-4}. Excluding the region near the front ($0.05 \le x \le 3.5$) the picture is different:

N	100	200	400	800	1600
upwind L^1	9.6×10^{-3}	4.3×10^{-3}	2.1×10^{-3}	1.1×10^{-3}	5.3×10^{-4}
Lax–Wendroff L^1	1.1×10^{-3}	2.7×10^{-4}	6.1×10^{-5}	1.5×10^{-5}	3.8×10^{-6}

with observed orders 1.0 for the upwind scheme and 2.0 for Lax–Wendroff in the smooth region. Thus the formal accuracy is reached away from the front, and the global error is controlled by the neighbourhood of the discontinuity, where all three schemes converge at an order below one. A single global error norm therefore says little about how the scheme carries a front, and a benchmark for hyperbolic heat conduction should report the error in a window around the front and away from it separately.

26.4 A caution about the reference solution

The exact solution above was obtained by numerical Laplace inversion, and the choice of inversion algorithm matters at the accuracy required here. At $x = 1$, $t = 5$ the Talbot method with default settings gave 1.76126, whereas the de Hoog and Stehfest methods give 1.761515 and 1.761515; the finite-difference solution at $N = 800$ agrees with the latter to 2×10^{-6}. A reference solution with an error of 3×10^{-4} would have hidden the second-order convergence of Lax–Wendroff at $N \gtrsim 400$. Inversion results used as benchmarks should be cross-checked with a second algorithm (Section 25.2).

27 Applications

The theories reviewed here are used in four broad application areas. For each, we state which regime of the parameter space is relevant, based on the characteristic numbers derived in the previous sections.

27.1 Laser ultrasonics and short-pulse heating

Laser-generated ultrasound exploits the thermal stress produced by pulsed heating to generate elastic waves [201]. For nanosecond pulses on bulk metals, $\omega\tau_q \ll 1$ and the classical coupled theory with surface heating is adequate for the generated ultrasound; the thermal diffusion length over the pulse duration, $\sqrt{\alpha t_p}$, sets the source depth. For femtosecond–picosecond pulses, the heat-carrier relaxation (electron–phonon coupling, in metals) occurs on time scales where hyperbolic or two-temperature models become relevant, and thin-film heating is modelled with lagging, fractional or memory-dependent laws; analyses of bi-layers under pulsed heating combine these [116]. The benchmark thermal shock of Section 16 shows what to look for: the stress peak at the elastic front and its sensitivity to the heat law, here about 20 % between Fourier and Lord–Shulman for the parameters used.

27.2 Micro- and nano-electromechanical resonators

Thermoelastic damping is a leading loss mechanism in flexural micro-resonators. The classical Zener relaxation (Section 6.4) captures the dependence on thickness and frequency, and the extension to beams and plates by Lifshitz and Roukes [138] is standard. The corrections reviewed here enter in two ways. For ultra-thin structures and very high frequencies, lagging and non-Fourier effects change the damping; DPL-based thermoelastic damping in nonlocal nanoplates [121], couple-stress micro-plates [122] and rotating nano-beams with spatiotemporal nonlocality [114] are examples. And at small sizes nonlocal and gradient elasticity modify the vibration frequencies on which the damping depends. Two cautions follow from our analysis: the form of the nonlocal thermoelastic constitutive law (form A or B, Section 22) changes the predicted damping near the elastic cut-off, and the lag time enters only through $\omega\tau$, so unless the frequency is within one or two decades of $1/\tau_q$, the classical result is adequate.

27.3 Piezoelectric devices, sensors and semiconductors

Thermopiezoelectric coupling affects bulk and surface acoustic wave devices, energy harvesters and sensors, and photothermal effects in piezoelectric semiconductors add carrier dynamics [91, 184, 185]. In the quasi-static, charge-free regime the effect of the electric field on the thermoelastic waves is the exact renormalisation (68); designs that rely on thermal sensing via the pyroelectric effect depend on the sign convention discussed there. Interface problems (substrate–film, electrode–crystal) are the subject of the energy-ratio literature [95, 96, 99, 103, 104], where Section 21 suggests the checks to apply.

27.4 Biological tissue and heterogeneous media

Heat transfer in biological tissue is classically described by the bioheat equation of Pennes [202], which adds blood perfusion to the Fourier equation. Non-Fourier models have been applied to tissue, based on reports of large relaxation times in nonhomogeneous materials [142, 143]; the interpretation of the corresponding measurements has been debated, and the DPL and fractional models have been fitted to such data [46, 124]. For water-like media $\omega_* = \rho c c_1^2/k \approx 10^{13}$ rad/s (with $\rho \approx 10^3$ kg/m^3, $c \approx 4\times10^3$ J/kg K, $c_1 \approx 1.5\times10^3$ m/s, $k \approx 0.6$ W/m K), far above any frequency of interest, and the thermoelastic coupling is weak, so the interest in tissue is in the thermal field itself rather than in thermoelastic waves.

27.5 Geophysical and porous media

Porous and double-porous thermoelastic models with two temperatures are applied to wave propagation in saturated and partially saturated solids [102, 105, 191]; here the additional slow waves, strongly attenuated, carry fluid and heat. We note that the validity of the generalized heat laws in such media, in which the relevant time scales are controlled by pore-scale processes, is largely untested.

27.6 Summary

In all four areas the single most informative dimensionless number is $\omega\tau$ (frequency times the relevant lag), and the second is ω/ω_*. The regime map in Section 31 makes this quantitative.

28 A quantitative guide to application regimes

Section 27 lists application areas. Here the dimensionless groups that decide the heat law are evaluated for six representative scenarios, so that the reader can see at a glance which of the models of this review is relevant. The table is generated by `scripts/table_scenarios.py` from the rounded handbook inputs of Table 3 and a lag $\tau_q = 1\,\mathrm{ps}$; $\delta_T = \sqrt{2\chi/\omega}$ is the thermal penetration depth and $\lambda = c_1/f$ the acoustic wavelength.

Table 9: Dimensionless groups for six scenarios ($T_0 = 300\,\mathrm{K}$, $\tau_q = 1\,\mathrm{ps}$). The assessment column applies the rule: Fourier is adequate if $\omega\tau_q < 0.1$ and $\omega/\omega_* < 0.1$; a lag law may matter if $\omega\tau_q < 1.5$ but then the kinetic validity of the model must be checked; otherwise the continuum heat law itself is questionable.

Scenario	Material	f (Hz)	ω/ω_*	$\omega\tau_q$	δ_T (m)	λ (m)	Assessment
Ultrasonic testing	Aluminium	1.0×10^{7}	1.6×10^{-4}	6.3×10^{-5}	1.8×10^{-6}	6.2×10^{-4}	Fourier adequate
GHz acoustics	Aluminium	1.0×10^{9}	1.6×10^{-2}	6.3×10^{-3}	1.8×10^{-7}	6.2×10^{-6}	Fourier adequate
Picosecond ultrasonics	Aluminium	1.0×10^{11}	1.6×10^{0}	6.3×10^{-1}	1.8×10^{-8}	6.2×10^{-8}	lag may matter; check kinetics
Silicon beam, $h = 1\,\mu$m, at f_Z	Silicon	1.4×10^{8}	1.1×10^{-3}	9.1×10^{-4}	4.5×10^{-7}	6.1×10^{-5}	Fourier adequate
Silicon film, $h = 20\,$nm, at f_Z	Silicon	3.6×10^{11}	2.7×10^{0}	2.3×10^{0}	9.0×10^{-9}	2.5×10^{-8}	continuum law questionable
Copper film, $h = 50\,$nm, at f_Z	Copper	7.3×10^{10}	2.6×10^{0}	4.6×10^{-1}	2.3×10^{-8}	6.2×10^{-8}	lag may matter; check kinetics

28.1 Reading the table

- *Ultrasonic testing and GHz acoustics (aluminium).* With $\omega\tau_q$ between 6×10^{-5} and 6×10^{-3} and ω/ω_* between 1.6×10^{-4} and 1.6×10^{-2}, classical coupled thermoelasticity is adequate. The thermal penetration depth ($1.8\,\mu$m at 10 MHz, $0.18\,\mu$m at 1 GHz) is much smaller than the acoustic wavelength ($620\,\mu$m and $6.2\,\mu$m), so the wave sees the temperature field as a thin boundary layer at interfaces and as a small bulk damping.
- *Picosecond ultrasonics (aluminium, 100 GHz).* Here $\omega/\omega_* = 1.6$ and $\omega\tau_q = 0.63$ and the wavelength (62 nm) is comparable with the thermal penetration depth (18 nm). This is the regime of the regime map (Section 31) where lag and coupling effects could be visible, and also where carrier mean free paths are comparable with the wavelength, so the interpretation of any deviation from the classical result requires a kinetic or atomistic cross-check.
- *Silicon beam, $h = 1\,\mu$m.* At the Zener frequency $f_Z = 1.4\times10^{8}\,$Hz, $\omega\tau_q \approx 10^{-3}$ and the Fourier law with the exact flexural result Eq. (29) is adequate. The thermal penetration depth ($0.45\,\mu$m) is a sizeable fraction of the thickness, which is why the damping peaks here.
- *Silicon film 20 nm and copper film 50 nm.* At the thermal relaxation frequency of the film, $\omega\tau_q$ is 2.3 and 0.46 and ω/ω_* is 2.7 and 2.6. These are the regimes in which a continuum heat law with bulk parameters has to be treated with caution: the film thickness is comparable to or below the carrier mean free path, boundary scattering reduces the effective conductivity, and the relaxation frequency of the continuum model is not reliable.

28.2 Consequences for the choice of theory

1. The classical coupled theory suffices for essentially all ultrasonic, MEMS-scale-above-micron and geophysical-scale applications, and it is the one to use as a baseline in any case.
2. Lag models can only matter in the lower right-hand corner of Table 9: frequencies above about 10^{10} Hz or thicknesses below about 100 nm. In that corner the continuum assumptions are also stressed, so any claim of a lag effect needs an independent test.
3. The Guyer–Krumhansl and nonlocal models (Sections 4 and 22) are the first refinement once size effects are suspected, because they carry a length scale; lag models do not.
4. The electromechanical and magnetic couplings (Section 23) change the stiffness at all frequencies, and are therefore relevant over the whole table, even where the heat law is not.

29 Worked examples

This section shows how the results of the review are used for four practical questions. All numbers follow from the rounded handbook inputs of Table 3 at $T_0 = 300$ K (script `scripts/worked_examples.py`); they illustrate the procedure and should be redone with measured parameters for a specific specimen.

29.1 Example 1: Does a lag law matter for ultrasound in aluminium?

Question. A 1 GHz longitudinal wave propagates in aluminium; should a Cattaneo (Lord–Shulman) law with $\tau_q = 1$ ps be used instead of Fourier's law?

Step 1: characteristic numbers. From Table 3: $c_1 = 6198$ m/s, $\varepsilon = 0.027$, $\omega_* = \rho c c_1^2/k = 3.94 \times 10^{11}$ rad/s. The lag enters through $\tau_q \omega_* = 0.39$, so the thermal front speed $c_T = \sqrt{k/(\rho c \tau_q)} = 9.9$ km/s is $1.6\,c_1$.

Step 2: where the wave sits on the dispersion curves. At 1 GHz the frequency ratio is $\omega/\omega_* = 1.6 \times 10^{-2}$ and $\omega\tau_q = 6.3 \times 10^{-3}$. The Debye loss of Eq. (27) is $Q^{-1} = \varepsilon x/(1+x^2) = 4.3 \times 10^{-4}$ with $x = \omega/\omega_*$; at 1 MHz it is 4.3×10^{-7}.

Step 3: what the lag changes. The lag modifies the response through $\omega\tau_q$. By the truncation analysis of Section 10 the corrections to the thermal branch are at the 10^{-4} level or below for $\omega\tau_q \sim 6 \times 10^{-3}$ (the first-order relative error scales as $(\omega\tau)^2/4$), so the Fourier result is adequate. The lag would matter only at frequencies of order ω_* or above, i.e. tens of GHz, where in addition the phonon mean free path is comparable with the wavelength and the kinetic closure of Section 4 is not valid.

Conclusion. For ultrasonic and MHz–GHz applications in aluminium the parabolic (Biot) theory is the correct model, and a lag law adds parameters without adding accuracy.

29.2 Example 2: Thermoelastic damping of a silicon nanobeam

Question. At which frequency does the thermoelastic loss peak for a silicon beam of thickness h, and how large is it?

Step 1. The diffusivity is $\chi = k/\rho c = 9.2 \times 10^{-5}$ m^2/s. The Zener frequency is $f_Z = \pi\chi/(2h^2)$,

which gives 1.4×10^6 Hz for $h = 10\,\mu$m, 1.4×10^8 Hz for $1\,\mu$m, 1.4×10^{10} Hz for 100 nm and 3.6×10^9 Hz for $h = 200$ nm (a value in between).

Step 2. The relaxation strength is $\Delta = E\alpha_T^2 T_0/c_v = 2.0 \times 10^{-4}$, so the peak loss is $Q_{\max}^{-1} \approx 0.494\,\Delta \approx 1.0 \times 10^{-4}$ (Section 7). This corresponds to a quality factor limit of about 10^4 at the peak, which is a significant limit for sub-micron resonators with intrinsic Q above 10^4.

Step 3: check validity. A micron-size silicon beam at $f_Z \sim 10^8$ Hz has $\omega\tau_q \sim 10^{-3}$ for $\tau_q \sim 1$ ps, so a lag law is not required. For $h \lesssim 100$ nm, bulk values of k are no longer reliable because the phonon mean free path in silicon at room temperature (hundreds of nanometres) exceeds the thickness, and the Guyer–Krumhansl or kinetic description (Section 4) is the relevant model, with a thickness-dependent effective conductivity as input.

29.3 Example 3: Energy partition at an aluminium–copper interface

Question. What fraction of a normally incident longitudinal wave in aluminium is reflected at a welded copper interface (elastic limit)?

The impedances are $Z_1 = \rho c_1 = 1.67 \times 10^7$ and $Z_2 = 4.07 \times 10^7$ kg m^{-2}s^{-1} (aluminium, copper). Then $r = (Z_1 - Z_2)/(Z_1 + Z_2) = -0.417$, $t = 2Z_1/(Z_1 + Z_2) = 0.583$, $R = r^2 = 0.174$ and $T = (Z_2/Z_1)t^2 = 0.826$, with $R + T = 1$ to four digits. The sign of r shows a phase reversal of the reflected displacement wave (impedance increases). When thermal coupling is included, the reflected and transmitted amplitudes acquire corrections of order ε and additional thermal waves are generated; the energy bookkeeping must use the net-flux conservation of Section 21, not the sum of mode-wise energy fractions.

29.4 Example 4: Is a heat pulse in steel hyperbolic?

Question. A surface of steel is heated at $t = 0$. After how long does the Fourier and the Cattaneo solution differ, and by how much in position?

For steel $\chi = 1.3 \times 10^{-5}$ m^2/s. The Fourier diffusion length is $\sqrt{\chi t} = 1.1 \times 10^{-7}$ m at $t = 1$ ns and 3.6×10^{-6} m at $1\,\mu$s. The Cattaneo front speed for $\tau_q = 1$ ps is $\sqrt{\chi/\tau_q} = 3.6$ km/s, so the front has travelled only 3.6×10^{-9} m in one lag time. The two solutions therefore differ only for $t \lesssim$ a few τ_q and within a few nanometres of the surface, where continuum theory and a bulk k are already questionable. Resolving the difference requires a time resolution well below a picosecond, so any such experiment needs ultrashort-pulse (sub-picosecond) heating and detection.

29.5 Lessons

(i) Compute ω_*, ε and $\omega\tau_q$ before choosing a model; the regime map of Section 31 shows at a glance whether lag effects can matter. (ii) For devices below about 100 nm the first deviation from the Fourier picture is size-related, not lag-related. (iii) The elastic interface calculation is a necessary baseline for the thermoelastic one.

30 Experimental evidence and parameter identification

A theory of wave propagation is only as useful as its parameters. This section reviews which experiments probe which model parameters, what they can and cannot discriminate, and how to read published values of lag times and fractional orders.

30.1 Second sound and ballistic heat pulses

The direct signature of wave-like heat transport is second sound, a propagating temperature wave. It was predicted for dielectric crystals at low temperature [29, 31] and observed in solid helium [27] and in sodium fluoride [28, 203], in narrow temperature windows where momentum-conserving phonon collisions dominate over resistive ones. A further observation in graphite above 100 K [204] extends the window to a much higher temperature in a material of high anisotropy. The conditions of these experiments, a hydrodynamic phonon window with $\tau_N \ll \tau_R$, are exactly those under which the Guyer–Krumhansl law (Section 4) is the appropriate description. They do not demonstrate that a Cattaneo or lag law with a single relaxation time describes heat conduction in common solids at room temperature.

30.2 Ultrafast optical methods

Pump–probe reflectivity measures electron–lattice energy exchange on femtosecond to picosecond timescales [205] and picosecond ultrasonics generates and detects coherent acoustic pulses in thin films [206]. In the first, the two-temperature picture of Section 9 is the relevant continuum model, and the measured quantity is an electron–phonon coupling time rather than τ_q. In the second, the heat input launches a strain pulse whose amplitude and shape depend on the thermal diffusion within the first tens of nanometres; thermoelastic coupling is then the source, not a loss mechanism. Time-domain thermoreflectance [207] extracts the conductivity of films and interfaces by fitting a diffusive model with interface resistance, and a deviation from that model at short delay is the usual place where a lag or ballistic correction is claimed.

30.3 Resonator quality factors

Micro- and nano-resonators provide the cleanest tests of thermoelastic damping because Q can be measured against temperature and thickness. The quantitative signature, Eqs. (28) and (29), is a loss peak at $f_Z = \pi\chi/(2h^2)$ with the scaling $f_Z \propto h^{-2}$ and a height proportional to Δ. Verification therefore requires a sweep of thickness or temperature and not a single value, and requires subtracting anchor and surface losses, which are not described by the models of this review. A mismatch can be attributed to a lag or size effect only when these contributions have been bounded.

30.4 Heterogeneous media and biological tissue

The relaxation times τ_q of order seconds reported for non-homogeneous materials, including processed biological tissue [142, 143], reflect heat transfer through a heterogeneous microstructure and are effective macroscopic parameters. Their interpretation as a molecular relaxation

time is not warranted, and the observed values have been disputed in the literature. For tissue the parabolic Pennes bioheat equation [202] remains the baseline against which any hyperbolic correction must be assessed.

30.5 How to read a published lag time

1. Identify what was measured: a response time of a temperature signal, a conductivity, or a quality factor.
2. Identify the model used to extract τ and whether other parameters were fitted simultaneously. A fit of two or more lag parameters from a single curve is usually not identifiable (compare Section 10).
3. Compare $\omega\tau$ with the validity limits of the model (Section 10 for truncated lag laws, Section 4 for the kinetic basis).
4. Check whether a simpler model (Fourier with interface resistance, or Guyer–Krumhansl) explains the same data.

A model parameter that cannot be measured independently of the model that defines it is a fitting constant and not a material property, and papers that tabulate such constants for many materials without an independent test provide limited guidance.

31 Comparative synthesis and decision guide

31.1 When does the heat law matter? A regime map

The question that most users face is whether a generalized heat law will change their prediction. We answer it for the damping and speed of the elastic-type branch by computing the relative difference between Lord–Shulman and classical predictions over the plane of the two relevant dimensionless numbers, the frequency ω/ω_* and the lag $\tau_0\omega_*$ (Figure 24). The coupling is taken as $\varepsilon = 0.02$, typical of metals (Table 3). Three conclusions can be read off. (i) The heat law affects the *damping* much earlier than the *speed*: a 10 % change in Q^{-1} at $\omega \gtrsim \omega_*$ requires $\omega\tau_0 \gtrsim 0.05$–$0.1$, and a 1 % change requires roughly $\omega\tau_0 \gtrsim 0.005$, whereas the phase speed changes by less than 1 % over most of the plane. (ii) At frequencies far below ω_* the difference is negligible unless the lag is very large, because the thermal field is then adiabatic over a period. (iii) In the neighbourhood of $\omega\tau_0 = 1$ differences of tens of percent in Q^{-1} appear. Consequently, a paper that reports a "strong effect of the relaxation time on the phase speed" for realistic lags and ultrasonic frequencies should be examined for the choice of nondimensionalisation: a nondimensional lag chosen for plotting convenience can be far larger than any physical value.

31.2 Taxonomy of models and combinations

Table 10 places the models discussed in this review according to the structure of their heat law (the function $G(s)$), the elastic constitutive law, and the additional fields. A published model is a point in the product of these three choices; the number of models in the literature is large because the product is large, not because there are many physically distinct mechanisms.

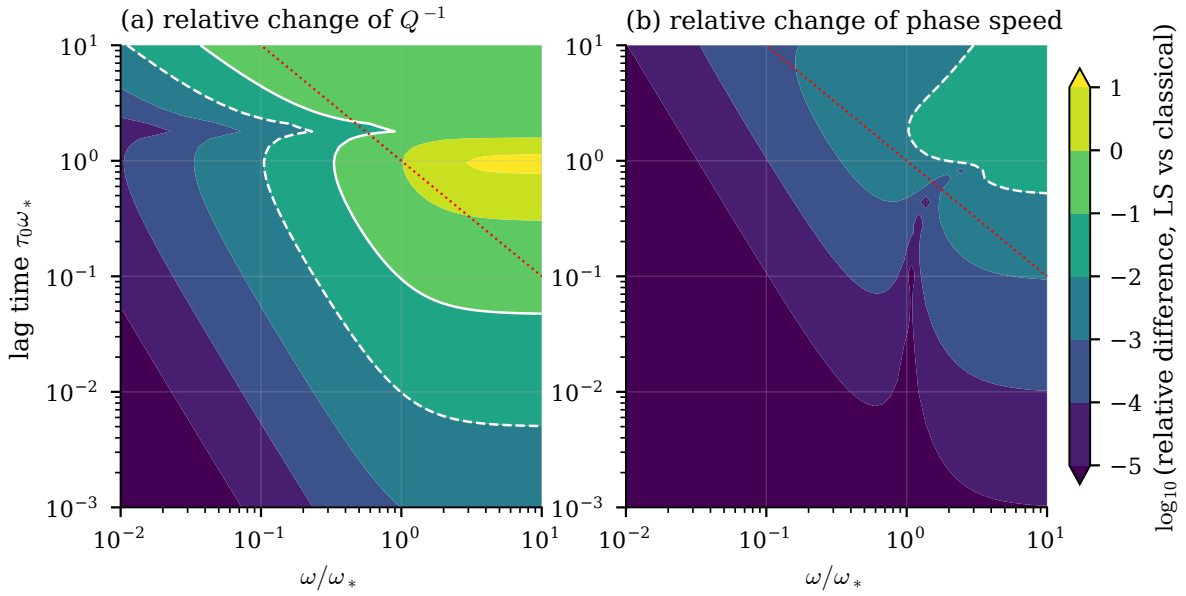


Figure 24: Relative difference between the Lord–Shulman and classical predictions for the elastic-type branch ($\varepsilon = 0.02$): (a) damping Q^{-1}, (b) phase speed. Dashed and solid white contours mark 1 % and 10 % differences; the dotted red line is $\omega\tau_0 = 1$. Near $\tau_0\omega_* \approx 1$ the thermal and elastic speeds coincide ($c_T = c_1$), the branches hybridise, and the labelling "elastic-type" is ambiguous; features in that region should be read qualitatively.

Table 10: Taxonomy. A model is specified by one entry from each of the three groups.

Axis	Options (representative references)
Heat law $G(s)$	Fourier [1]; Cattaneo/LS [32]; Green–Lindsay (modifies coupling) [33]; GN I–III [36, 38]; DPL [40]; TPL [42]; MGT [71]; fractional [56]; MDD [59]; Guyer–Krumhansl [30]; two-temperature variants [62].
Elastic law	Local; Eringen nonlocal [15, 48]; gradient [17, 18]; couple stress [179]; micropolar [13]; Klein–Gordon type [128]; viscoelastic (Kelvin–Voigt) [185].
Added fields	Piezoelectric [11]; semiconductor/photothermal [184]; magnetic [58]; diffusion [119]; voids/porosity [76, 77, 191].

31.3 Comparative table

Table 11 summarises the models along the criteria that determine when each is appropriate.

31.4 Pitfalls checklist

The following issues occur repeatedly in the literature and were each examined in this review.

1. **The $\tau_T \geq \tau_q/2$ condition belongs to the second-order-in-τ_q truncation** of DPL, not to the first-order template; the first-order version is stable for all $\tau_T > 0$ (Section 9).
2. **DPL $\to$ Lord–Shulman** needs $\tau_T \to 0$ at fixed τ_q, not $\tau_q \to 0$; $\tau_T = \tau_q$ gives exactly Fourier's law.
3. **TPL $\to$ DPL** needs $k^* \to 0$; with $k^* \neq 0$ the low-frequency limit changes qualitatively.
4. **Fractional order $\alpha \to 2$ is the wave equation, not Fourier's law.**

Table 11: Comparative summary of heat-conduction models for thermoelastic waves.

Model	Extra parameters	Finite front speed	Well-posed when	Main caution
Classical	0	no	always	Instantaneous thermal signal.
Lord–Shulman	1 (τ_0)	yes (c_T)	$\tau_0 > 0$	Lag also multiplies the coupling term.
Green–Lindsay	2 (τ_0, τ_1)	yes	$\tau_0 \geq \tau_1 \geq 0$	Discontinuities at fronts.
GN II	1 (k^*)	yes (c_*)	$k^* > 0$	No dissipation.
GN III	2 (k, k^*)	no (strict)	both > 0	Not an improved Cattaneo law.
DPL, 1st order	2	no (strict)	no instability found numerically	Both-limits diffusive; $\tau_T = \tau_q$ is Fourier.
DPL, 2nd order in τ_q	2	depends	$\tau_T \geq \tau_q/2$	Ill-posed otherwise.
TPL	4	no (strict)	see [44]	Many parameters; k^* changes low-ω limit.
MGT	2–3	yes (c_T)	parameter conditions	Third order in time.
Fractional	1 order + lag	depends on α	model-dependent	$\alpha \to 2$ is not Fourier.
MDD	1–3	variant-dependent	model-dependent	High-frequency limit depends on kernel.

5. **GN-III does not have a finite thermal front speed**; it is wave-like at low and diffusive at high frequency.
6. **The classical theory is adiabatic at low and isothermal at high frequency**; labels on speeds must say which modulus c_1 refers to.
7. **Branch identification**: follow roots by continuity; nearest-to-ω/c_1 selection fails when branches hybridise.
8. **DPL is diffusive at both ends**; "wave-like" only at intermediate frequency.
9. **The sign of the effect of a hyperbolic heat law on a pulse depends on the boundary condition** (prescribed flux or temperature).
10. **MDD forms differ in their high-frequency limit**; check $s\kappa(s)$ as $s \to \infty$.
11. **Energy ratios in lossy media**: the sum of mode fractions need not equal one because of interference; the net flux must be continuous.
12. **Eringen calibration**: ξ depends on the calibration target (zone boundary versus long wavelength).
13. **Gradient sign**: a positive gradient coefficient gives stiffening dispersion, the opposite of lattice softening.
14. **Nonlocal thermoelastic form** (smoothing the thermal stress or not) changes damping near the cut-off.
15. **Talbot inversion fails for delayed transforms** for times before the delay; set the solution to zero ahead of the front.
16. **Piezo sign conventions**: the sign of ep determines whether pyroelectricity increases or decreases thermal stress.
17. **Report dimensionless groups** so that nondimensional lags can be related to physical values.

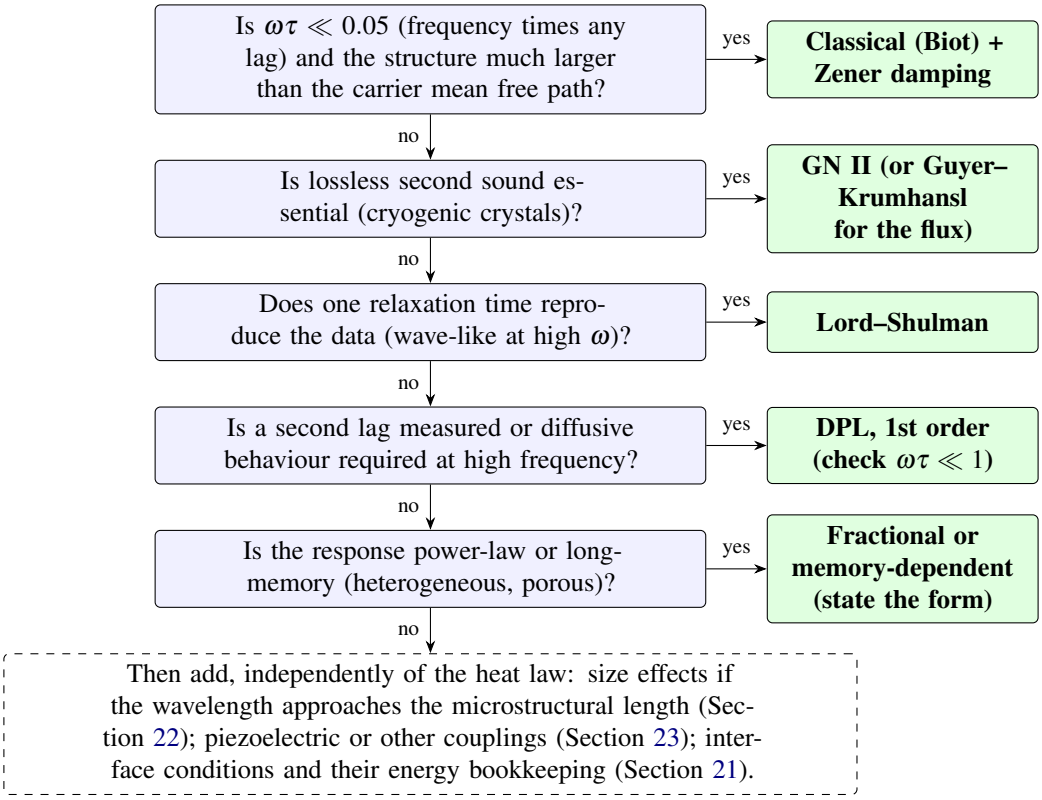


Figure 25: Decision sequence for choosing a starting model. The threshold $\omega\tau \ll 0.05$ follows from the regime map of Figure 24.

31.5 Decision guide

Figure 25 gives a decision sequence for choosing a starting model. It is a guide, not a substitute for checking against data.

32 Reporting standards for new work

The comparisons made in this review were often limited not by the models but by what the original papers report. This section proposes a short set of reporting standards that would make future results directly comparable. They follow from the pitfalls collected in Section 31 and from the checks of Sections 25 and 26.

32.1 What a paper on generalized thermoelastic waves should state

1. **The heat-flux law as an explicit parameter set.** Give the model in the template of Section 3 (or state where it departs from it) so that $G(s)$, the polynomial of Eq. (30) and the limit $\omega \to \infty$ follow without reading the derivation. For lagging models give the truncation order.
2. **Which limits were verified.** State the reductions that were tested numerically or symbolically, for example Fourier, Lord–Shulman and the elastic limit $\varepsilon \to 0$, and give the numerical agreement achieved.
3. **A non-dimensional parameter table.** Report ε, ω/ω_*, $\omega\tau_q$ (and $\omega\tau_T$, α, $\xi\omega_*/c_1$ as relevant) for every figure. A figure for which these are not given cannot be placed on the regime

map of Section 31.

4. **The material source and the status of the parameters.** Say whether a lag time or fractional order is measured, estimated by a stated argument, or assigned for illustration. Illustrative values should not be presented in a table with measured ones.
5. **Branch handling.** For dispersion curves state how roots were selected and followed (Section 5), and mark regions of hybridisation.
6. **Energy statements.** If energy ratios are reported, give the energy flux used (Section 3.5), state whether the ratios sum to one, and if they do not, whether the cross terms between modes were included (Section 21).
7. **Inversion and discretisation.** For transient results state the Laplace-inversion method and the cross-check, or the discretisation and Courant number, and give the error in a window around the front separately from the global error (Section 26).
8. **Boundary and interface conditions.** State the thermal condition (insulated, isothermal, convective) and, at interfaces, the continuity conditions for the heat flux and for the temperature or the thermal displacement.

32.2 Claims that need an independent test

- A lag time or fractional order extracted from a single curve with several free parameters.
- A statement that a model predicts a "new wave" without a check of the number of roots of the dispersion polynomial in each frequency range and of their causality (Section 5).
- A reflection or transmission result that agrees with an energy ratio sum of one only because the thermal wave has been dropped.
- A size effect obtained from a nonlocal law whose length parameter has been chosen by trial rather than calibrated to a lattice or atomistic dispersion curve (Section 22).

32.3 Limitations of this review

The review treats linear waves about a uniform reference state and does not cover nonlinear thermoelasticity, radiative or reactive heat transfer, or the full literature on bounded-domain well-posedness. The numerical checks use rounded handbook values and non-dimensional parameter sets chosen to separate the effects discussed; they illustrate mechanisms and should be repeated with measured parameters before being used for design. The classified catalogue in the appendix is organised by title keywords and does not assess the content of the individual papers. Statements that rest on the cited literature rather than on the computations reported here are identified by their citations.

33 Open problems and research agenda

33.1 Theory

1. **A classification theorem for the template.** For the first-order template we have shown numerically that DPL is stable for all positive lags and identified when the thermal front has finite speed (the behaviour of $sG(s)$ at large s). A complete analytical statement of well-posedness, decay and front structure for the coupled TPL and MGT systems with all

parameters positive, including the conditions on $(\tau_q, \tau_T, \tau_v, k, k^*)$ and the role of the coupling ε, is still lacking in a form directly usable by applied authors.

2. **Thermodynamics of truncated lag models.** The time-differential DPL and TPL models raise questions of thermodynamic compatibility and continuous dependence [44, 45]; a systematic account of which truncations admit an energy functional and entropy inequality, and what restrictions this imposes on the lags, would resolve the confusion about the $\tau_T \geq \tau_q/2$ condition (Section 9).
3. **Memory-dependent and fractional models.** Energy functionals, thermodynamic admissibility, uniqueness and finite-speed properties for the memory-dependent and fractional laws are not available in the generality of the template models. The high-frequency limit of the MDD law, which depends on the kernel (Section 11), suggests that a classification of kernels by $s\kappa(s)$ at large s would be valuable.
4. **Interface conditions for lagged and memory models.** Thermodynamically consistent interface conditions with thermal resistance, for lagged and memory heat laws, including the entropy production at the interface (Section 21).
5. **Mode interference in dissipative media.** A systematic framework for apportioning energy flux among generated waves in lossy media, so that energy ratios have an unambiguous meaning.

33.2 Combined effects

6. **Joint size and memory effects.** Combinations of nonlocal or gradient elasticity with hyperbolic or memory heat laws are widely computed but rarely validated against lattice dynamics or atomistic simulation. A benchmark in which a lattice model with heat carriers supplies the reference dispersion would show which combination is justified.
7. **Calibration of length scales.** The calibration target (zone boundary versus long wavelength) changes ξ by about one third (Section 22); systematic calibration protocols for thermoelastic problems are missing.
8. **Nonlocal forms of thermoelastic coupling.** The two forms (smoothed or unsmoothed thermal stress) differ near the cut-off; the thermodynamically consistent form should be derived from a nonlocal free energy.
9. **Surface and interface thermoelasticity at the nanoscale.** Surface stress, surface heat capacity and Kapitza resistance modify both surface waves and interface partition in ways not captured by the bulk models treated here.

33.3 Experiments and data

10. **Measurement of both branches.** Experiments that measure the elastic-type and the thermal-type branch in the same solid across the crossover frequency would discriminate among heat laws; at present most evidence is indirect.
11. **Independent determination of lags.** The relaxation times and lags are generally obtained by fitting; independent measurement (for example from pump–probe experiments together with ultrasonic damping) would allow a test of the models rather than a fit.
12. **Thermoelastic damping at ultra-high frequency.** Resonators operating at gigahertz frequencies reach the regime where lagged corrections matter; systematic data on damping versus frequency and size would calibrate them.

33.4 Computation

13. **Benchmark suite.** A set of agreed benchmark problems (the heat pulse, thermal shock, Rayleigh wave and interface problems of this review) with reference solutions at specified precision would improve the reliability of numerical codes.
14. **Efficient memory operators.** Fast algorithms for fractional and memory-dependent operators in time-domain codes, with error control.

34 Conclusions

Generalized thermoelasticity is not a collection of unrelated theories but a family organised by the function $G(s)$ that relates heat flux to temperature gradient, combined with choices of elastic law and added fields. This organisation makes three kinds of result possible. First, exact statements about reductions and limits, including a clarification that the often-quoted DPL stability condition belongs to a specific truncation and that GN-III has no finite front speed. Second, a single dispersion polynomial that gives comparative speeds, attenuation and thermoelastic damping, and that shows that the heat law affects damping long before it affects speed, and only for $\omega\tau \gtrsim 0.05$. Third, problem-level insights: finite fronts and surface jumps in heat pulses, the structure of thermoelastic Rayleigh waves, the renormalisation by piezoelectric coupling, calibration dependence of nonlocal lengths, and the distinction between net energy flux and mode-by-mode fractions at interfaces in lossy media. Many of the open questions concern validation rather than derivation: independent measurement of lags, joint validation of size and memory effects against lattice dynamics, and benchmark solutions. We hope that the unified notation, the verified reductions, the pitfalls checklist and the decision guide help readers choose, justify and check models, and that the open problems direct effort where it is most needed.

A Derivation and verification of the dispersion polynomial

With $\partial_x \to \mathrm{i}k$ and $\partial_t \to s = -\mathrm{i}\omega$, the momentum and energy balances (25) for the template give the linear system

$$\left(\rho s^2 + \rho c_1^2 k^2\right)\hat{u} + \mathrm{i}\beta k\,\hat{\theta} = 0, \tag{71}$$

$$T_0\beta\, s\,\mathrm{i}k\,\hat{u} + \left(s\rho c + G(s)k^2\right)\hat{\theta} = 0, \tag{72}$$

where the second equation uses $\mathrm{i}k\,\hat{q} = -\mathrm{i}k\,G\,\mathrm{i}k\,\hat{\theta} = Gk^2\hat{\theta}$. Its determinant is $(\rho c_1^2 k^2 + \rho s^2)(s\rho c + Gk^2) + \beta^2 T_0\, s\, k^2$ (since $(\mathrm{i}k)(\mathrm{i}k) = -k^2$), which is (26) because $\rho s^2 = -\rho\omega^2$. The identity was checked symbolically with all template parameters kept as symbols (script `check_dispersion.py`); a first version of our own implementation had the opposite sign for the coupling term, and the symbolic comparison detected it immediately, which is the reason we advocate the check.

Unified form for LS and GL. Multiplying by $1 + \tau_0 s$ for $G = k/(1 + \tau_0 s)$ gives (30) with $\tau^\dagger = \tau_0$; the Green–Lindsay system of (33) gives the same polynomial with $\tau^\dagger = \tau_1$, since the thermal-stress term carries $(1 + \tau_1 s)$.

Limits. Setting $\beta \to 0$ decouples the roots into $K = -s^2/c_1^2$ and $K = -s\rho c/G$; for $G = k$ the latter is $k^2 = \mathrm{i}\omega/\alpha$ with phase speed $\sqrt{2\alpha\omega}$.

Nonlocal forms. If the stress law is $(1-\xi^2\partial_x^2)\sigma = c\,u_{,x} - \beta\theta$ (form A of Section 22), the momentum balance becomes $[\rho s^2(1+\xi^2 K) + \rho c_1^2 K]\hat{u} + \mathrm{i}\beta k\hat{\theta} = 0$ and the determinant is $[\rho s^2(1+\xi^2 K) + \rho c_1^2 K](s\rho c + GK) + \beta^2 T_0 sK = 0$; for form B the last term is multiplied by $1+\xi^2 K$.

B Characteristic numbers and parameter estimates

For a given solid and a given heat-flux model, the dimensionless groups that control the problem are collected in Table 12. Note that $c_T/c_1 = 1/\sqrt{\tau_0\omega_*}$: the thermal front is faster than the

Table 12: Dimensionless groups for thermoelastic wave problems.

Group	Definition	Meaning
ε	$\beta^2 T_0/(\rho^2 c c_1^2)$	adiabatic–isothermal modulus difference
ω_*	$\rho c c_1^2/k$	frequency at which conduction over a wavelength matches the period
$\omega\tau$	$\omega\tau_q$	frequency times flux lag
c_T/c_1	$\sqrt{k/(\rho c\tau_0)}/c_1 = 1/\sqrt{\tau_0\omega_*}$	thermal-to-elastic front speed
τ_T/τ_q		lag ratio (DPL)
$\xi\omega_*/c_1$		nonlocal length in thermal units
K^2	$e^2/(\varepsilon_e c)$	electromechanical coupling

elastic one for $\tau_0\omega_* < 1$ and slower for $\tau_0\omega_* > 1$, and branch hybridisation occurs at $\tau_0\omega_* \approx 1$ (Figure 24). With the values of Table 3 and a lag time of one picosecond, $\tau_0\omega_* \approx 0.4$ for aluminium, i.e. a thermal front somewhat faster than the elastic one.

C Reproducibility: scripts and what each one verifies

All numerical results of this review are produced by Python scripts supplied with the manuscript. Table 13 lists them with the checks they perform.

Limits of verification. The scripts verify the mathematics of the models stated here against their own field equations and limiting cases. They do not verify that a model describes a given material; that requires comparison with experiment. Material values in Table 3 are rounded handbook-type numbers used for illustration.

D Classified catalogue of the cited literature

This appendix lists every work in the reference database, grouped by the subject of its title. Works that are not discussed in the main text are included as further reading for the corresponding topic. The grouping is automatic and by title keywords only; it is an aid to locating the

literature and does not summarise the content of any paper. Within each group the entries are ordered by year. A work whose title touches several topics appears once, under the first matching group.

D.1 Second sound, phonon transport and kinetic theory

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D.2 Dual-phase-lag, three-phase-lag and delay models

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D.3 Green–Naghdi, Green–Lindsay and extended thermodynamics

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E Glossary and summary of working formulas

E.1 Abbreviations

BG	Bleustein–Gulyaev (shear-horizontal piezoelectric surface wave)
CD, classical	Fourier heat conduction with Biot's coupled thermoelasticity (parabolic)
DPL	dual-phase-lag heat conduction
EIT	extended irreversible thermodynamics
GK	Guyer–Krumhansl heat conduction (phonon hydrodynamics)
GL	Green–Lindsay thermoelasticity (two relaxation times in the constitutive law)
GN I, II, III	Green–Naghdi types I, II, III (thermal displacement formulation)
LS	Lord–Shulman thermoelasticity (Cattaneo law with one relaxation time)
LR	Lifshitz–Roukes (exact flexural thermoelastic damping)
MDD	memory-dependent derivative
MGT	Moore–Gibson–Thompson heat conduction (third order in time)
SH, SV	shear-horizontal, shear-vertical polarisations
TED	thermoelastic damping
TPL	three-phase-lag heat conduction

E.2 Working formulas

The following relations are used repeatedly. Each is derived or checked in the section named in the last column; the symbols are those of Table 1.

Quantity	Formula	Where
Thermoelastic coupling	$\varepsilon = \beta^2 T_0/(\rho^2 c c_1^2)$	Section 3
Thermal characteristic frequency	$\omega_* = \rho c c_1^2/k$	Section 6
Adiabatic speed	$c_1\sqrt{1+\varepsilon}$ at $\omega \ll \omega_*$	Section 6.3
Thermal front speed (LS)	$c_T = \sqrt{k/(\rho c \tau_0)}$	Section 8
Lag–frequency number	$c_T/c_1 = 1/\sqrt{\tau_0 \omega_*}$	Appendix B
Debye loss (weak coupling)	$Q_x^{-1} \simeq \varepsilon x/(1+x^2)$, $x = \omega/\omega_*$	Section 6.4
Loss–dispersion sum rule	$\int Q_x^{-1} d\ln\omega = \pi(v_0 - v_\infty)/\bar{v}$	Section 5
Finite-front criterion	$sG(s) \to$ finite, non-zero as $s \to \infty$	Section 5
Zener loss, beam flexure	$Q^{-1} = \Delta\,\omega\tau_Z/(1+\omega^2\tau_Z^2)$, $\tau_Z = h^2/\pi^2\chi$	Section 7
Relaxation strength (beam / plane strain)	$\Delta = E\alpha_T^2 T_0/c_v$ / $\times(1+\nu)/(1-\nu)$	Section 19
Cattaneo mode transition	$k_c = 1/(2\sqrt{\chi\tau})$	Section 17
Cubic-crystal coupling	$\varepsilon_{\boldsymbol{n}} = \beta_0^2 T_0/(c_v M_{\boldsymbol{n}})$	Section 14
Welded interface (elastic)	$r = (Z_1 - Z_2)/(Z_1 + Z_2)$, $R + T = 1$	Section 21
Truncation accuracy (1st order)	relative error $\simeq (\omega\tau)^2/4$ for small $\omega\tau$	Section 10
GK high-frequency limit	$k_w^2 \to i\omega\tau_R/(3\ell^2)$	Section 4

E.3 Where to start for common questions

- *Is the heat law relevant at my frequency?* Compute ω_* and $\omega\tau$ (Section 31); the regime map shows when LS and classical differ.

- *Which model has finite speed?* See the comparative table in Section 31 and the criterion in Section 5.
- *How large is thermoelastic damping in a resonator?* Use Eq. (29) with the plane-strain factor when the width exceeds several thicknesses, and check kh (Section 19).
- *How do I validate a code?* Use the checks in Sections 25 and 26, including the sum rule and the elastic limits.

F Derivation of the weak-coupling relaxation form

This appendix derives the Debye-type loss used throughout (Eq. (27)) directly from the dispersion polynomial, and shows that it is the response of a standard linear solid. We use the classical theory in units $\rho = c_1 = c = k = 1$, so $\omega_* = 1$ and $s = -i\omega$.

F.1 Perturbation of the elastic root

The polynomial of Eq. (30) with Fourier heat flux reads

$$(K+s^2)(s+K)+\varepsilon sK = 0 \quad \Longleftrightarrow \quad K^2 + K\,(s+s^2+\varepsilon s)+s^3 = 0. \tag{73}$$

For $\varepsilon = 0$ the elastic root is $K = -s^2 = \omega^2$. Writing $K = -s^2 + \delta$ and keeping terms linear in ε and δ gives $\delta\, s(1-s) = \varepsilon s^3$, hence

$$K = -s^2 + \frac{\varepsilon s^2}{1-s} = \omega^2\Big[1 - \frac{\varepsilon}{1+i\omega}\Big], \qquad k \simeq \omega\Big[1 - \frac{\varepsilon}{2(1+i\omega)}\Big]. \tag{74}$$

In dimensional form $\omega \to \omega/\omega_*$ in the denominator. The real and imaginary parts of k give

$$\frac{\omega}{\operatorname{Re} k} \simeq 1 + \frac{\varepsilon}{2}\,\frac{1}{1+x^2}, \qquad Q_x^{-1} = \frac{2\operatorname{Im} k}{\operatorname{Re} k} \simeq \frac{\varepsilon x}{1+x^2}, \quad x = \frac{\omega}{\omega_*}, \tag{75}$$

that is, the phase speed falls from $c_1(1+\varepsilon/2)$ at $x \ll 1$ to c_1 at $x \gg 1$, and the loss peaks at $x = 1$ with the value $\varepsilon/2$. The numerical solution of the full polynomial gives a peak of 0.0501 at $\varepsilon = 0.1$ (Section 12), and the ratio $Q_{\max}^{-1}/\varepsilon = 0.4999998$ at small ε (script `fig_zener_q.py`).

F.2 Interpretation as a standard linear solid

The result is equivalent to a frequency-dependent modulus. From $\rho\omega^2/K = M(\omega)$,

$$M(\omega) \simeq M_{\text{iso}}\Big[1 + \frac{\varepsilon}{1+i\omega/\omega_*}\Big], \tag{76}$$

where $M_{\text{iso}} = \rho c_1^2$. The limits are the adiabatic modulus $M_{\text{iso}}(1+\varepsilon)$ at low and the isothermal modulus M_{iso} at high frequency, and the relaxation time is $1/\omega_*$. This is the standard linear solid with relaxation strength ε and relaxation time $\tau = 1/\omega_*$, whose loss is $\operatorname{Im}M/\operatorname{Re}M \simeq \varepsilon x/(1+x^2)$. Zener's flexural result Eq. (28) is the same relaxation form with strength Δ and relaxation time τ_Z determined by transverse heat flow across the thickness.

F.3 Consistency with the sum rule

Integrating the loss over $\ln\omega$ gives $\int_0^\infty \varepsilon x/(1+x^2)\,d\ln x = \pi\varepsilon/2$, and the change in phase speed between the two limits is $\varepsilon/2$ in units of c_1. The two satisfy the Kramers–Kronig sum rule Eq. (21), with the exact coupled solution checked numerically in Section 5.

F.4 Range of validity

The expansion assumes $\varepsilon \ll 1$ and is accurate to $O(\varepsilon^2)$. For $\varepsilon = 0.1$ the exact peak is 0.0501 against 0.0500, but the bar-mode losses at small k differ from the first-order form by about 13% because of the difference between temporal and spatial damping (Section 5). For lag laws, ω_* is replaced by a function of $\omega\tau$ and the form is no longer a single relaxation; the comparison of Figs. 10 and 17 then applies.

G Parameter register for the figures

In line with the reporting standards of Section 32, Table 14 lists the parameters behind every computed figure. Quantities are in the non-dimensional units of the scripts ($\rho = c_1 = c_v = 1$, lengths in c_1/ω_*, times in $1/\omega_*$, so $k_{\mathrm{cond}} = 1$) unless stated otherwise. "Branch" indicates the branch plotted and "tracked" that continuity in the frequency or wavenumber was used.

The values of ε used in the figures (0.02 to 0.3) are larger than most solids in Table 3 ($\varepsilon \lesssim 0.03$) so that the effects are visible; results scale with ε at weak coupling. The values of $\tau\omega_*$ are likewise chosen to bring the lag effects into the plotted range, and the corresponding physical frequencies are those of Table 9.

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Table 13: Scripts and the identities or limits they verify.

Script	Content and checks
`check_dispersion.py`	Symbolic determinant of the field equations equals the dispersion polynomial (26) (difference zero).
`check_reductions.py`	Reductions of $G(s)$: DPL→LS, TPL→DPL, GN-II/III, large-s limits.
`check_fronts.py`	LS and GL front speeds from the polynomial agree with (32) to four digits.
`check_gn2.py`	GN-II speeds are non-dispersive and equal the formula.
`check_thermopiezo.py`	Thermopiezoelectric determinant is proportional to the renormalised thermoelastic polynomial (68).
`fig_dispersion.py`, `fig_zener_q.py`	Dispersion and damping; weak-coupling Debye result confirmed ($Q^{-1}/\varepsilon = 0.4999998$).
`fig_stability.py`	Growth rates for the 1st- and 2nd-order DPL truncations.
`fig_heatpulse.py`	Cattaneo pulse by Talbot inversion; Fourier limit and front exactness.
`fig_shock.py`	Thermal shock of a coupled half-space; boundary conditions satisfied to machine precision.
`fig_rayleigh.py`	Thermoelastic Rayleigh root; $\varepsilon \to 0$ limit gives the elastic Rayleigh speed.
`interface_te.py`, `fig_interface_energy.py`	Elastic and thermoelastic interface; impedance limit to twelve digits; net-flux continuity to 10^{-15}.
`fig_lattice.py`, `fig_nonlocal_te.py`	Lattice calibration of ξ; nonlocal thermoelastic branches; $\xi \to 0$ and $\varepsilon \to 0$ limits.
`fig_fractional_mdd.py`	Caputo heat equation (erfc check) and MDD kernel transform ($\kappa(0) = 1/3$).
`fig_regime_map.py`, `bench_laplace.py`, `table_materials.py`	Regime map; Laplace-inversion benchmark; material coupling table.
`fig_lag_truncation.py`	Exact delay law versus first- and second-order truncations; relative errors; sign change of attenuation at $\omega\tau = \pi/2$.
`fig_flexTED.py`, `lamb_te.py`, `fig_lamb_te.py`	Zener versus Lifshitz–Roukes; plate determinant matches exact Rayleigh–Lamb to 10^{-10} in the elastic limit and the Lifshitz–Roukes loss (plane-strain Δ) with ratio 1.000 at small kh.
`fig_frac_coupled.py`, `fig_bar_modes.py`, `fig_gk.py`	Coupled fractional dispersion; clamped-bar modal loss; Guyer–Krumhansl thermal branch (high-frequency limit confirmed to 0.2%).
`fig_aniso.py`, `fig_reflection.py`, `check_bg.py`	Cubic-crystal coupling by direction; free-surface reflection (elastic limit to 10^{-10}); Bleustein–Gulyaev closed forms versus secular determinant.
`bench_schemes.py`, `check_kk.py`	Finite-difference orders for the Cattaneo front; Kramers–Kronig sum rule to 10^{-6}.
`worked_examples.py`, `table_scenarios.py`, `gen_catalogue.py`, `gen_refs_new.py`	Numbers of the worked examples; application-regime table; classified catalogue of the references; assembly of `refs_new.bib` (Crossref records and author-library entries, de-duplicated by DOI and title).

Table 14: Parameters used for the computed figures.

Script (figure)	Models	Parameters
`fig_dispersion`	classical, LS, DPL ($\tau_T = \tau_q/2$)	$\varepsilon = 0.1$, $\tau\omega_* = 0.25$ (thermal front $2c_1$); branches tracked
`fig_zener_q`	classical, LS, DPL	$\varepsilon = 0.02$ and 0.1 for the Debye check; lag panels $\varepsilon = 0.1$, $\tau\omega_* = 0.25$
`fig_stability`	DPL, first and second order in τ_q	$\varepsilon = 0.1$, $\tau_q\omega_* = 1$, τ_T/τ_q varied
`fig_thermal_family`	Fourier, LS, DPL, GN-II, GN-III, MGT	uncoupled; $k^* = 0.25$ where used, $\tau_q = 1$, $\tau_T = 0.5\tau_q$
`fig_lag_truncation`	delay, 1st, 2nd order	uncoupled, $\tau = 1$
`fig_fractional_mdd`	Caputo, MDD	$\alpha = 0.5, 1, 1.5, 2$ in panel (a); $\rho c = k = \tau = 1$ in panel (c)
`fig_frac_coupled`	fractional Cattaneo	$\varepsilon = 0.1$, $\tau\omega_* = 0.25$, $\alpha = 1, 0.8, 0.6$ and classical
`fig_heatpulse`	Fourier, LS	uncoupled, $\alpha = \tau = k = q_0 = 1$
`fig_shock`	classical, LS, DPL	$\varepsilon = 0.1$, boundary-temperature step, 1D half-space
`fig_rayleigh`	classical, LS	$\varepsilon = 0.05$, $\nu = 0.3$, $\tau\omega_* = 0.25$; insulated and isothermal
`fig_lamb_te`	classical	$\varepsilon = 0.05$, $\nu = 0.3$, $h = 20\,c_1/\omega_*$, insulated; tracked
`fig_flexTED`	Zener, Lifshitz–Roukes	normalised by Δ and ω_Z
`fig_bar_modes`	classical, LS	$\varepsilon = 0.1$, $\tau\omega_* = 0.25, 1$; tracked
`fig_reflection`	classical, LS	$\varepsilon = 0.1$, $\nu = 0.3$, $\tau\omega_* = 0.25$, $\omega/\omega_* = 0.1, 1$; insulated
`fig_interface`	elastic welded interface	impedance ratio $Z_2/Z_1 \in [10^{-1.5}, 10^{1.5}]$; closed forms
`fig_interface_energy`	LS, two media	medium 1: $\rho = c_1 = c_v = k = 1$, $\varepsilon = 0.1$, $\tau = 0.25$; medium 2: $\rho = 2$, $c_1 = 1.2$, $c_v = 0.8$, $k = 0.6$, $\varepsilon = 0.3$, $\tau = 0.15$
`fig_nonlocal`, `fig_lattice`	Eringen, gradient, lattice	$\xi = \ell = 0.5$ in units of the length scale in the axis labels; lattice fit $\xi^2/a^2 = 1/4 - 1/\pi^2$
`fig_nonlocal_te`	nonlocal thermoelastic, forms A, B	$\varepsilon = 0.1$, $\xi = 0.3\,c_1/\omega_*$, classical heat flux
`fig_regime_map`	LS versus classical	$\varepsilon = 0.02$; $\omega/\omega_* \in [10^{-2}, 10]$, $\tau\omega_* \in [10^{-3}, 10]$
`fig_aniso`	classical (cubic)	rounded handbook values; $T_0 = 300\,\mathrm{K}$
`fig_gk`	Fourier, Cattaneo, GK	uncoupled; $\chi = \tau_R = 1$, $\ell^2 = 0.05, 0.3$